

EE/CE 252/251

Signals and Systems

with

Dr. Naveed R. Butt

@

Habib University

Signal Basics III

Models

Common signals

Complex Numbers

Quick revision

Examples

Some practice
problems

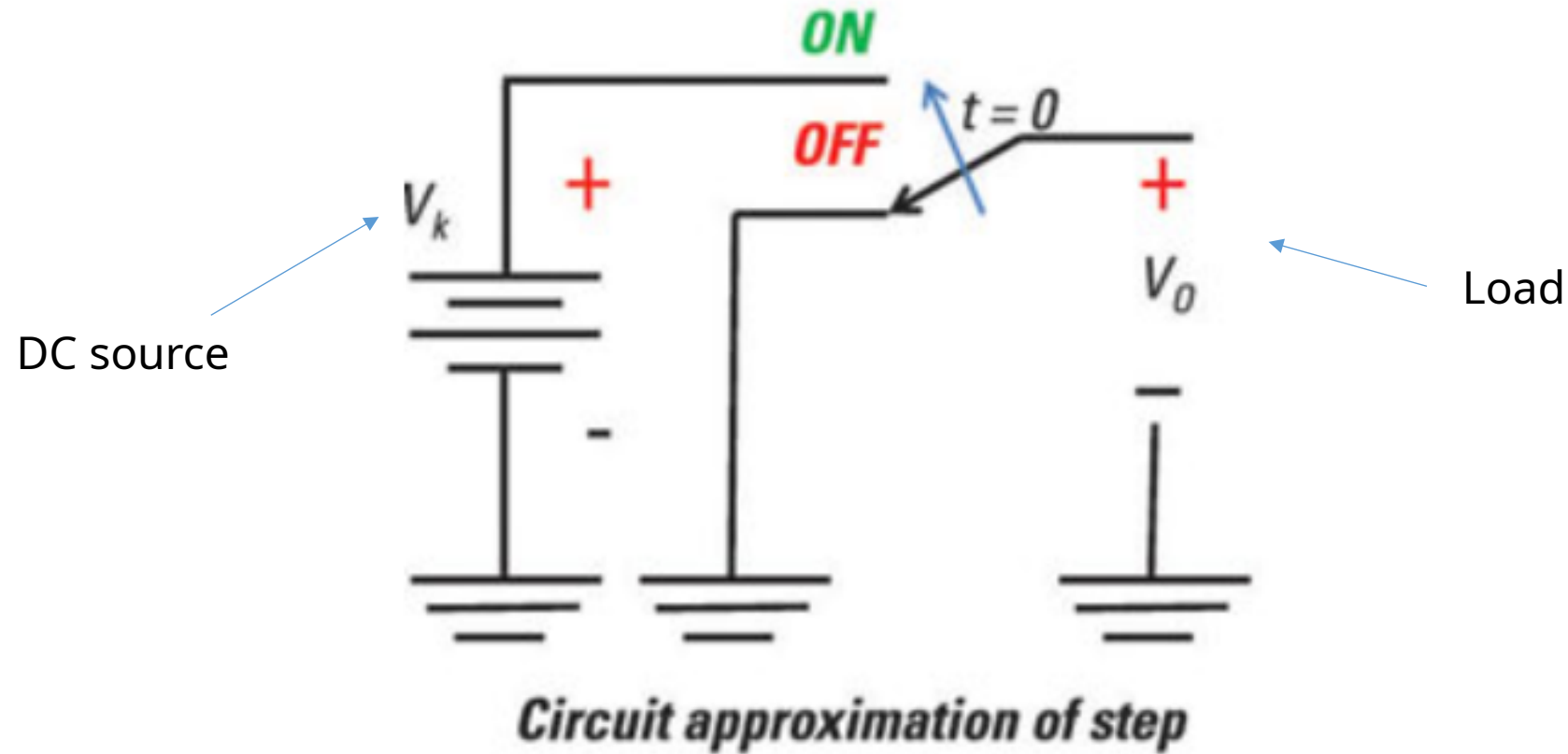
We now look at some signals that show up again and again and again and...

Unit Step

(kind of a SWITCH ON action)

Unit Step

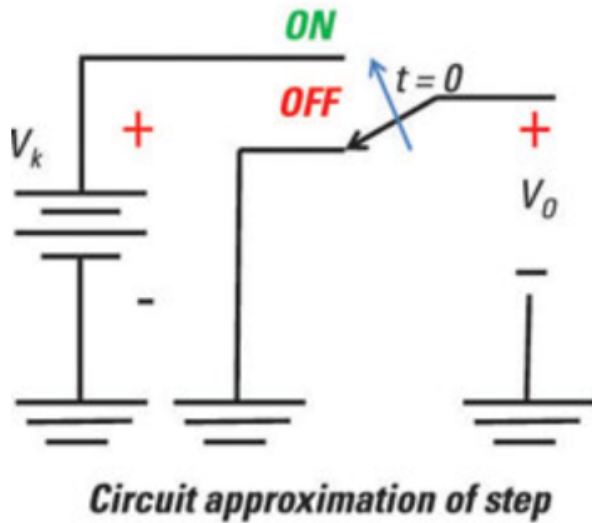
(kind of a SWITCH ON action)



Unit Step

Mathematically and graphically speaking...

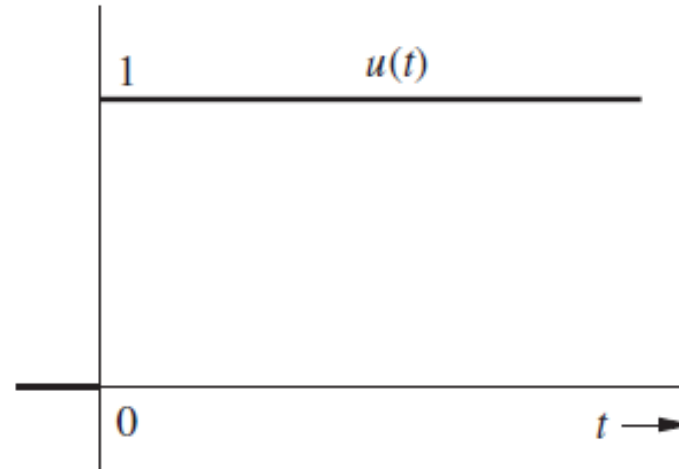
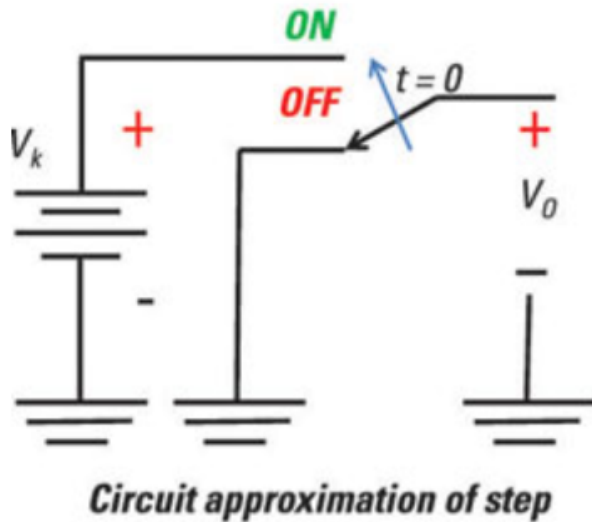
Continuous-Time Case



Unit Step

Mathematically and graphically speaking...

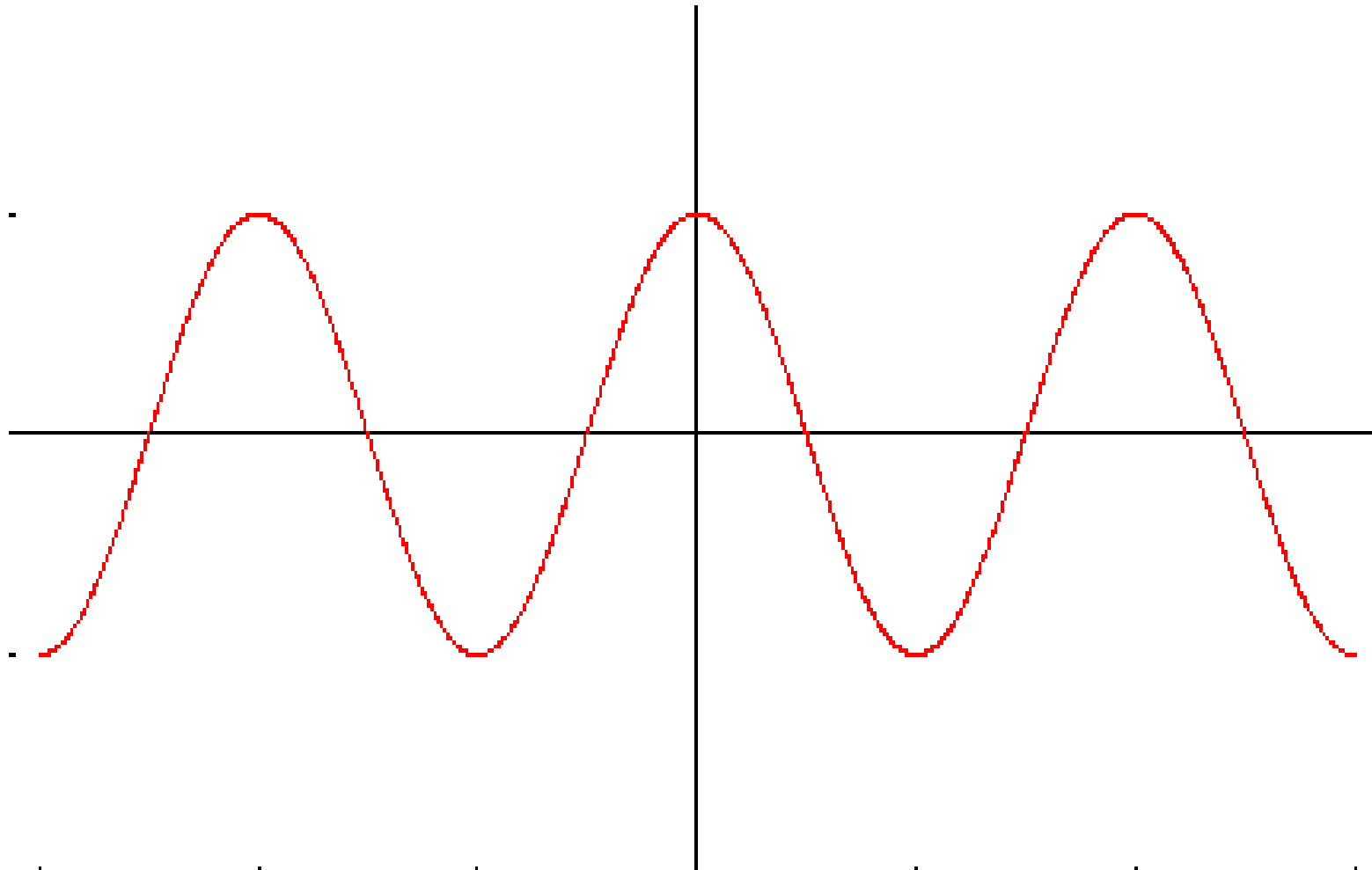
Continuous-Time Case



$$u(t) = \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases}$$

Q. How can we limit a signal so it doesn't start before $t = 0$?

$$x(t) = \cos(\omega_0 t)$$

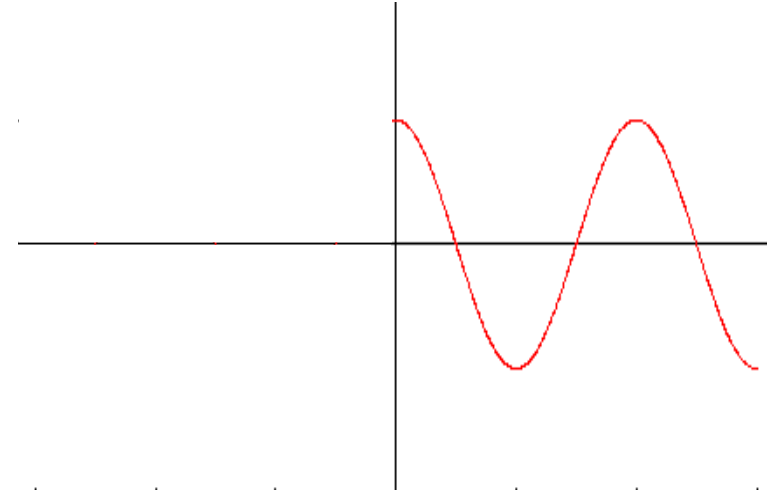


Method 1: Write it as a piece-wise function.

$$y(t) = \begin{cases} \cos(\omega t), & t \geq 0 \\ 0, & t < 0 \end{cases}$$

Method 1: Write it as a piece-wise function.

$$y(t) = \begin{cases} \cos(\omega t), & t \geq 0 \\ 0, & t < 0 \end{cases}$$

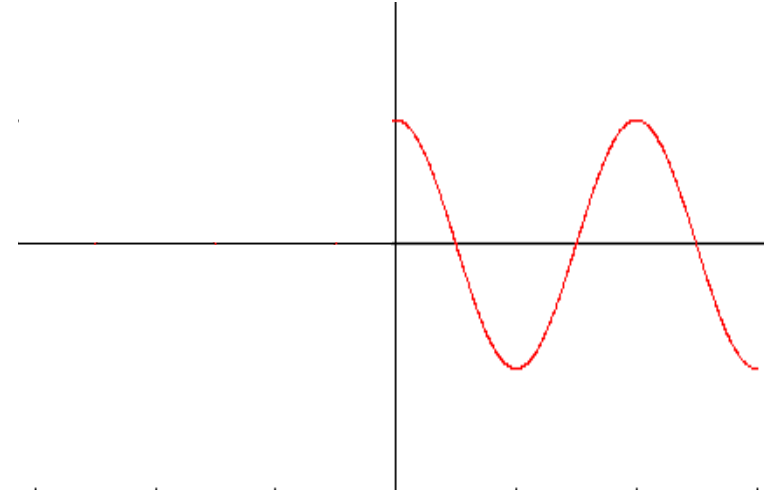


Method 1: Write it as a piece-wise function.

$$y(t) = \begin{cases} \cos(\omega t), & t \geq 0 \\ 0, & t < 0 \end{cases}$$

Method 2: Use step function!

Can you guess
how?

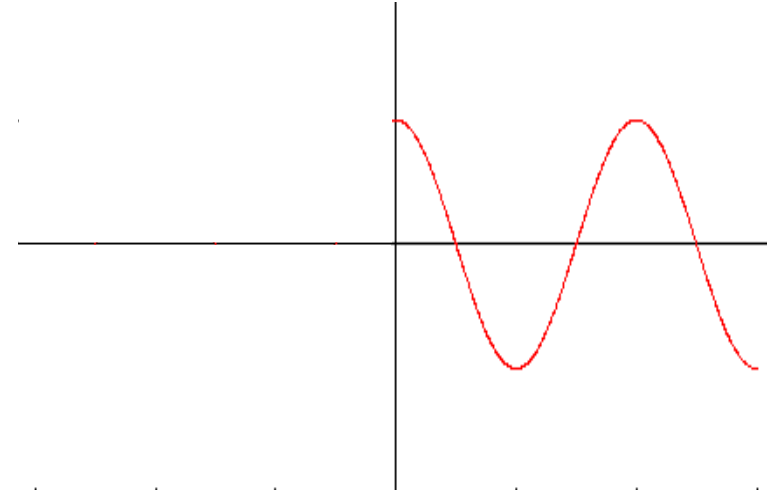


Method 1: Write it as a piece-wise function.

$$y(t) = \begin{cases} \cos(\omega t), & t \geq 0 \\ 0, & t < 0 \end{cases}$$

Method 2: Use step function!

$$y(t) = \cos(\omega t) \times u(t)$$

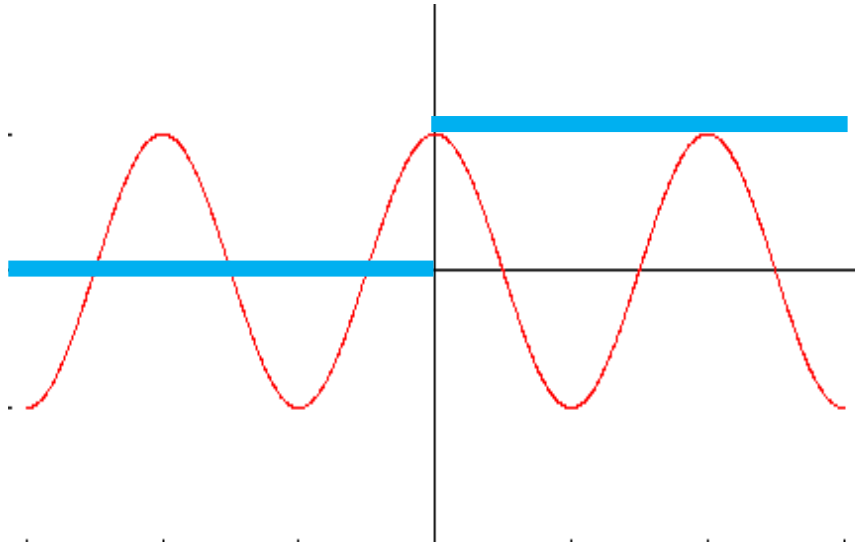
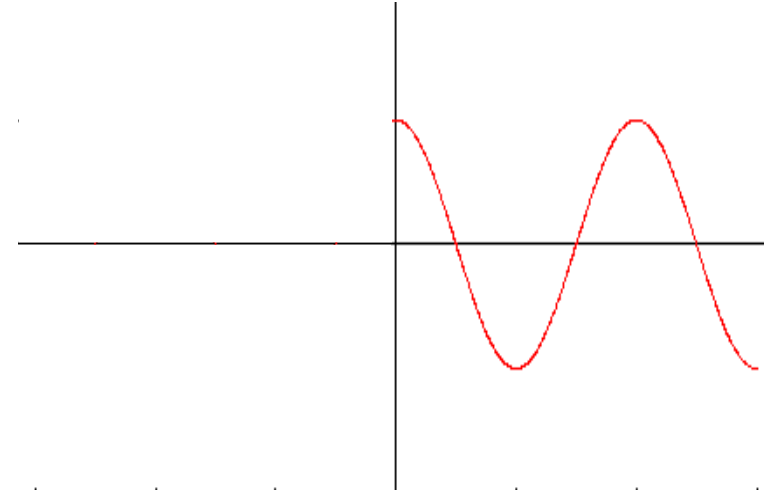


Method 1: Write it as a piece-wise function.

$$y(t) = \begin{cases} \cos(\omega t), & t \geq 0 \\ 0, & t < 0 \end{cases}$$

Method 2: Use step function!

$$y(t) = \cos(\omega t) \times u(t)$$

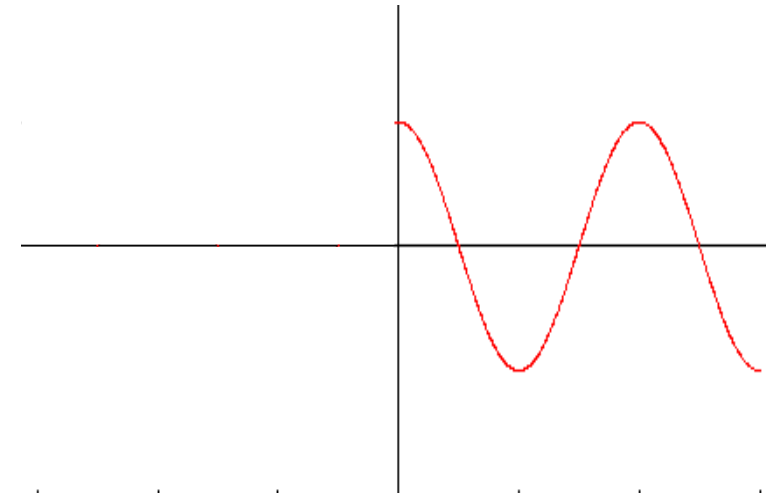
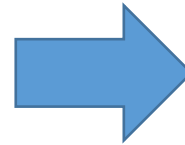
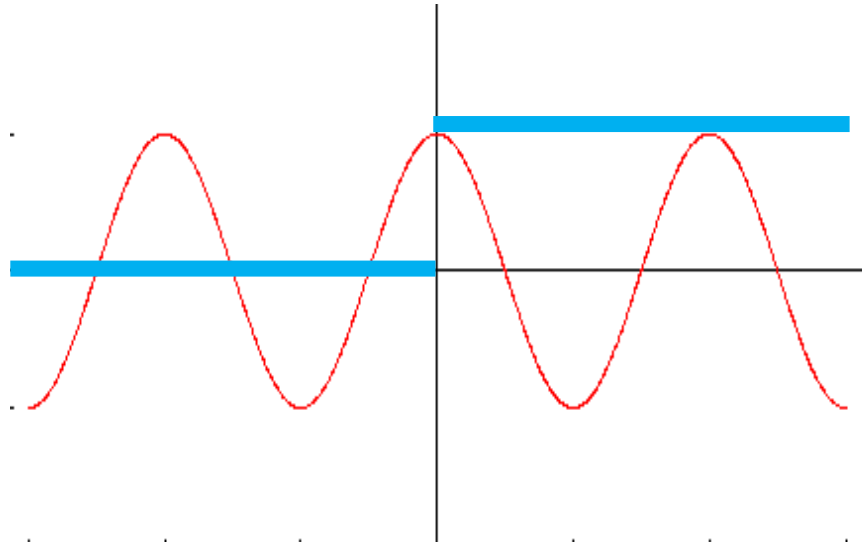
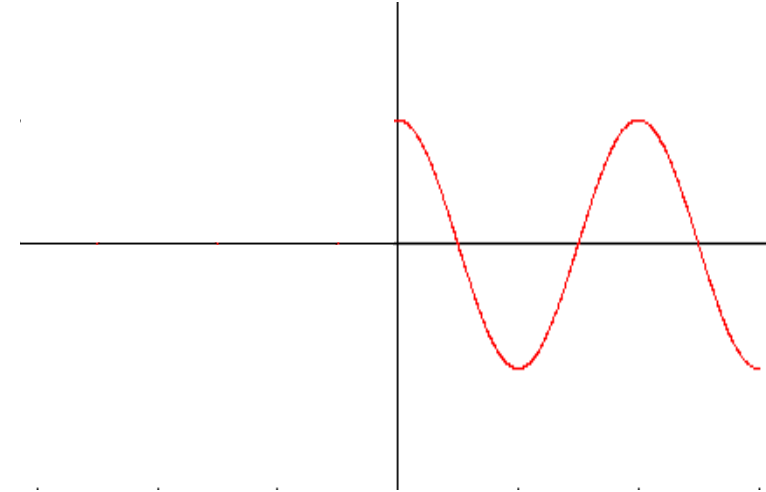


Method 1: Write it as a piece-wise function.

$$y(t) = \begin{cases} \cos(\omega t), & t \geq 0 \\ 0, & t < 0 \end{cases}$$

Method 2: Use step function!

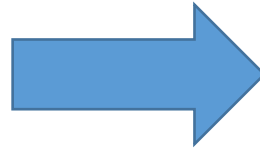
$$y(t) = \cos(\omega t) \times u(t)$$



Multiplication with unit step can help create piece-wise functions.

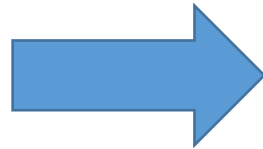
Multiplication with unit step can help create piece-wise functions.

$$x(t) = \begin{cases} e^{-at}, & t \geq 0 \\ 0, & t < 0 \end{cases}$$



Multiplication with unit step can help create piece-wise functions.

$$x(t) = \begin{cases} e^{-at}, & t \geq 0 \\ 0, & t < 0 \end{cases}$$



$$x(t) = e^{-at}u(t)$$

Operations on Unit Step

Operations on Unit Step

Can you write relation for and draw $u(t - T)$?

Operations on Unit Step

Can you write relation for and draw $u(t - T)$?

$$u(t - T) = \begin{cases} 1, & t - T \geq 0 \\ 0, & t - T < 0 \end{cases}$$

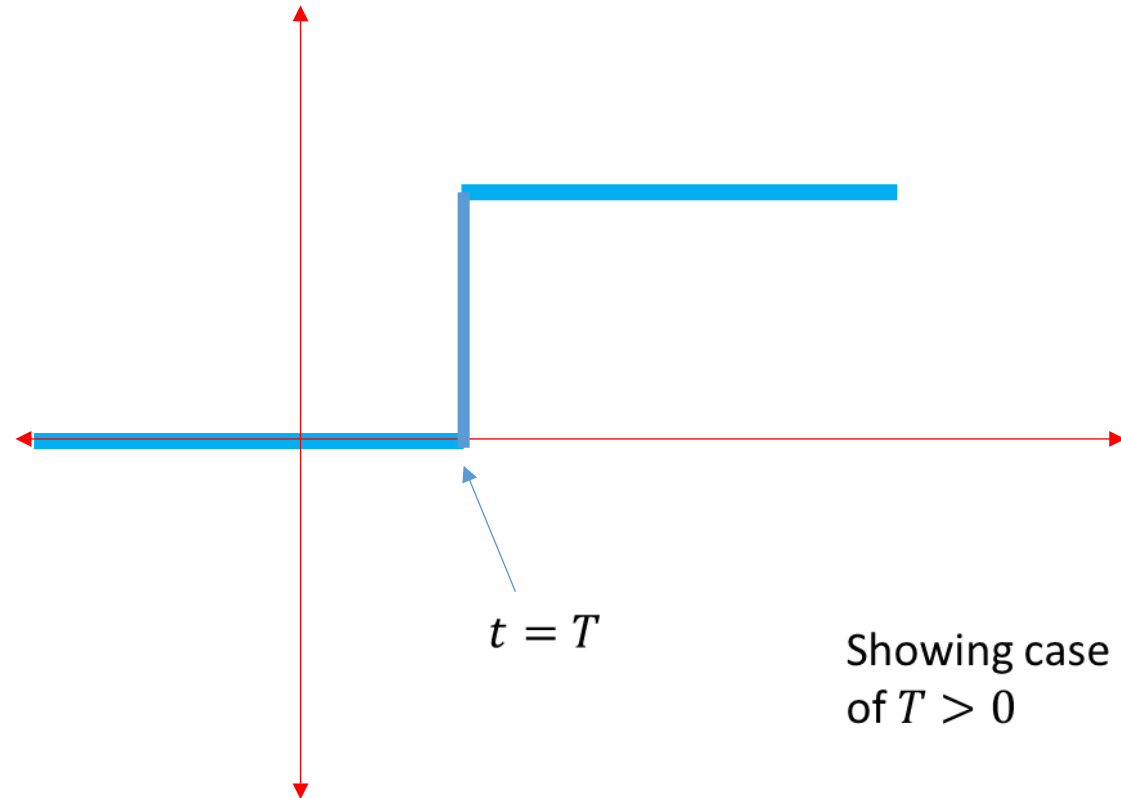
$$u(t - T) = \begin{cases} 1, & t \geq T \\ 0, & t < T \end{cases}$$

Operations on Unit Step

Can you write relation for and draw $u(t - T)$?

$$u(t - T) = \begin{cases} 1, & t - T \geq 0 \\ 0, & t - T < 0 \end{cases}$$

$$u(t - T) = \begin{cases} 1, & t \geq T \\ 0, & t < T \end{cases}$$



Showing case
of $T > 0$

Operations on Unit Step

Can you write relation for and draw $u(-t)$?

Operations on Unit Step

Can you write relation for and draw $u(-t)$?

$$u(-t) = \begin{cases} 1, & -t \geq 0 \\ 0, & -t < 0 \end{cases}$$

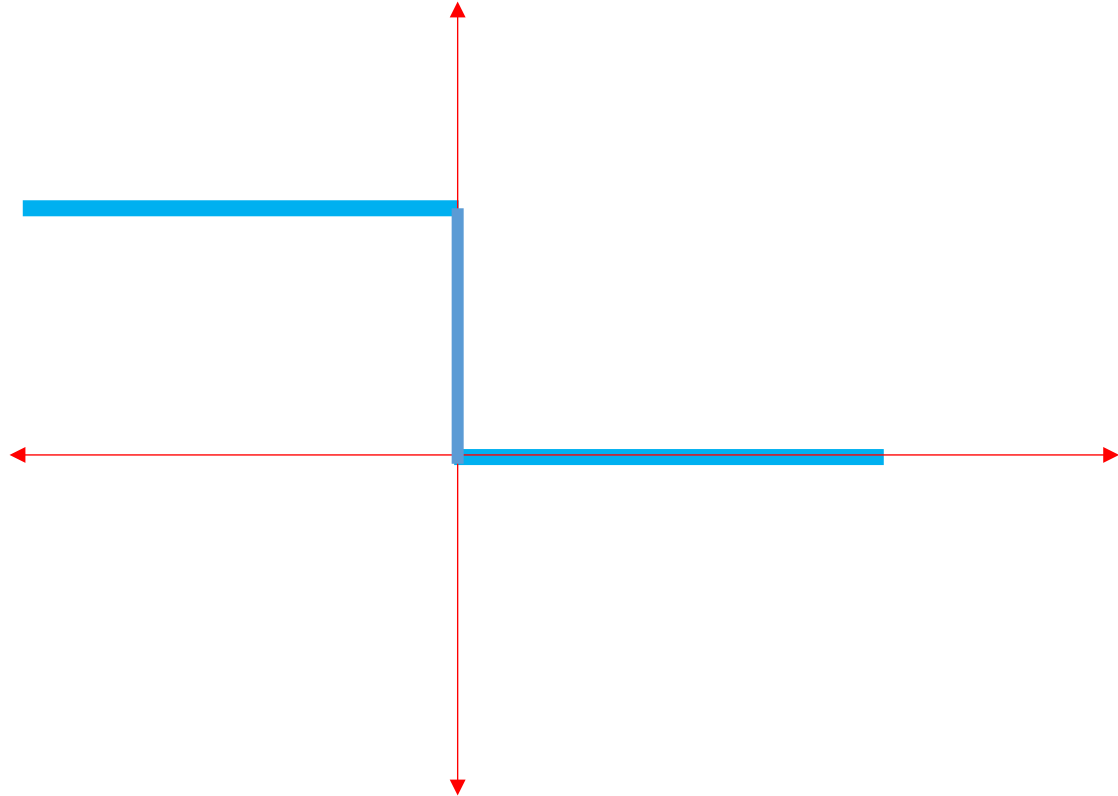
$$u(-t) = \begin{cases} 1, & t \leq 0 \\ 0, & t > 0 \end{cases}$$

Operations on Unit Step

Can you write relation for and draw $u(-t)$?

$$u(-t) = \begin{cases} 1, & -t \geq 0 \\ 0, & -t < 0 \end{cases}$$

$$u(-t) = \begin{cases} 1, & t \leq 0 \\ 0, & t > 0 \end{cases}$$



Operations on Unit Step

Can you write relation for and draw $u(2t)$?

Operations on Unit Step

Can you write relation for and draw $u(2t)$?

$$u(2t) = \begin{cases} 1, & 2t \geq 0 \\ 0, & 2t < 0 \end{cases}$$

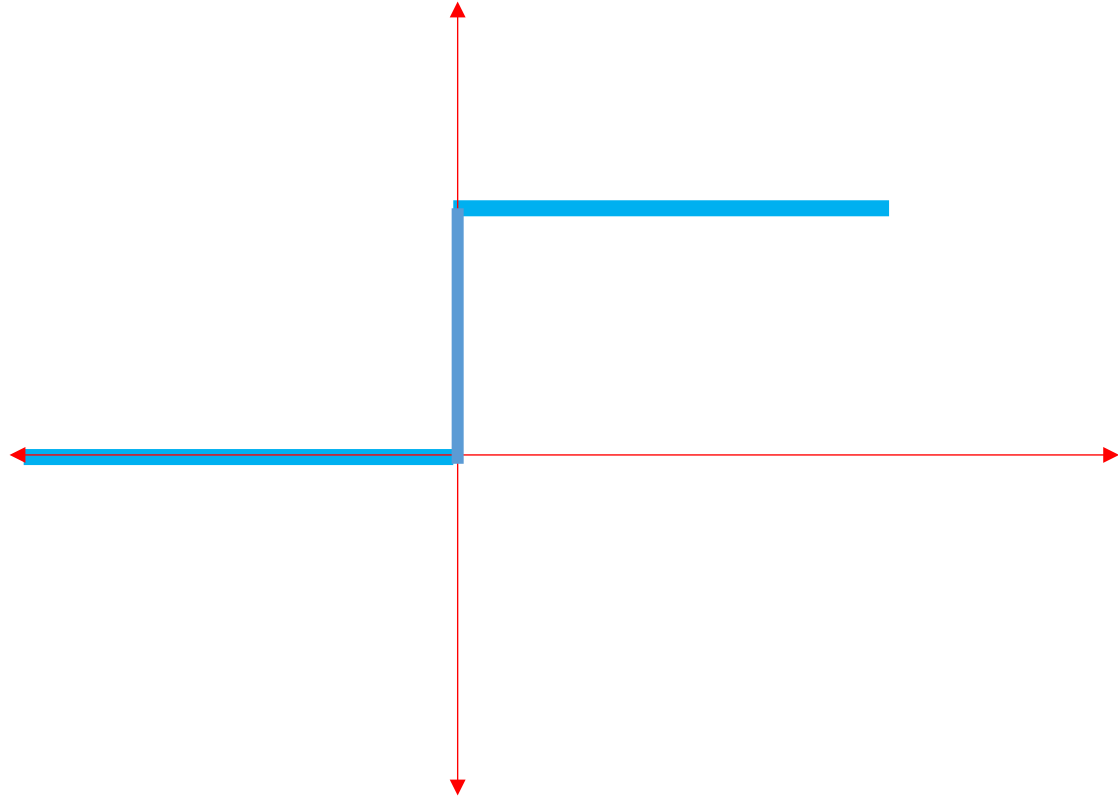
$$u(2t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

Operations on Unit Step

Can you write relation for and draw $u(2t)$?

$$u(2t) = \begin{cases} 1, & 2t \geq 0 \\ 0, & 2t < 0 \end{cases}$$

$$u(2t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

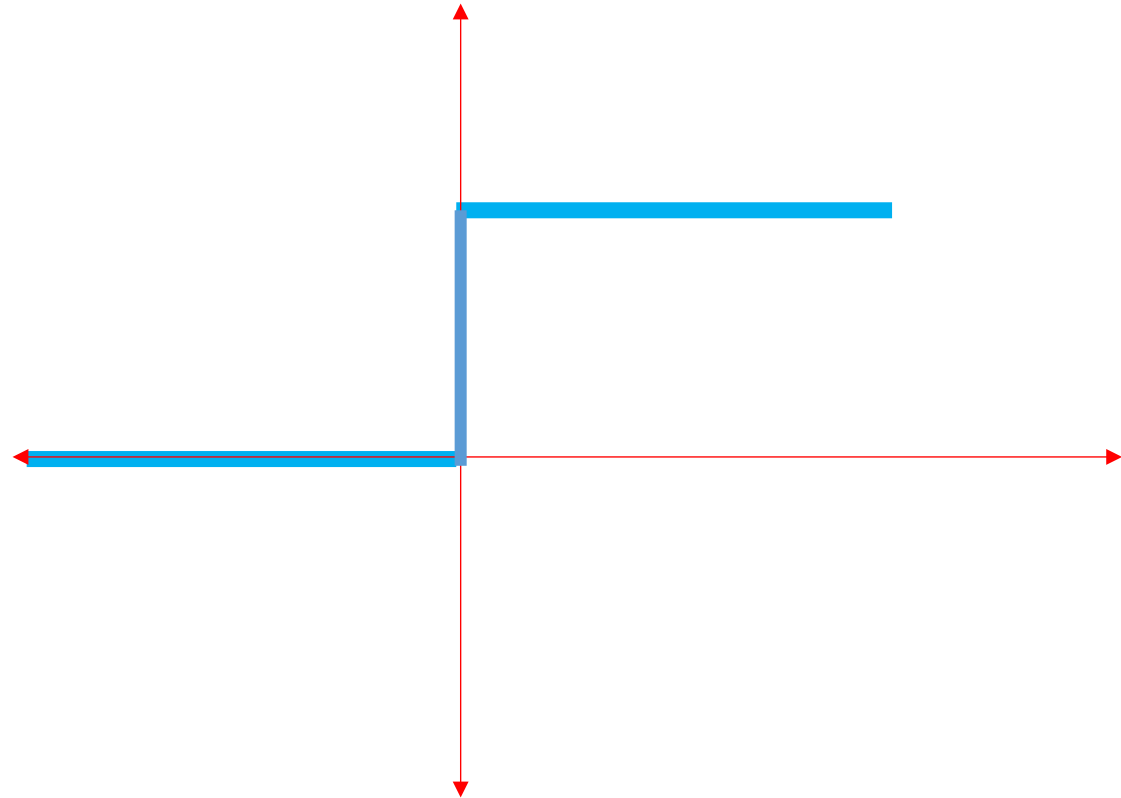


Operations on Unit Step

Can you write relation for and draw $u(2t)$?

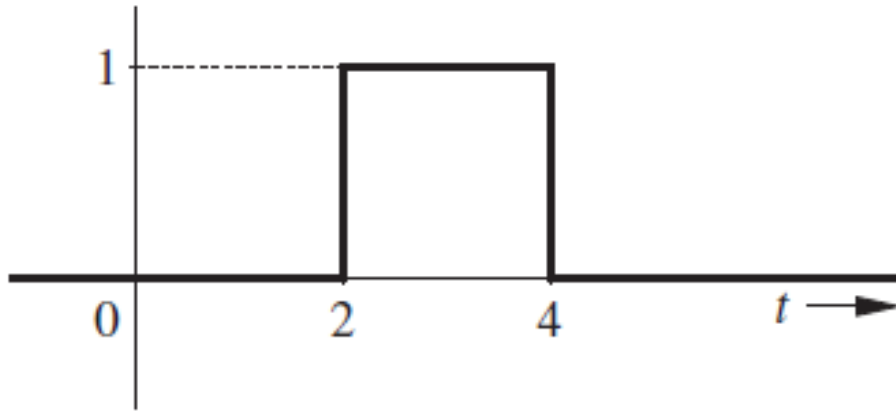
$$u(2t) = \begin{cases} 1, & 2t \geq 0 \\ 0, & 2t < 0 \end{cases}$$

$$u(2t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

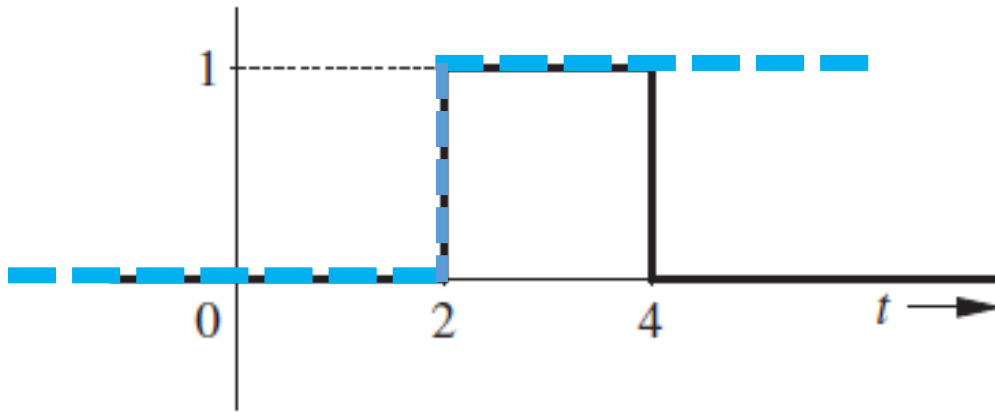


Lesson: Unit step is invariant to compression and expansion!

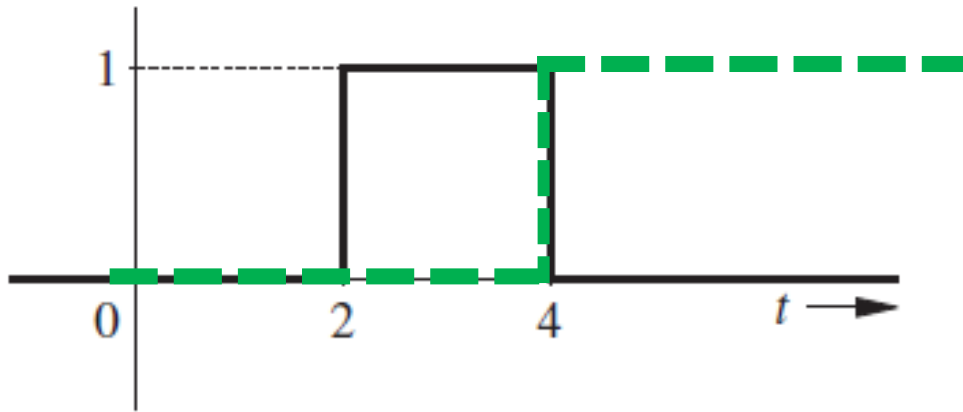
Can you make off-on-off
(rectangular pulse) using two Unit
Step functions?



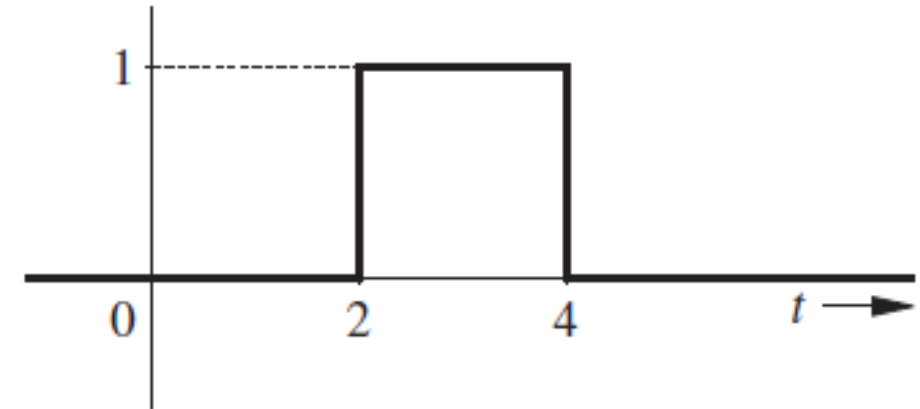
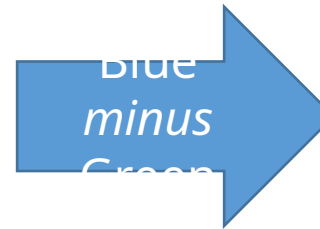
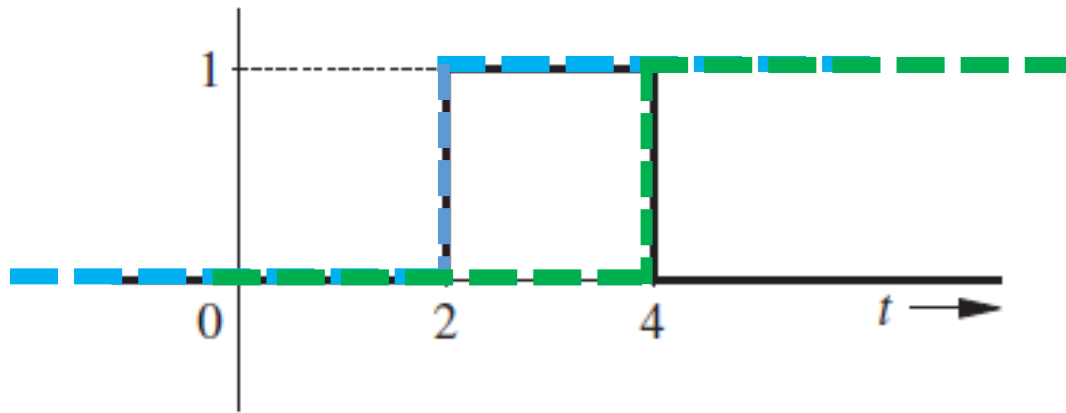
Can you make off-on-off
(rectangular pulse) using two Unit
Step functions?



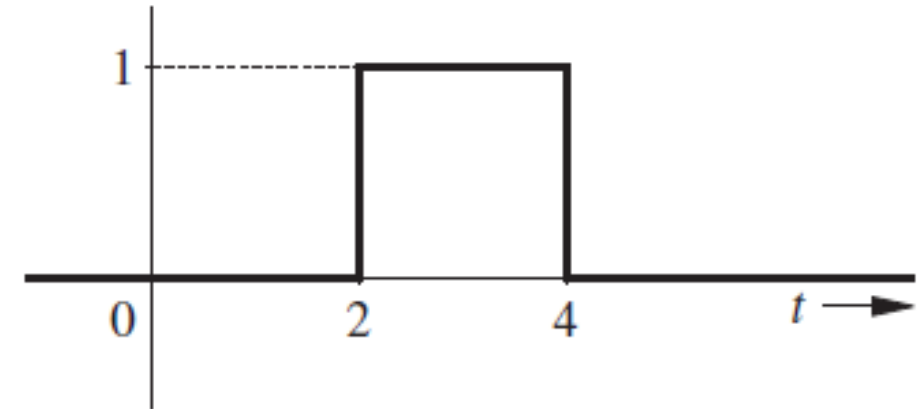
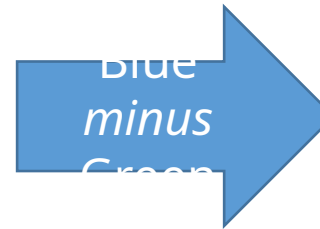
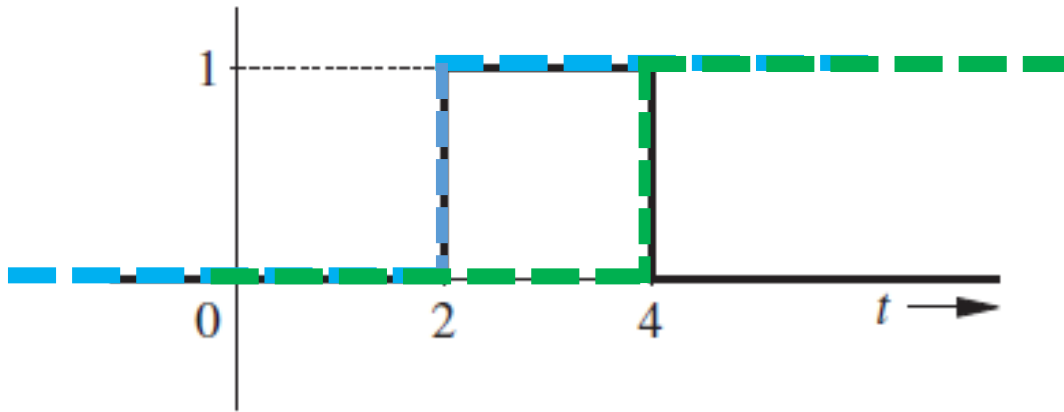
Can you make off-on-off
(rectangular pulse) using two Unit
Step functions?



Can you make off-on-off
(rectangular pulse) using two Unit
Step functions?

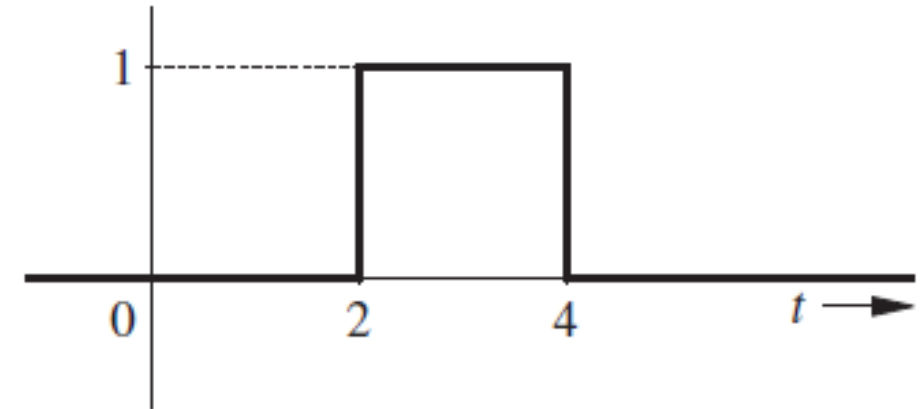
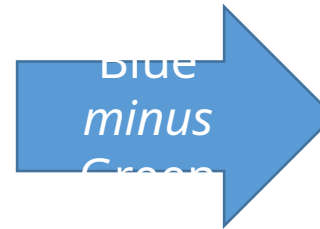
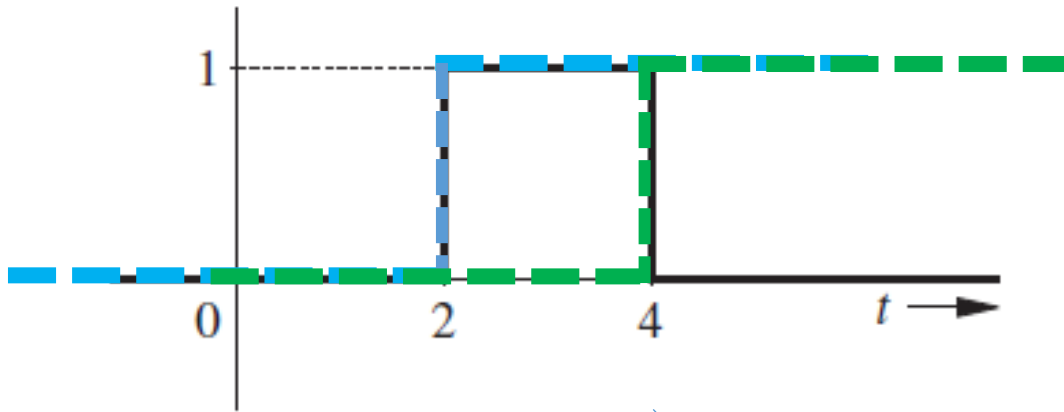


Can you make off-on-off
(rectangular pulse) using two Unit
Step functions?



Can you write
this
mathematically
?

Can you make off-on-off
(rectangular pulse) using two Unit
Step functions?



Can you write
this
mathematically
?

$$x(t) = u(t-2) - u(t-4)$$

Can you find the
area under the unit
step?

$$r(t) = \int_{-\infty}^t u(\tau) d\tau$$

Can you find the
area under the unit
step?

$$r(t) = \int_{-\infty}^t u(\tau) d\tau$$

$$= \int_0^t u(\tau) d\tau = t \text{ for } t \geq 0$$

Can you find the
area under the unit
step?

$$r(t) = \int_{-\infty}^t u(\tau) d\tau$$

$$= \int_0^t u(\tau) d\tau = t \text{ for } t \geq 0$$

$$= \int_{-\infty}^t u(\tau) d\tau = 0 \text{ for } t < 0$$

Can you find the area under the unit step?

$$r(t) = \int_{-\infty}^t u(\tau) d\tau$$

$$= \int_0^t u(\tau) d\tau = t \quad \text{for } t \geq 0$$

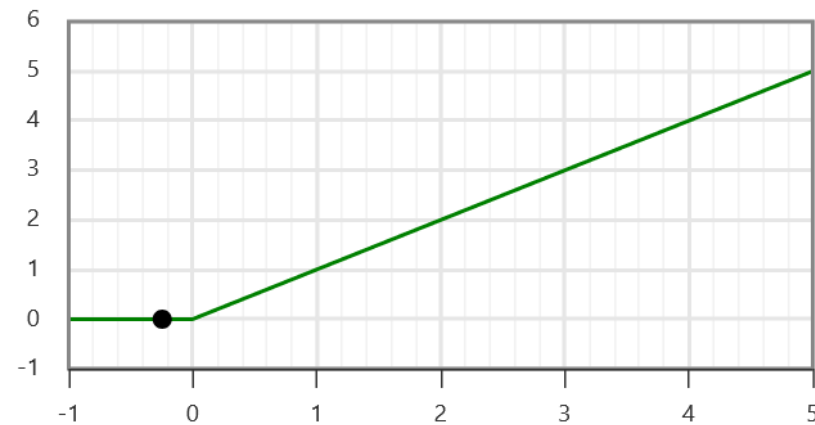
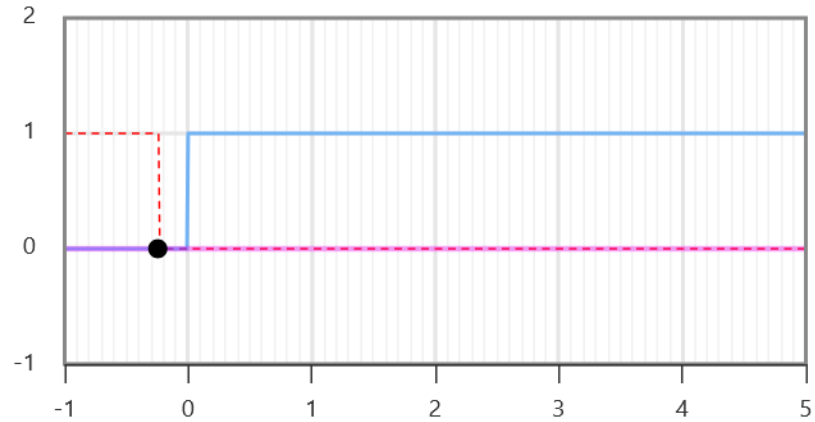
$$= \int_{-\infty}^t u(\tau) d\tau = 0 \quad \text{for } t < 0$$

- In fact, the integral of a unit step is another useful signal called *ramp function*, defined as:

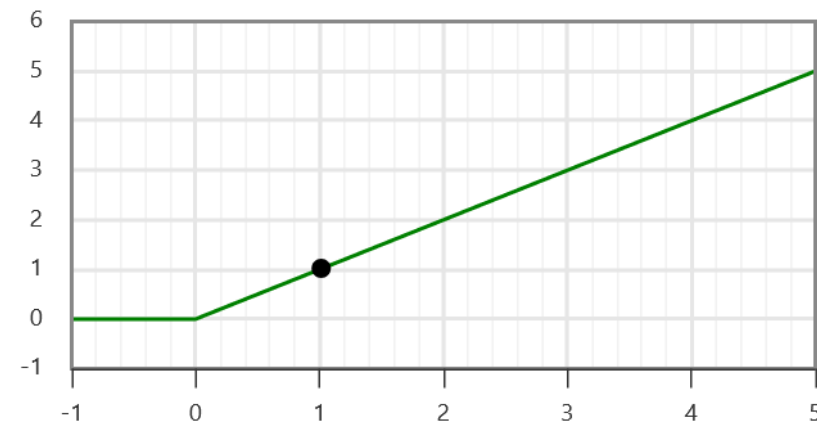
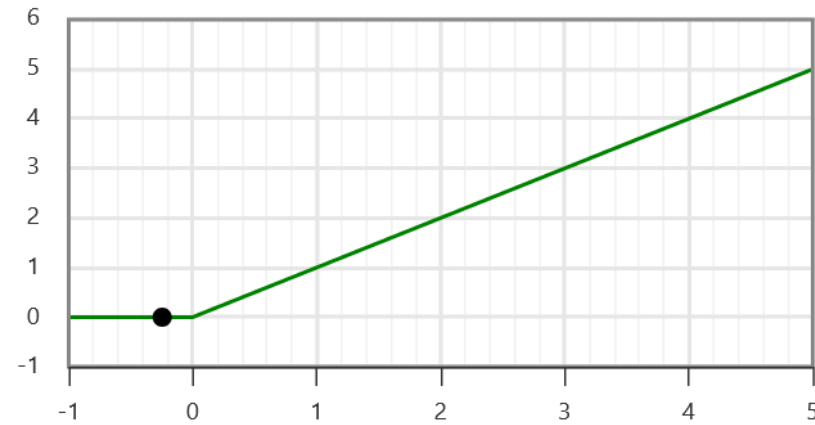
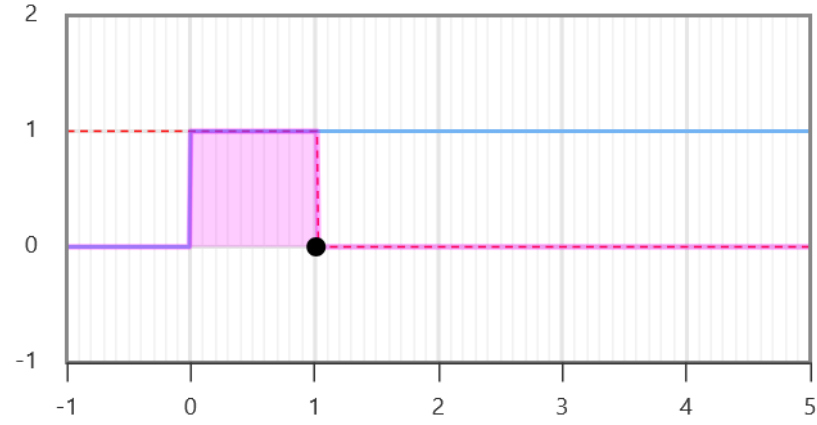
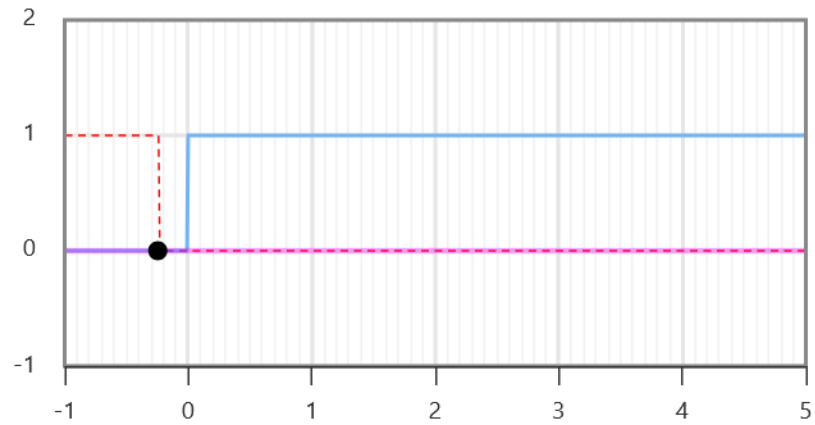
$$r(t) = \begin{cases} t, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

Visualizing integral of unit step and resulting ramp function

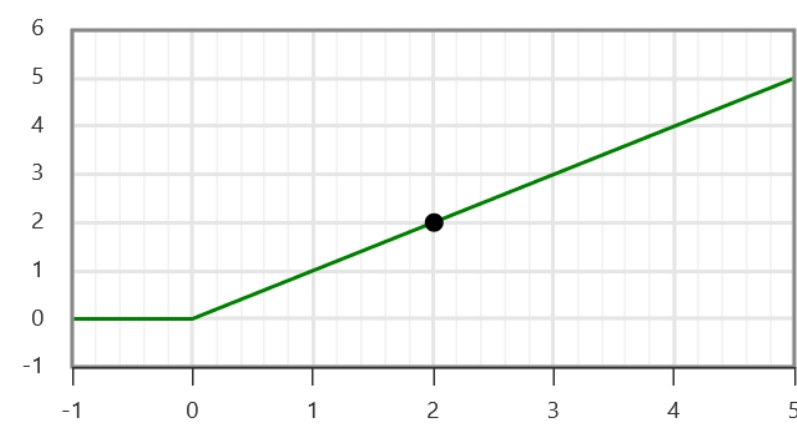
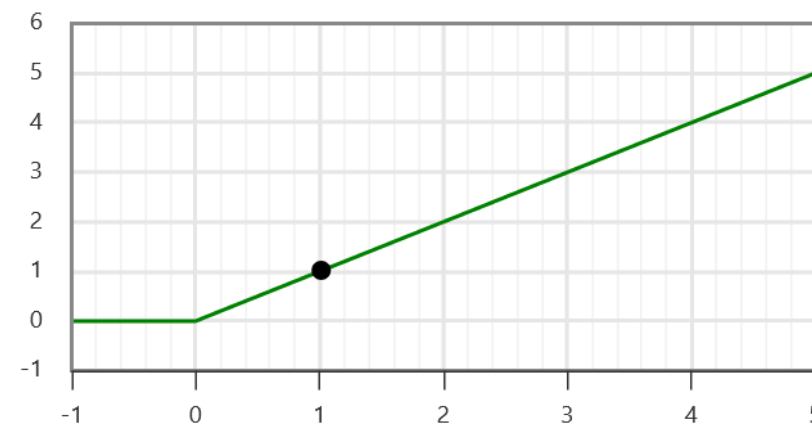
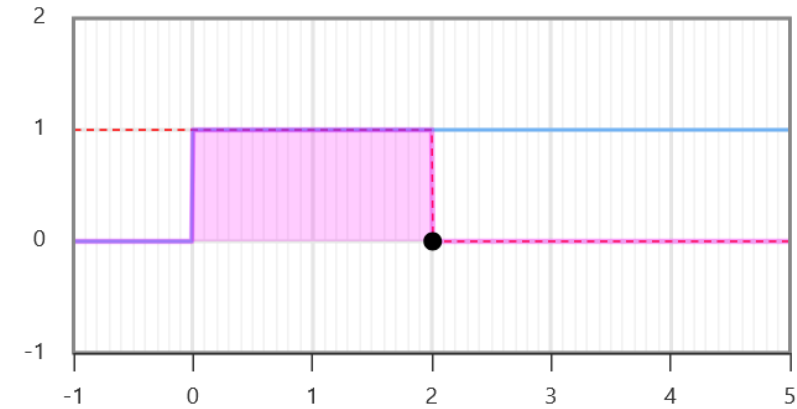
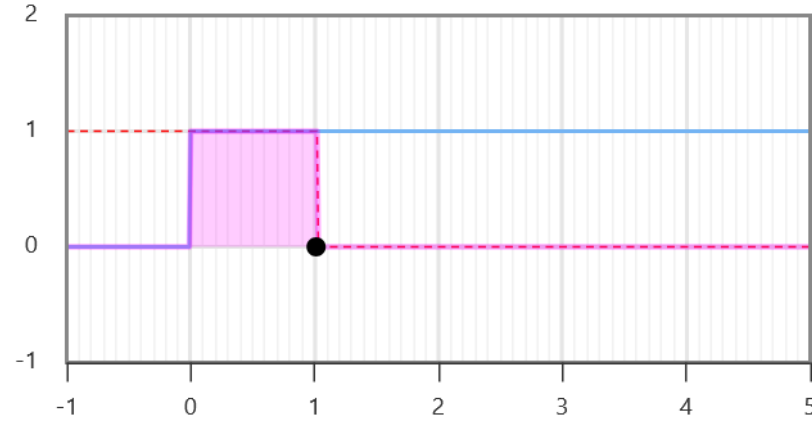
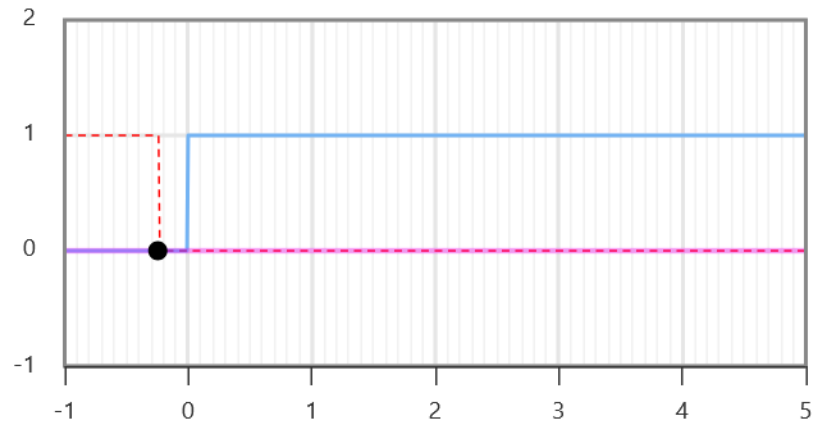
Visualizing integral of unit step and resulting ramp function



Visualizing integral of unit step and resulting ramp function

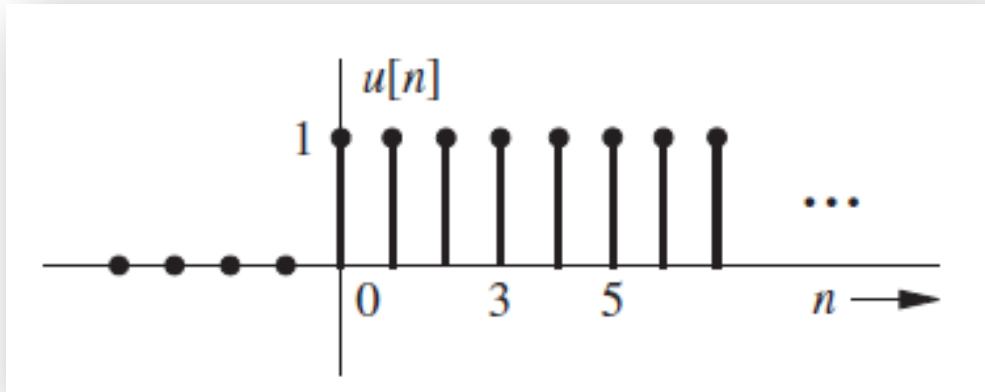


Visualizing integral of unit step and resulting ramp function



Step

Discrete-Time Case



$$u[n] = \begin{cases} 1 & \text{for } n \geq 0 \\ 0 & \text{for } n < 0 \end{cases}$$

2. Unit Impulse Function

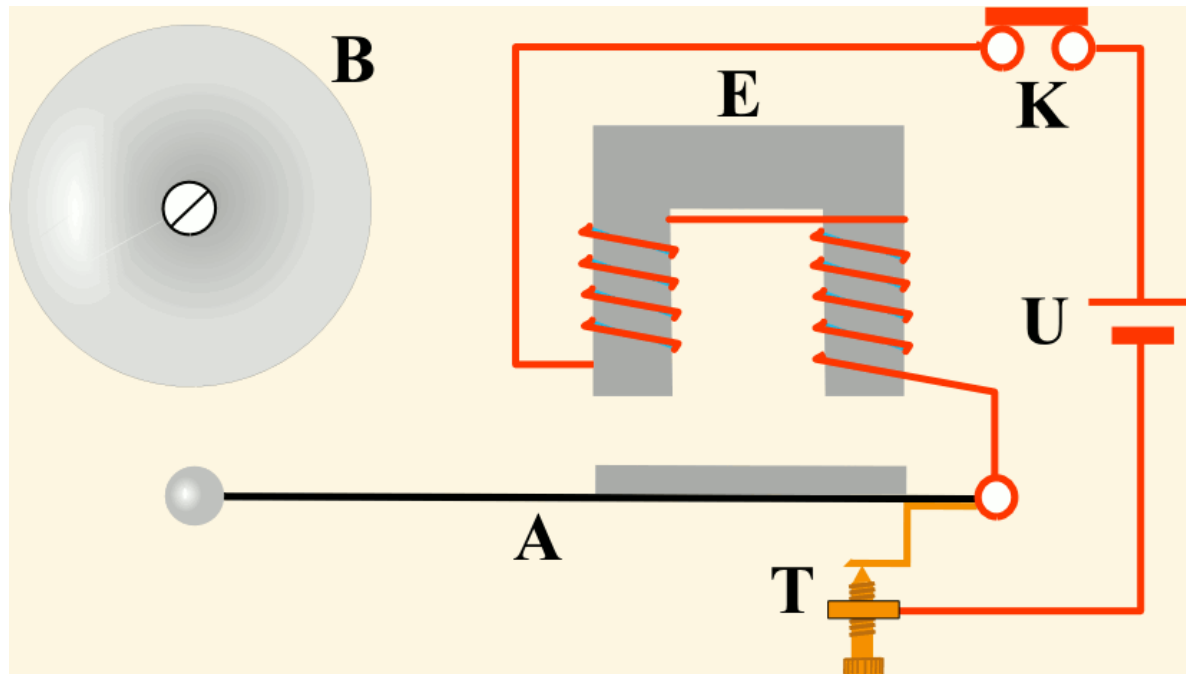
Aka Delta Function

Impulse

*A Momentary
Touch...*

Impulse

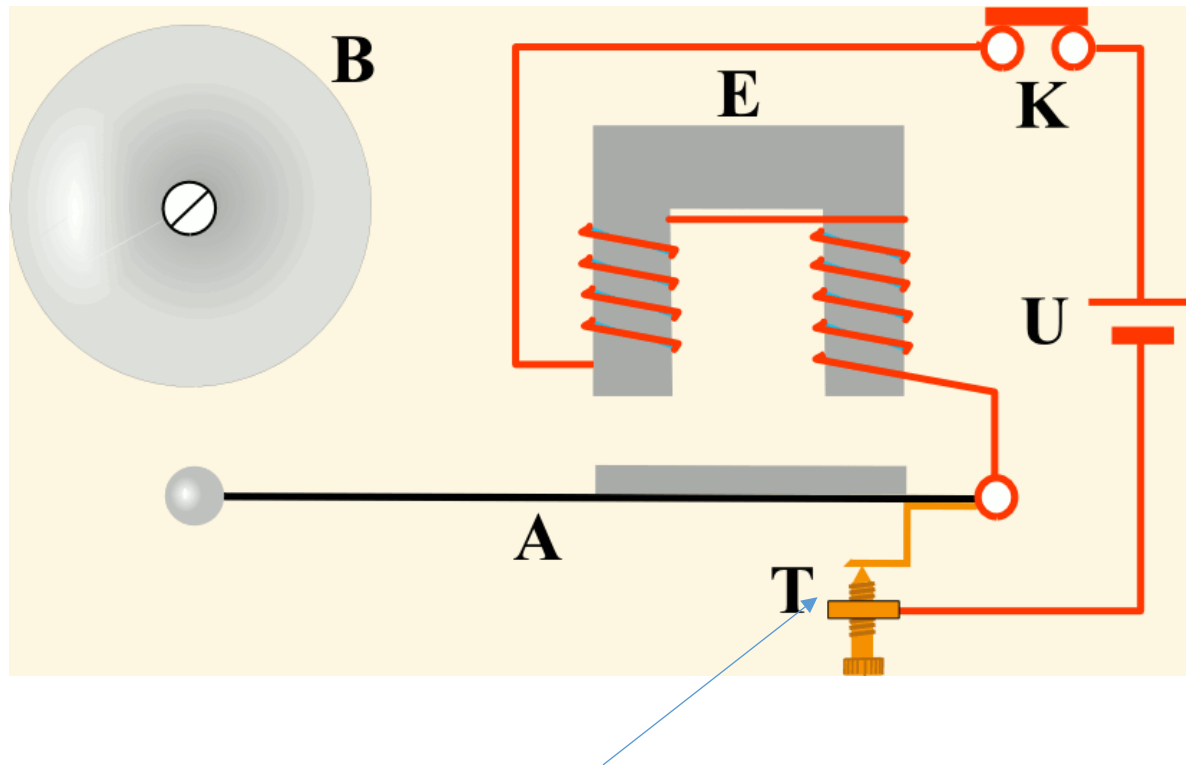
*A Momentary
Touch...*



Impuls

e

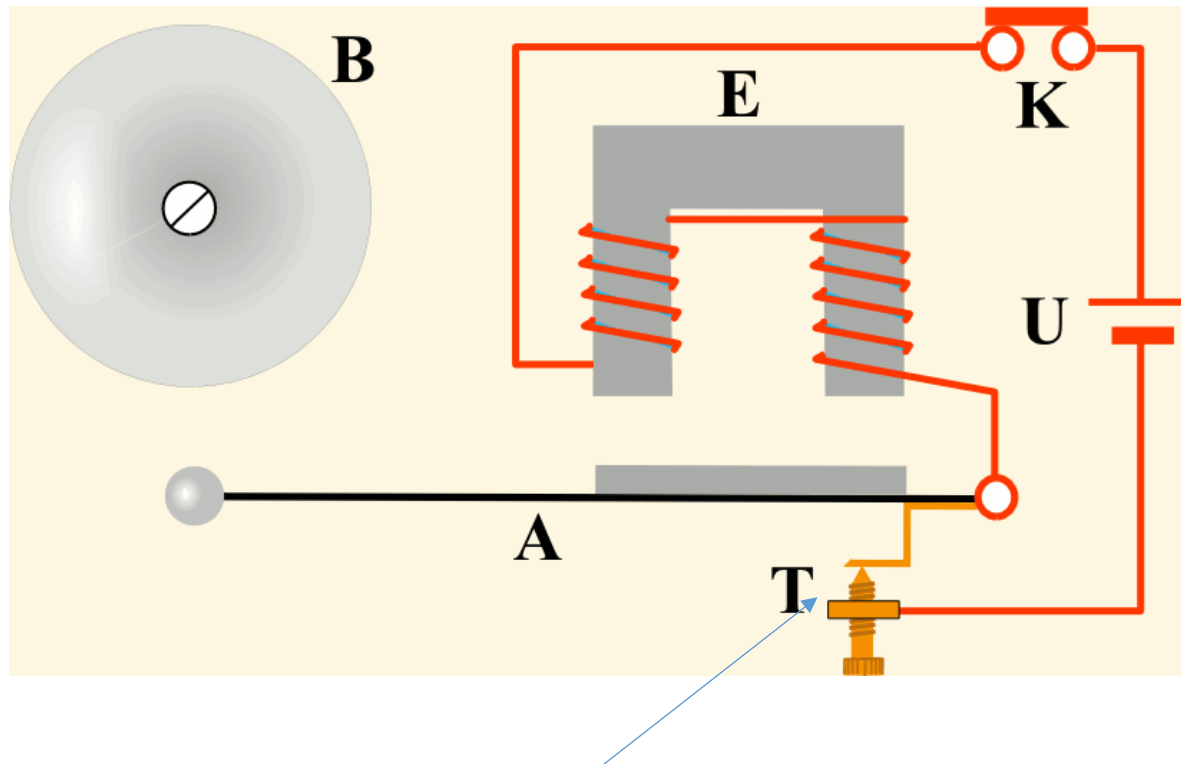
*A Momentary
Touch...*



A momentary touch! (as soon as the two touch, electromagnet is activated and pulls them apart)

Impulse

*A Momentary
Touch...*



A momentary touch! (as soon as the two touch, electromagnet is activated and pulls them apart)

While an embodied impulse function can be seen as a brief nudge/interrupt in a real system, the function is perhaps **most useful in its theoretical form** as we will see in this course.

**Impuls
e**

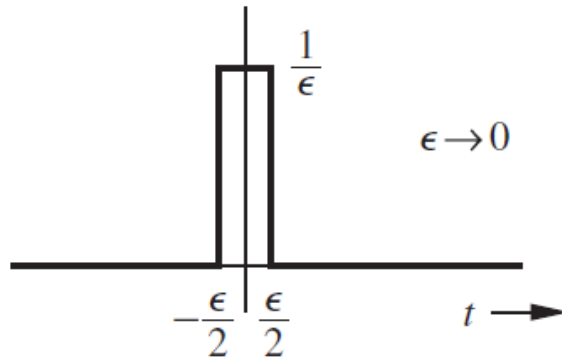
*Mathematically and graphically
speaking...*

**Continuous-Time
Case**

Impulse

Mathematically and graphically speaking...

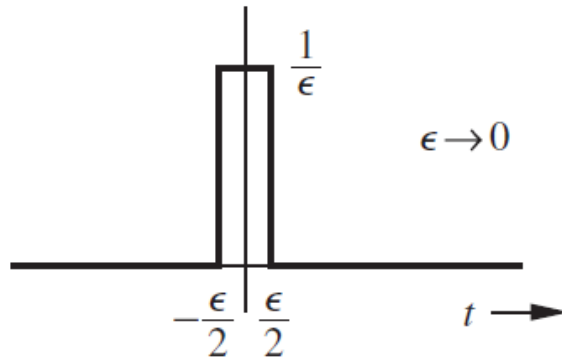
Continuous-Time Case



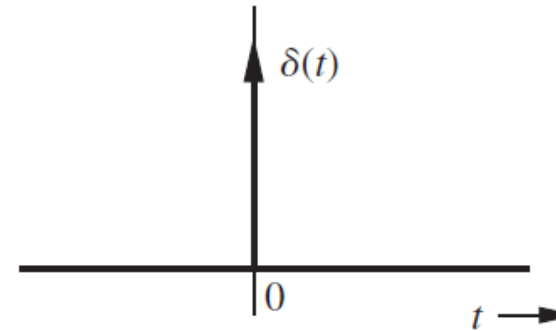
Impulse

Mathematically and graphically speaking...

Continuous-Time Case



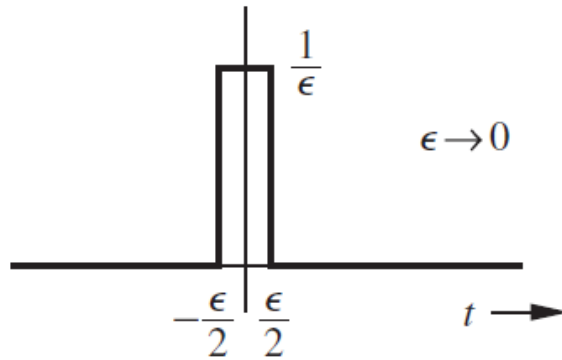
We normally draw it as ...



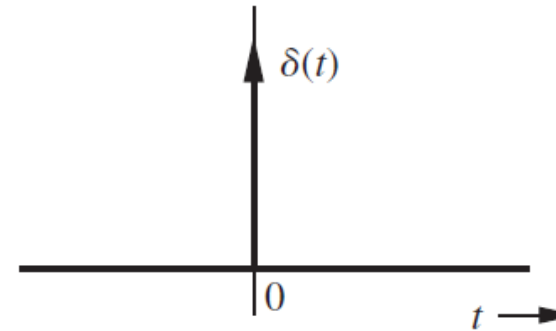
Impulse

Mathematically and graphically speaking...

Continuous-Time Case



We normally draw it as ...

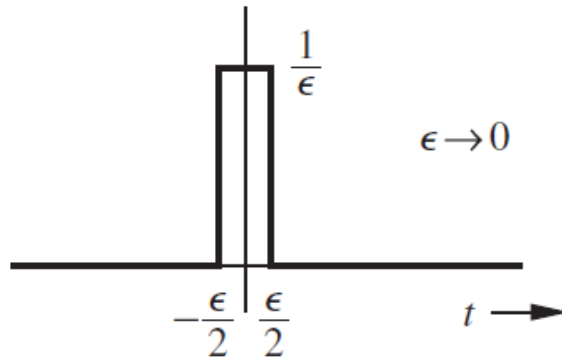


Two conditions that define a CT Impulse

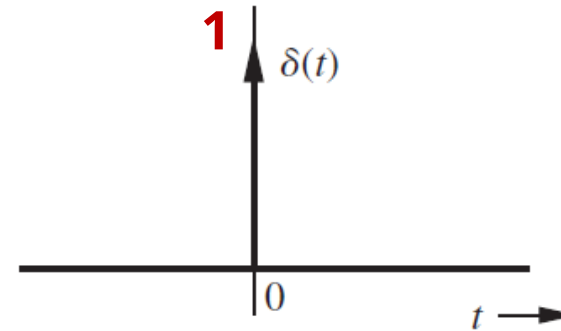
Impulse

Mathematically and graphically speaking...

Continuous-Time Case



We normally draw it as ...



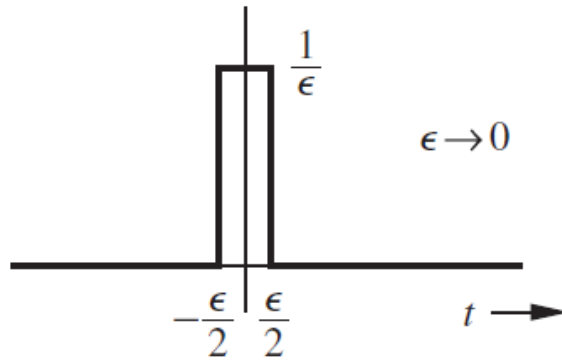
$$\delta(t) = 0 \quad t \neq 0 \quad \text{and} \quad \int_{-\infty}^{\infty} \delta(t) dt = 1$$

Two conditions that define a CT Impulse

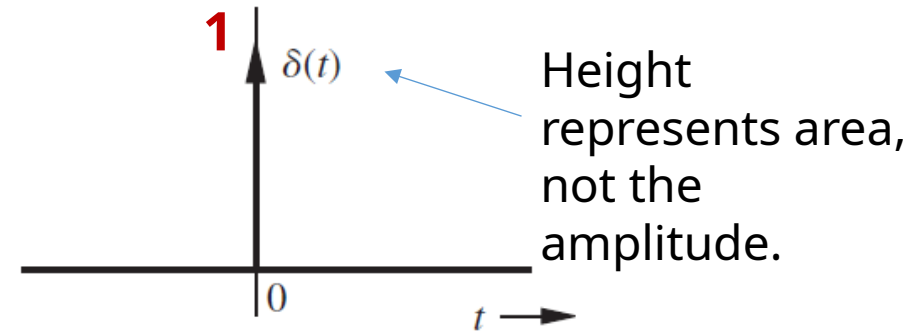
Impulse

Mathematically and graphically speaking...

Continuous-Time Case



We normally draw it as ...



$$\delta(t) = 0 \quad t \neq 0 \quad \text{and} \quad \int_{-\infty}^{\infty} \delta(t) dt = 1$$

Two conditions that define a CT Impulse

Operations on Impulse

Can you write relation
for and draw $\delta(t - T)$?

Operations on Impulse

Can you write relation for and draw $\delta(t - T)$?

$$\delta(t) = 0 \quad t \neq 0 \quad \xrightarrow{t \rightarrow t - T} \quad \delta(t - T) = 0 \quad t - T \neq 0 \quad \Rightarrow \quad \delta(t - T) = 0 \quad t \neq T$$

Operations on Impulse

Can you write relation for and draw $\delta(t - T)$?

$$\delta(t) = 0 \quad t \neq 0 \quad \xrightarrow{t \rightarrow t - T} \quad \delta(t - T) = 0 \quad t - T \neq 0 \quad \Rightarrow \quad \delta(t - T) = 0 \quad t \neq T$$

Are we missing something?

Operations on Impulse

Can you write relation for and draw $\delta(t - T)$?

$$\delta(t) = 0 \quad t \neq 0 \quad \xrightarrow{t \rightarrow t - T} \quad \delta(t - T) = 0 \quad t - T \neq 0 \quad \Rightarrow \quad \delta(t - T) = 0 \quad t \neq T$$

Are we missing something?

Since impulse is defined by two conditions, we need to consider the second as well.

Operations on Impulse

Can you write relation
for and draw $\delta(t - T)$?

$$\int_{-\infty}^{\infty} \delta(t) dt = 1 \quad \xrightarrow{t \rightarrow t - T} \quad ?$$

Operations on Impulse

Can you write relation for and draw $\delta(t - T)$?

$$\int_{-\infty}^{\infty} \delta(t) dt = 1 \quad \xrightarrow{t \rightarrow t - T} \quad \int_{-\infty - T}^{\infty - T} \delta(t - T) d(t - T) = \int_{-\infty}^{\infty} \delta(t - T) dt = 1$$

Operations on Impulse

Can you write relation for and draw $\delta(t - T)$?

$$\int_{-\infty}^{\infty} \delta(t) dt = 1 \quad \xrightarrow{t \rightarrow t - T} \quad \int_{-\infty - T}^{\infty - T} \delta(t - T) d(t - T) = \int_{-\infty}^{\infty} \delta(t - T) dt = 1$$

Can you think of why the area would remain unchanged after time-shift?

Operations on Impulse

Can you write relation for and draw $\delta(t - T)$?

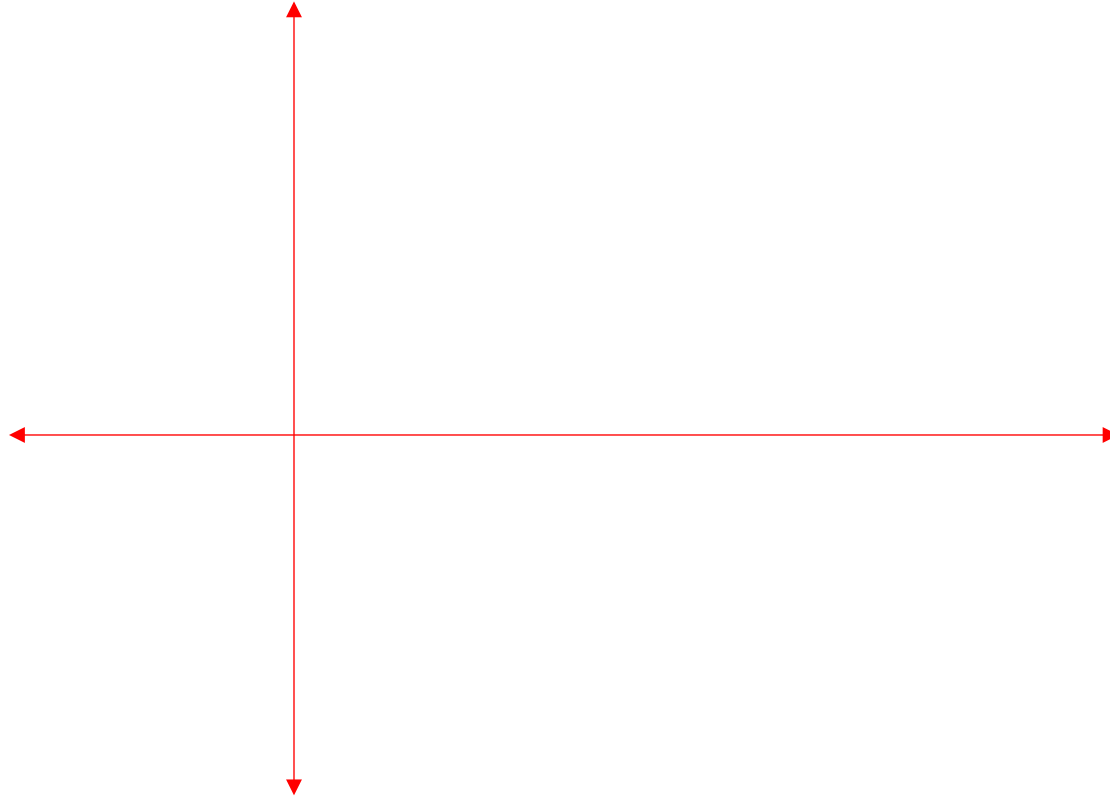
$$\int_{-\infty}^{\infty} \delta(t) dt = 1 \quad \xrightarrow{t \rightarrow t - T} \quad \int_{-\infty - T}^{\infty - T} \delta(t - T) d(t - T) = \int_{-\infty}^{\infty} \delta(t - T) dt = 1$$

Can you think of why the area would remain unchanged after time-shift?

Time-shift does not change signal amplitude or width!

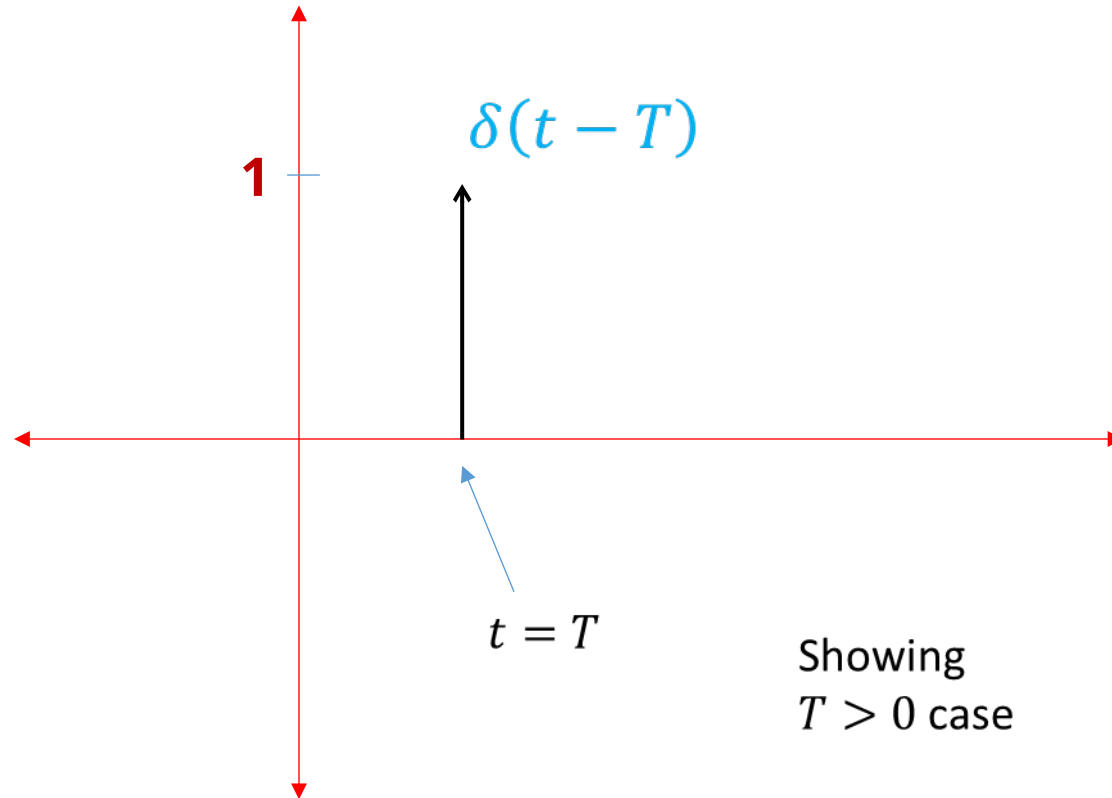
Operations on Impulse

Can you write relation
for and draw $\delta(t - T)$?



Operations on Impulse

Can you write relation for and draw $\delta(t - T)$?



Interesting Point

While $\int_{-\infty}^{\infty} \delta(t) dt = \int_{-\infty}^{\infty} \delta(t - 4) dt = 1$

$$\int_{-\infty}^3 \delta(t) dt \neq \int_{-\infty}^3 \delta(t - 4) dt$$

Interesting Point

While $\int_{-\infty}^{\infty} \delta(t) dt = \int_{-\infty}^{\infty} \delta(t - 4) dt = 1$

$$\int_{-\infty}^3 \delta(t) dt \neq \int_{-\infty}^3 \delta(t - 4) dt$$

Can you show why?

Operations on Impulse

Can you write relation
for and draw $\delta(2t)$?

Operations on Impulse

Can you write relation for and draw $\delta(2t)$?

$$\delta(t) = 0 \quad t \neq 0 \quad \xrightarrow{t \rightarrow 2t} \quad \delta(2t) = 0 \quad 2t \neq 0 \quad \rightarrow \quad \delta(2t) = 0 \quad t \neq 0$$

Operations on Impulse

Can you write relation for and draw $\delta(2t)$?

$$\delta(t) = 0 \quad t \neq 0 \quad \xrightarrow{t \rightarrow 2t} \quad \delta(2t) = 0 \quad 2t \neq 0 \quad \Rightarrow \quad \delta(2t) = 0 \quad t \neq 0$$

Since impulse is defined by two conditions, we need to consider the second as well.

Operations on Impulse

Can you write relation for and draw $\delta(2t)$?

$$\int_{-\infty}^{\infty} \delta(t) dt = 1 \quad \xrightarrow{t \rightarrow 2t} \quad ?$$

Operations on Impulse

Can you write relation for and draw $\delta(2t)$?

$$\int_{-\infty}^{\infty} \delta(t) dt = 1 \quad \xrightarrow{t \rightarrow 2t} \quad \int_{-2\infty}^{2\infty} \delta(2t) d(2t) = 1 \quad \Rightarrow \quad \int_{-\infty}^{\infty} \delta(2t) dt = \frac{1}{2}$$

Operations on Impulse

Can you write relation for and draw $\delta(2t)$?

$$\int_{-\infty}^{\infty} \delta(t) dt = 1 \quad \xrightarrow{t \rightarrow 2t} \quad \int_{-2\infty}^{2\infty} \delta(2t) d(2t) = 1 \quad \Rightarrow \quad \int_{-\infty}^{\infty} \delta(2t) dt = \frac{1}{2}$$

Can you think of why the area has been halved?

Operations on Impulse

Can you write relation for and draw $\delta(2t)$?

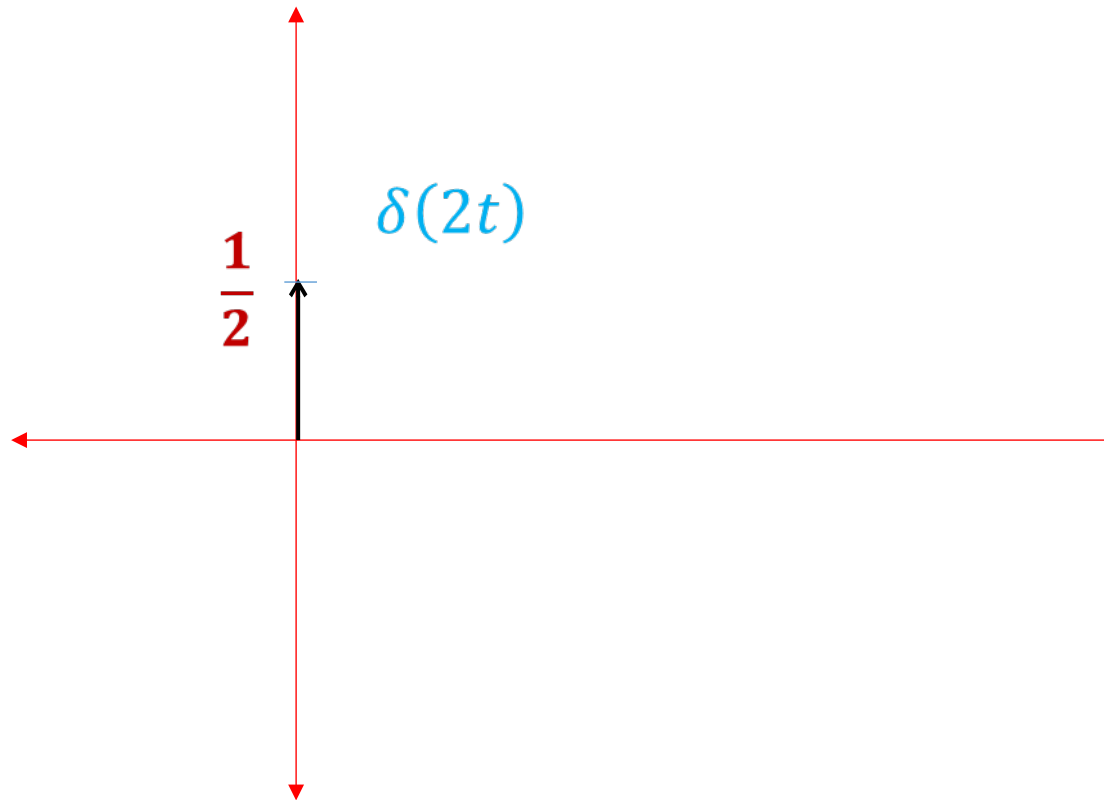
$$\int_{-\infty}^{\infty} \delta(t) dt = 1 \quad \xrightarrow{t \rightarrow 2t} \quad \int_{-\infty}^{\infty} \delta(2t) d(2t) = 1 \quad \Rightarrow \quad \int_{-\infty}^{\infty} \delta(2t) dt = \frac{1}{2}$$

Can you think of why the area has been halved?

Due to compression by a factor 2, the width of $\delta(2t)$ is half that of $\delta(t)$!

Operations on Impulse

Can you write relation for and draw $\delta(2t)$?



Operations on Impulse

Can you write relation for and draw an impulse that has been amplitude-scaled by α ?

Operations on Impulse

Can you write relation for and draw an impulse that has been amplitude-scaled by α ?

$\delta(t) = 0 \quad t \neq 0$ Multiplying both sides by α gives

$\int_{-\infty}^{\infty} \delta(t) dt = 1$ Multiplying both sides by α gives

Operations on Impulse

Can you write relation for and draw an impulse that has been amplitude-scaled by α ?

$$\delta(t) = 0 \quad t \neq 0$$

Multiplying both sides by α gives

$$\alpha\delta(t) = 0 \quad t \neq 0$$

$$\int_{-\infty}^{\infty} \delta(t) dt = 1$$

Multiplying both sides by α gives

$$\int_{-\infty}^{\infty} \alpha\delta(t) dt = \alpha$$

Operations on Impulse

Can you write relation for and draw an impulse that has been amplitude-scaled by α ?

$$\delta(t) = 0 \quad t \neq 0$$

Multiplying both sides by α gives

$$\alpha\delta(t) = 0 \quad t \neq 0$$

$$\int_{-\infty}^{\infty} \delta(t) dt = 1$$

Multiplying both sides by α gives

$$\int_{-\infty}^{\infty} \alpha\delta(t) dt = \alpha$$

So, amplitude-scaling impulse by α also scales its area to α

Operations on Impulse

Can you write relation for and draw an impulse that has been amplitude-scaled by α ?

$$\delta(t) = 0 \quad t \neq 0$$

Multiplying both sides by α gives

$$\alpha\delta(t) = 0 \quad t \neq 0$$

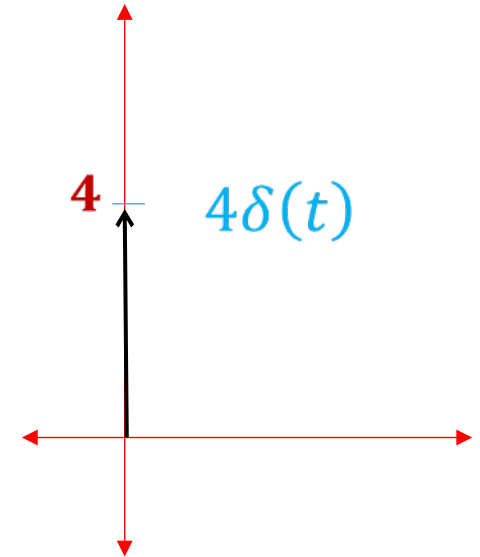
$$\int_{-\infty}^{\infty} \delta(t) dt = 1$$

Multiplying both sides by α gives

$$\int_{-\infty}^{\infty} \alpha\delta(t) dt = \alpha$$

So, amplitude-scaling impulse by α also scales its area to α

Example



Operations on Impulse

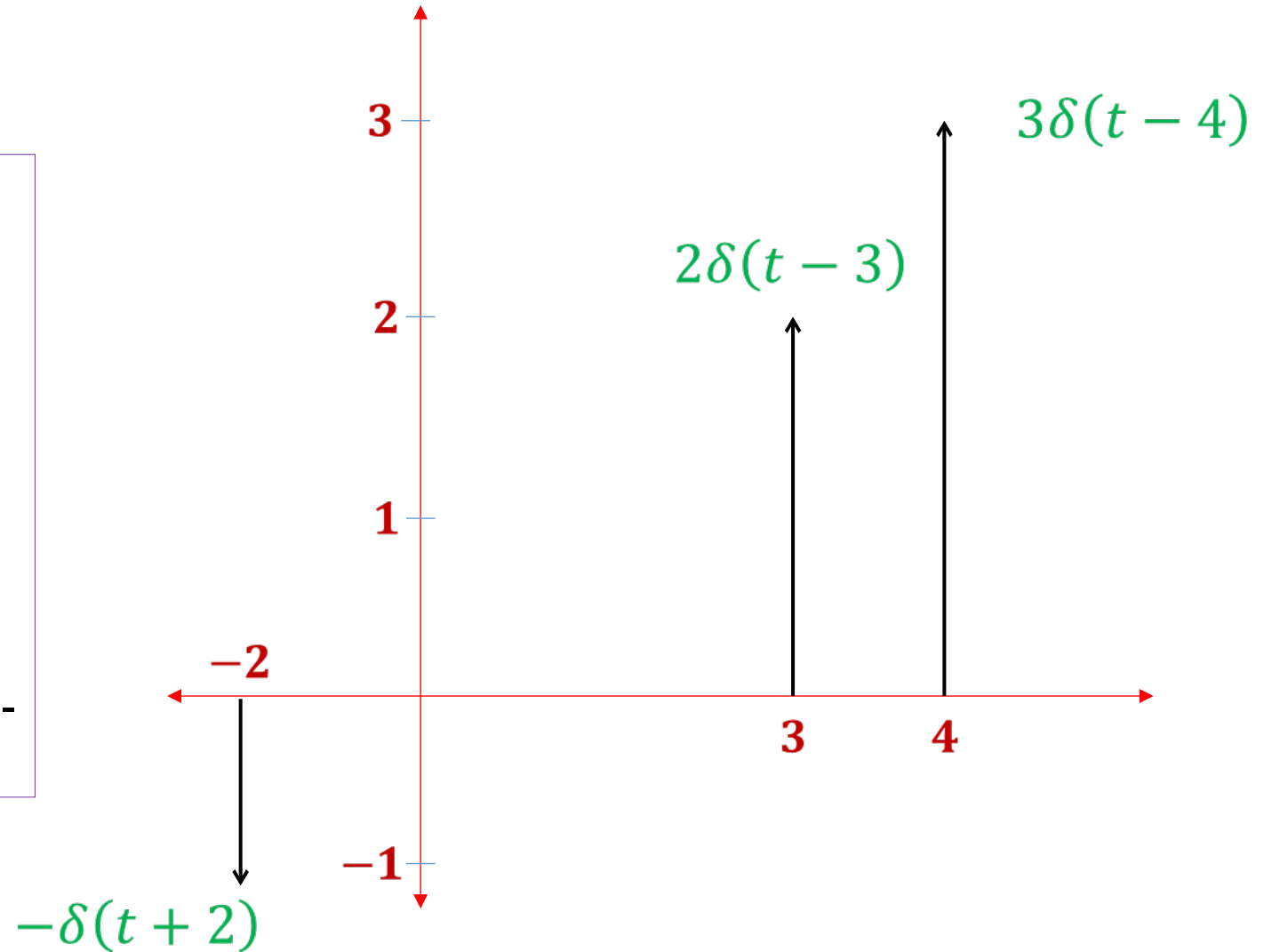
Can you draw the following impulses?

- Impulse amplitude-scaled by 3 and delayed by 4 units
- Impulse amplitude-scaled by 2 and delayed by 3 units
- Impulse amplitude-scaled by -1 and advanced by 2 units

Operations on Impulse

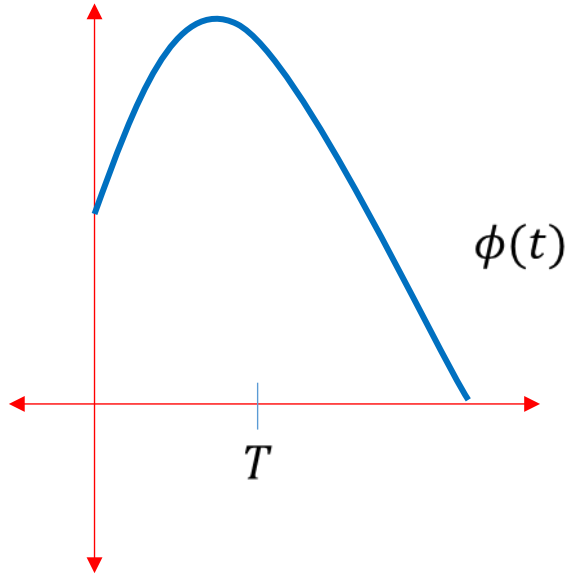
Can you draw the following impulses?

- Impulse amplitude-scaled by 3 and delayed by 4 units
- Impulse amplitude-scaled by 2 and delayed by 3 units
- Impulse amplitude-scaled by -1 and advanced by 2 units



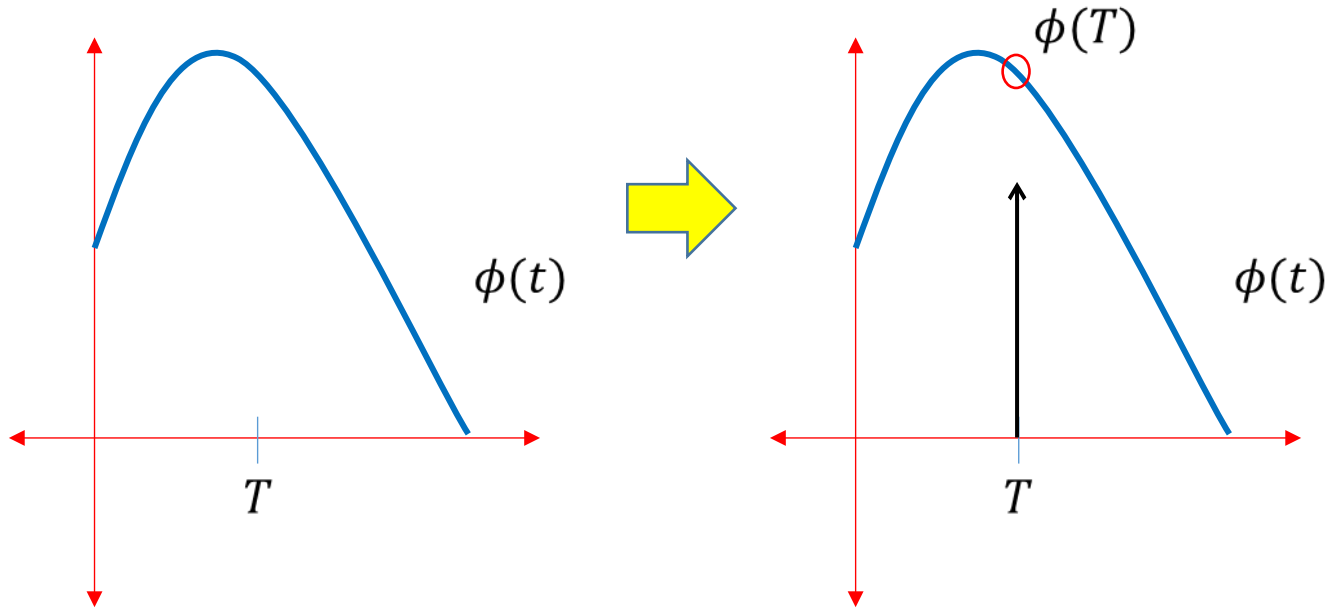
Multiplying a function by impulse

Given $\phi(t)$, can you plot and write a simplified expression for $\phi(t)\delta(t - T)$?



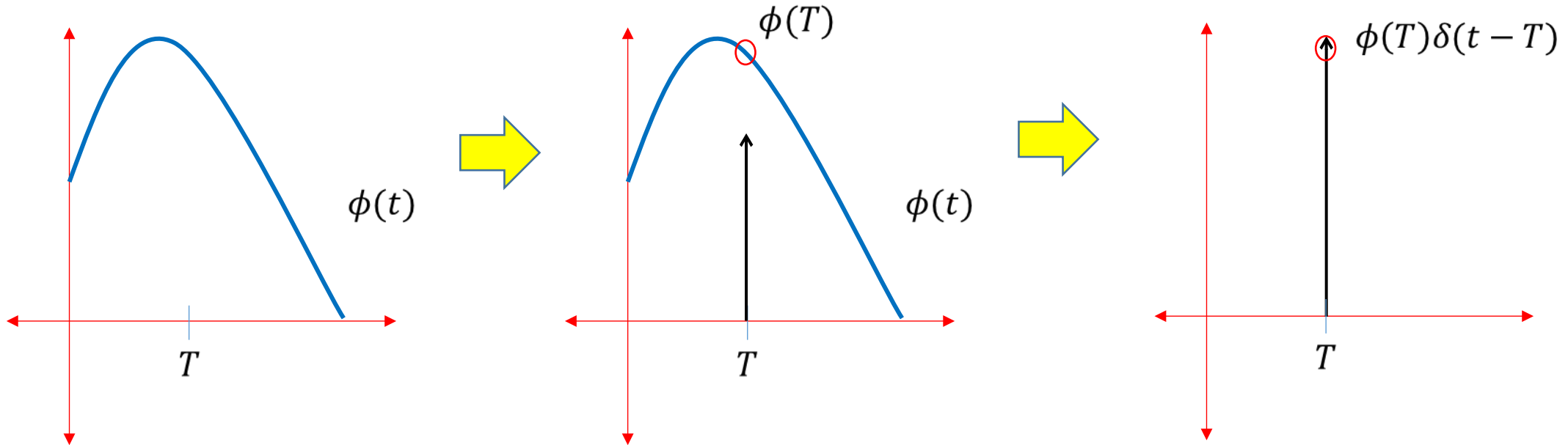
Multiplying a function by impulse

Given $\phi(t)$, can you plot and write a simplified expression for $\phi(t)\delta(t - T)$?



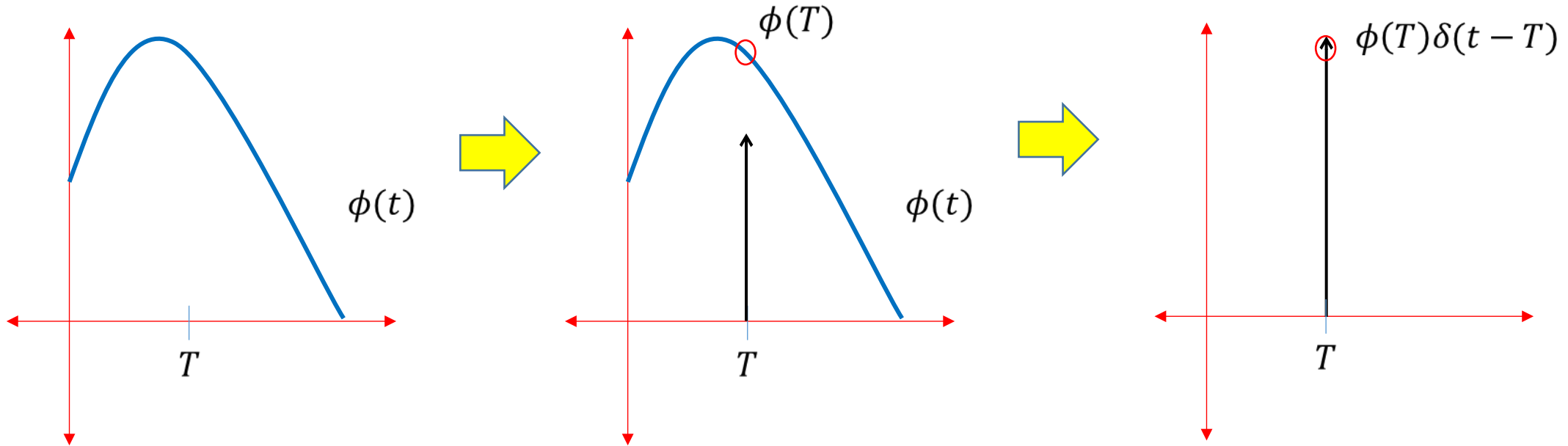
Multiplying a function by impulse

Given $\phi(t)$, can you plot and write a simplified expression for $\phi(t)\delta(t - T)$?



Multiplying a function by impulse

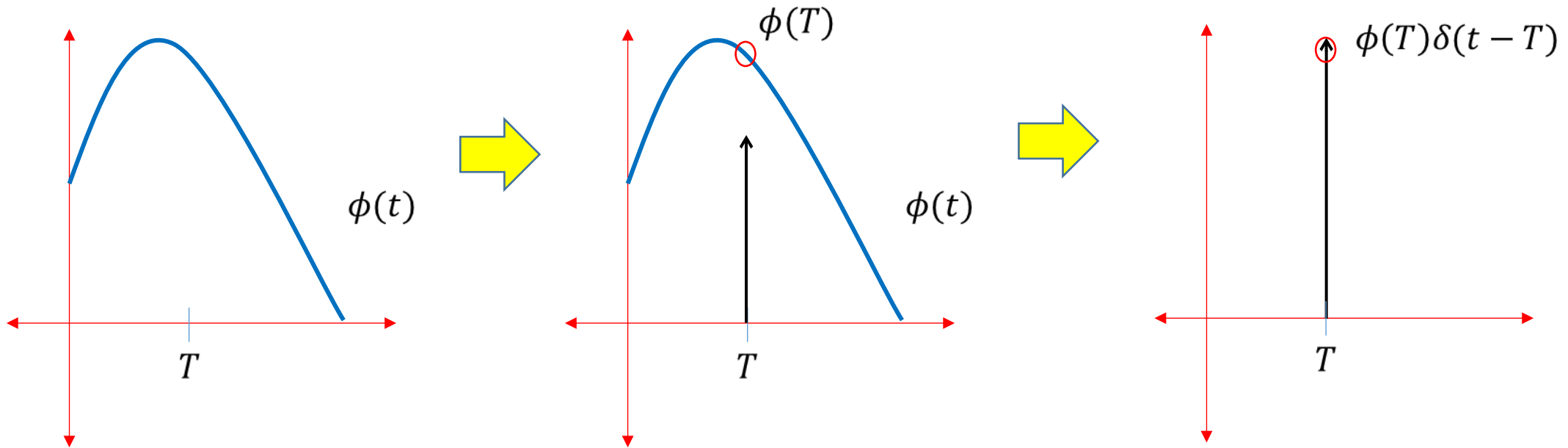
Given $\phi(t)$, can you plot and write a simplified expression for $\phi(t)\delta(t - T)$?



$$\phi(t)\delta(t - T) = \phi(T)\delta(t - T)$$

Multiplying a function by impulse

Given $\phi(t)$, can you plot and write a simplified expression for $\phi(t)\delta(t - T)$?

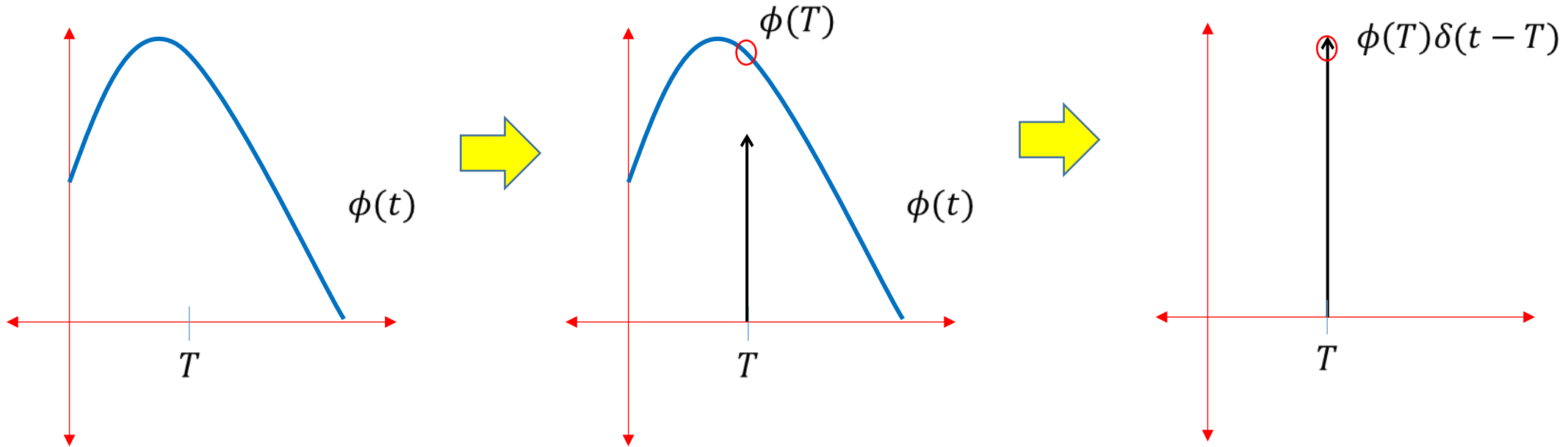


The only value of $\phi(t)$ that matters in this product is $\phi(T)$!

$$\phi(t)\delta(t - T) = \phi(T)\delta(t - T)$$

Multiplying a function by impulse

Given $\phi(t)$, can you plot and write a simplified expression for $\phi(t)\delta(t - T)$?



The only value of $\phi(t)$ that matters in this product is $\phi(T)$!

$$\phi(t)\delta(t - T) = \phi(T)\delta(t - T)$$

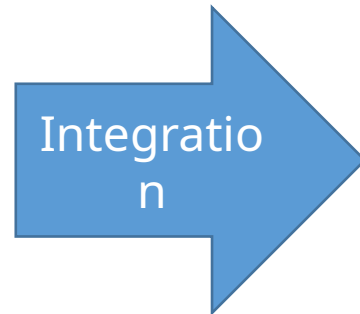
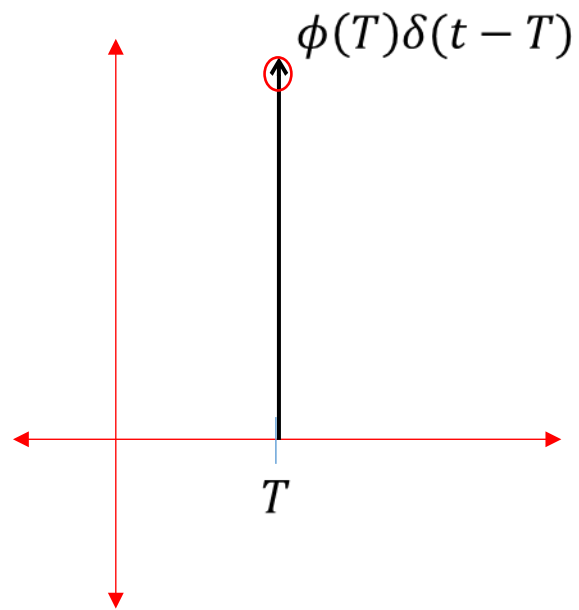
RHS is, in fact, an impulse at T amplitude-scaled to $\phi(T)$

Multiplying a function by impulse

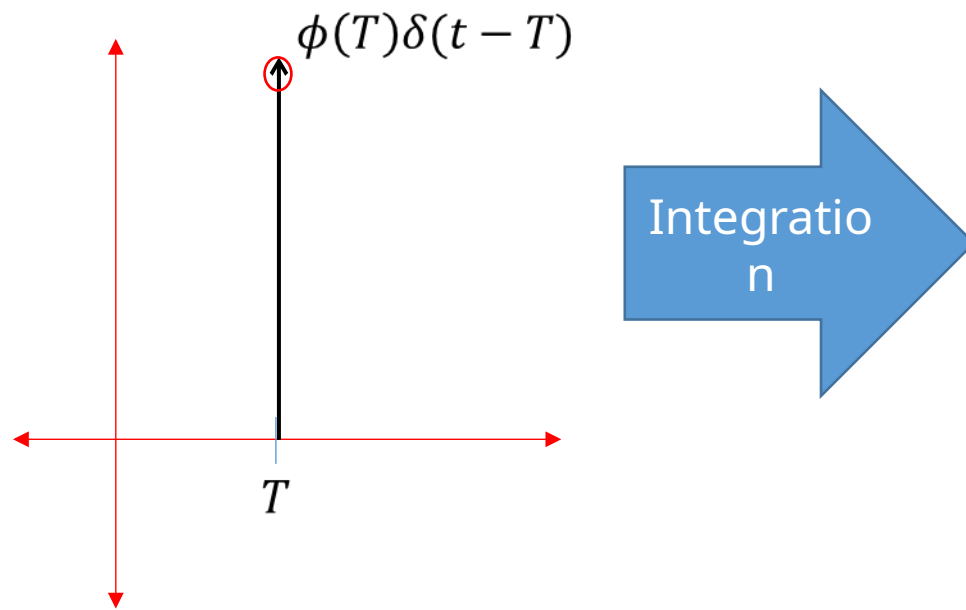
Given $\phi(t)$, can you plot and write a simplified expression for $\phi(t)\delta(t - T)$?

Q. What happens if we now integrate this product?

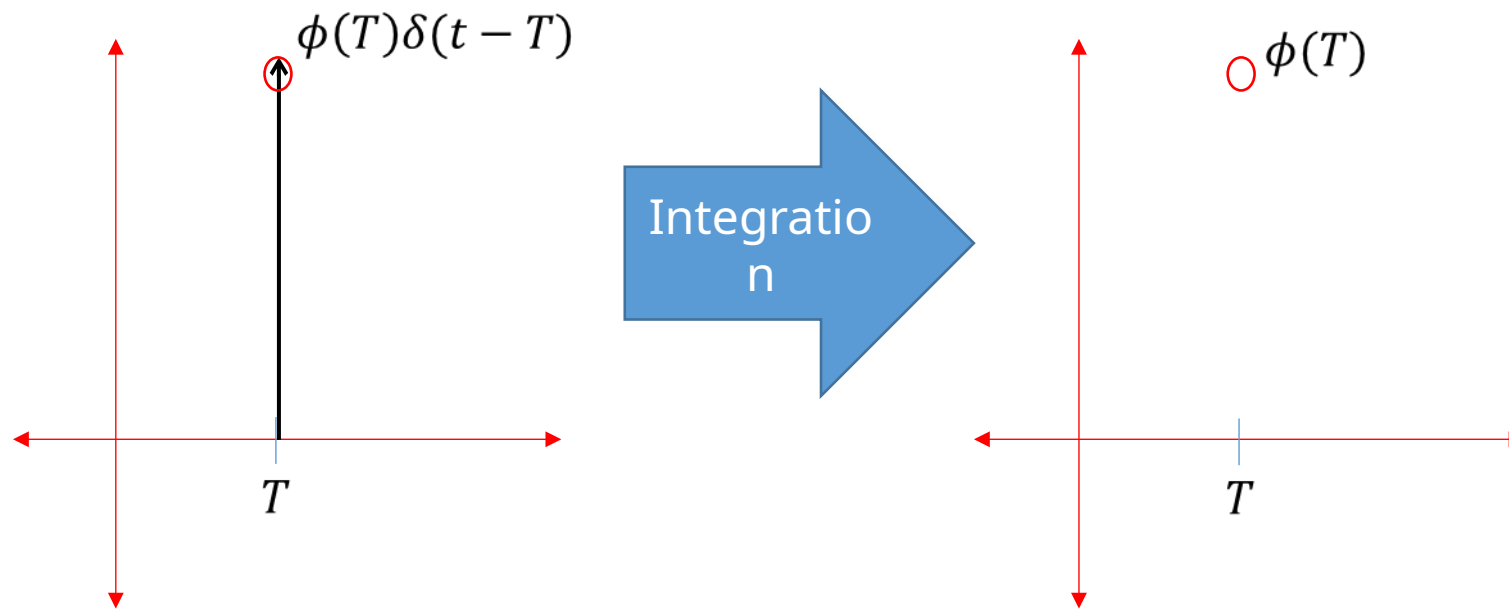
$$\phi(t)\delta(t - T) = \phi(T)\delta(t - T)$$



$$\int_{-\infty}^{\infty} \phi(t) \delta(t - T) dt =$$

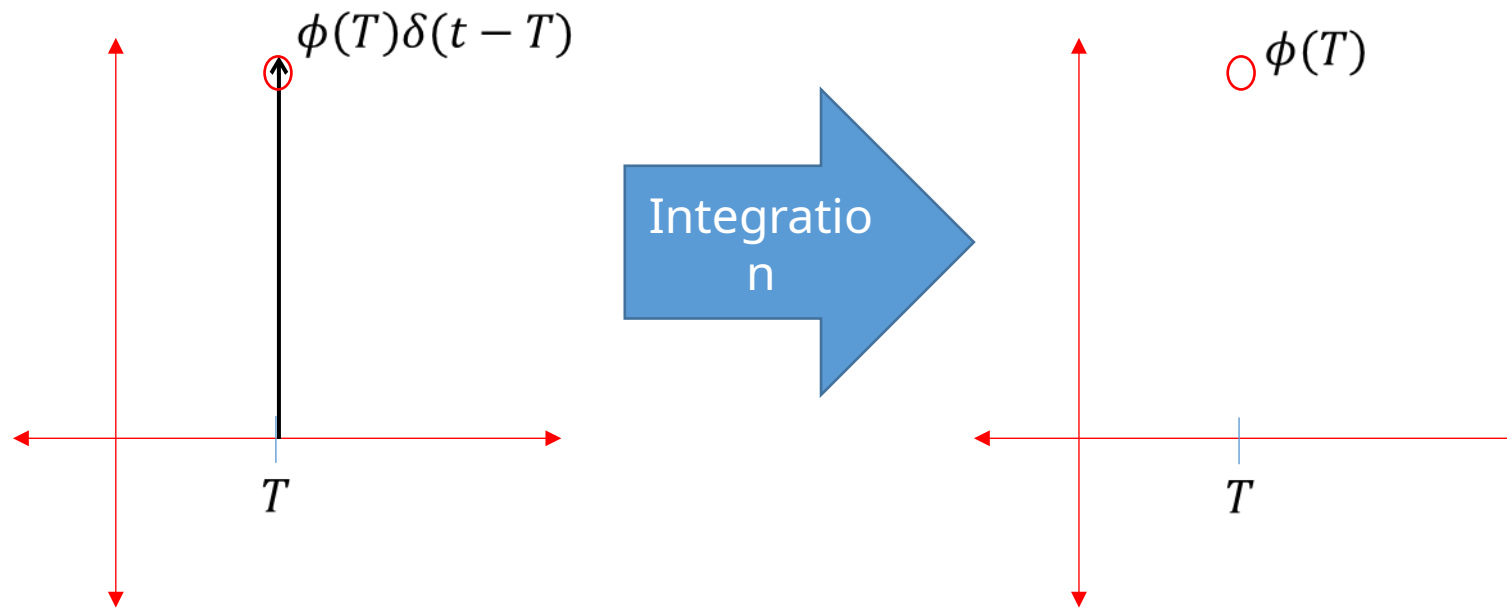


$$\int_{-\infty}^{\infty} \phi(t) \delta(t - T) dt = \phi(T) \int_{-\infty}^{\infty} \delta(t) dt = \phi(T)$$



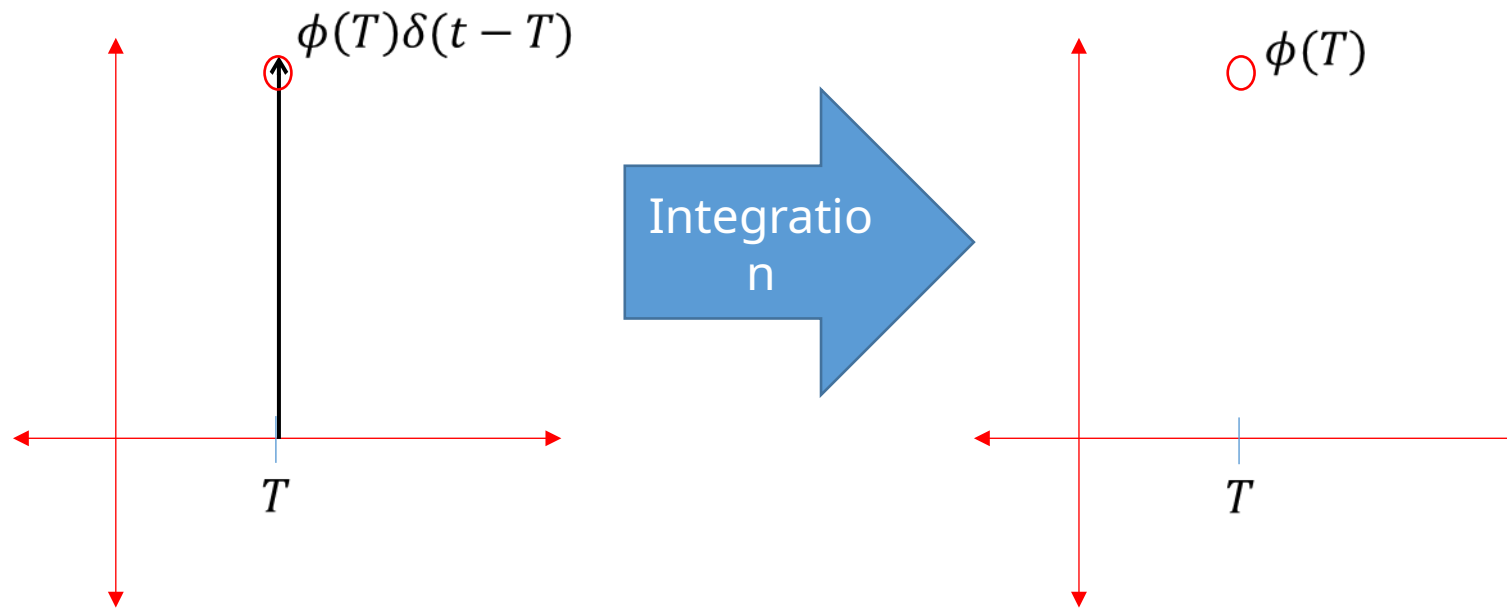
$$\int_{-\infty}^{\infty} \phi(t) \delta(t - T) dt = \phi(T) \int_{-\infty}^{\infty} \delta(t) dt = \phi(T)$$

So we get the value/sample of the function $\phi(t)$ at T .



$$\int_{-\infty}^{\infty} \phi(t) \delta(t - T) dt = \phi(T) \int_{-\infty}^{\infty} \delta(t) dt = \phi(T)$$

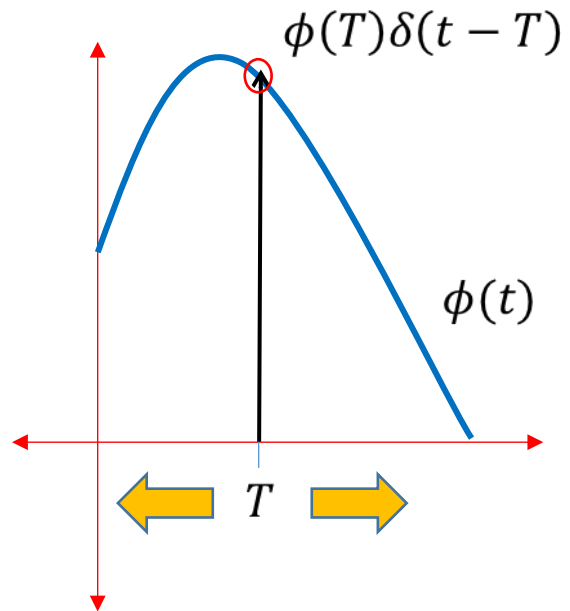
So we get the value/sample of the function $\phi(t)$ at T .



This is called the **sampling property of impulse function.**

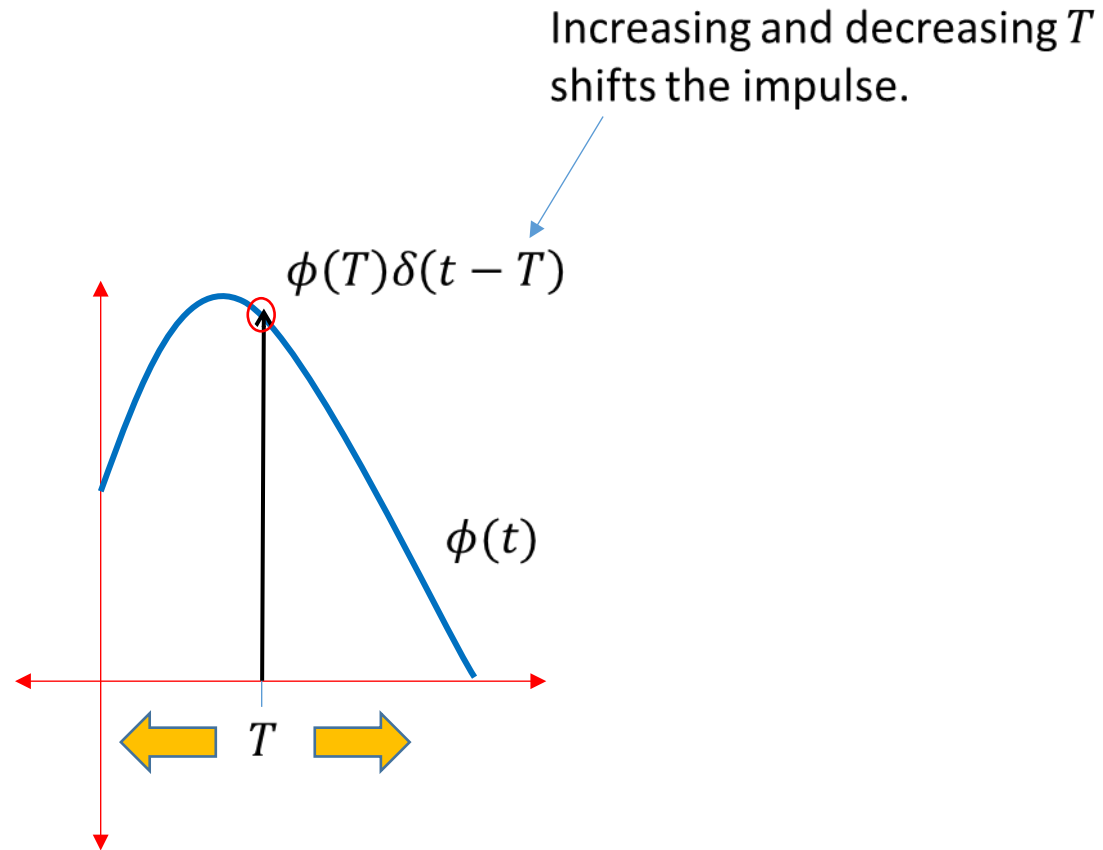
What happens as we change T ?

$$\phi(t)\delta(t-T) = \phi(T)\delta(t-T)$$



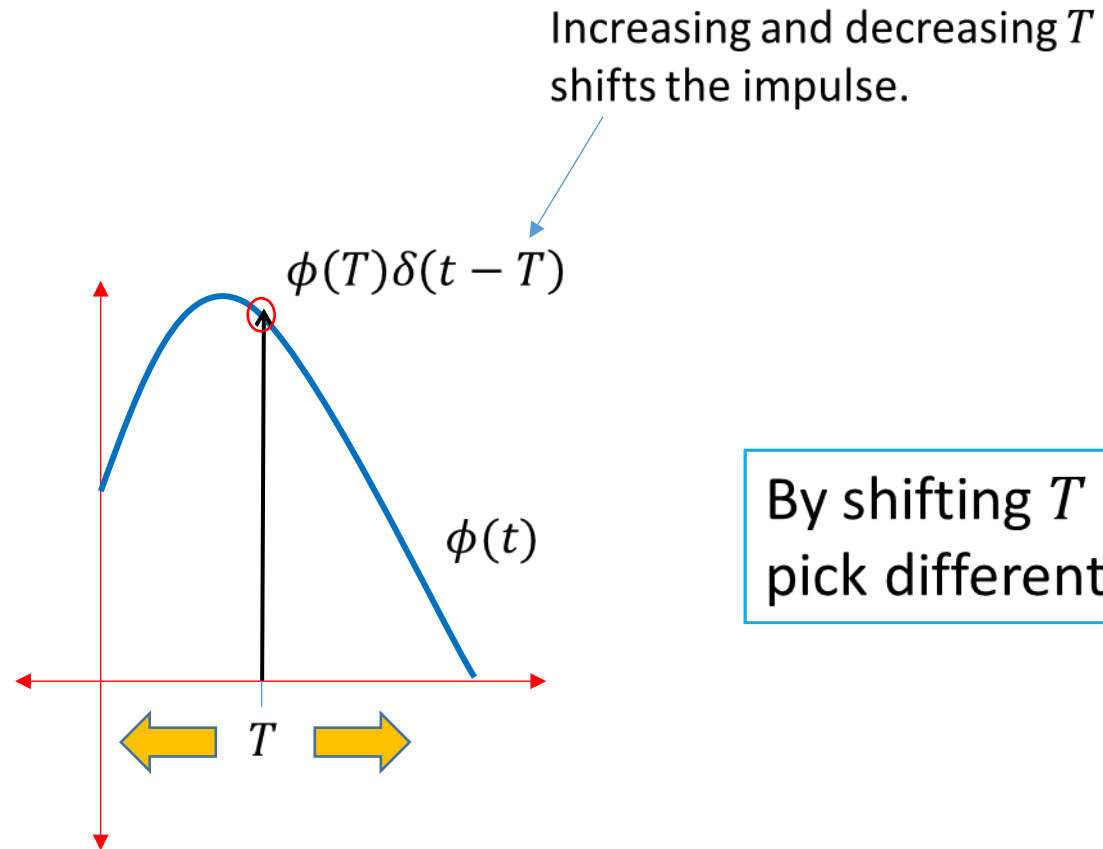
What happens as we change T ?

$$\phi(t)\delta(t-T) = \phi(T)\delta(t-T)$$



What happens as we change T ?

$$\phi(t)\delta(t-T) = \phi(T)\delta(t-T)$$



By shifting T to the right and left, we can pick different samples of the signal $\phi(t)$.

What happens as we change T ?

$$\phi(t)\delta(t-T) = \phi(T)\delta(t-T)$$

EXAMPLE

- Let $\phi(t) = t^3 + 3$, then
 - $\phi(t)\delta(t) = ?$
 - $\phi(t)\delta(t-1) = ?$
 - $\phi(t)\delta(t+1) = ?$

What happens as we change T ?

$$\phi(t)\delta(t-T) = \phi(T)\delta(t-T)$$

EXAMPLE

- Let $\phi(t) = t^3 + 3$, then
 - $\phi(t)\delta(t) = \phi(0)\delta(t) = 3\delta(t)$
 - $\phi(t)\delta(t-1) = \phi(1)\delta(t-1) = 4\delta(t-1)$
 - $\phi(t)\delta(t+1) = \phi(-1)\delta(t+1) = 2\delta(t+1)$

What happens as we change T ?

$$\phi(t)\delta(t-T) = \phi(T)\delta(t-T)$$

EXAMPLE

- Let $\phi(t) = t^3 + 3$, then
 - $\phi(t)\delta(t) = \phi(0)\delta(t) = 3\delta(t)$
 - $\phi(t)\delta(t-1) = \phi(1)\delta(t-1) = 4\delta(t-1)$
 - $\phi(t)\delta(t+1) = \phi(-1)\delta(t+1) = 2\delta(t+1)$

Tells you value of the
function at $t = -1$

What happens as we change T ?

$$\phi(t)\delta(t-T) = \phi(T)\delta(t-T)$$

EXAMPLE

- Let $\phi(t) = t^3 + 3$, then
 - $\phi(t)\delta(t) = \phi(0)\delta(t) = 3\delta(t)$
 - $\phi(t)\delta(t-1) = \phi(1)\delta(t-1) = 4\delta(t-1)$
 - $\phi(t)\delta(t+1) = \phi(-1)\delta(t+1) = 2\delta(t+1)$

Tells you value of the
function at $t = -1$

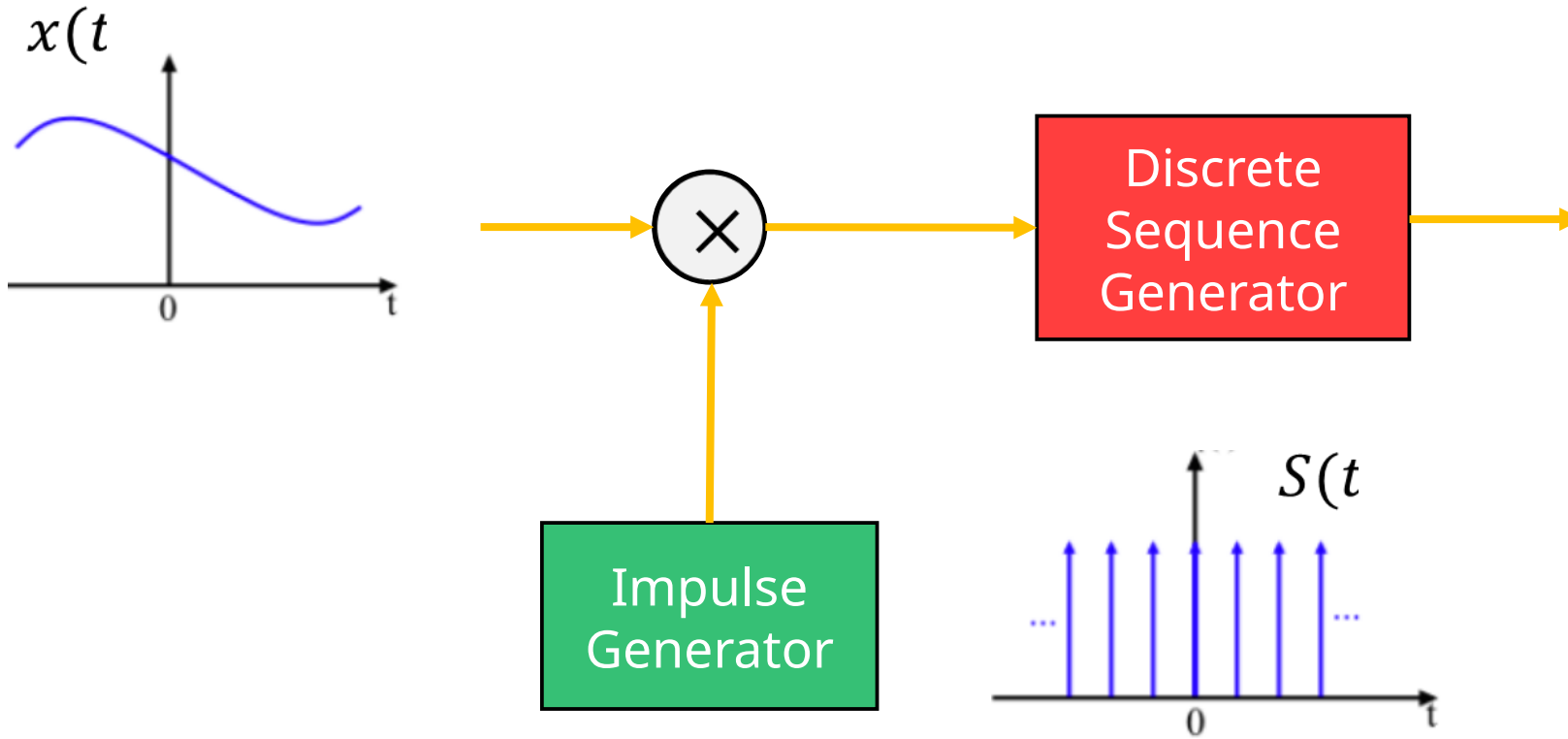
Tells you where to
place it when
plotting!

Impulse as an **IDEAL SAMPLER**

Multiplying a signal by a sequence of shifted impulses discretizes the signal in time.

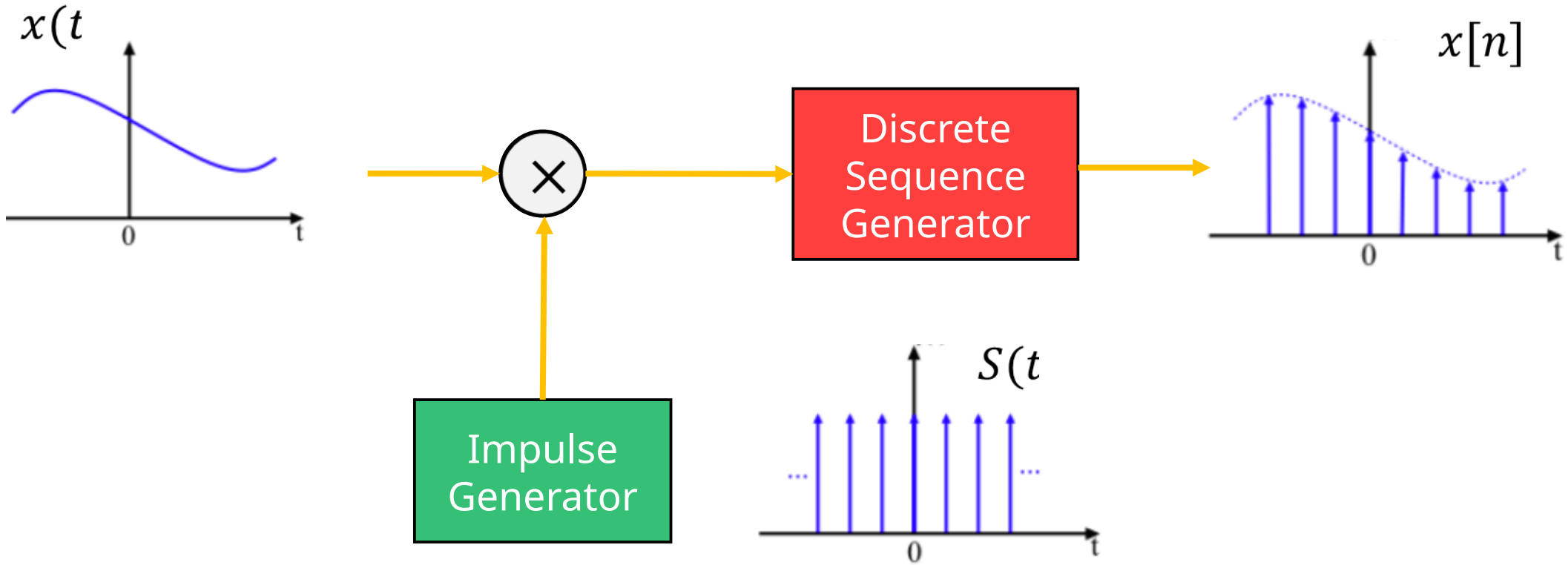
Impulse as an **IDEAL SAMPLER**

Multiplying a signal by a sequence of shifted impulses discretizes the signal in time.



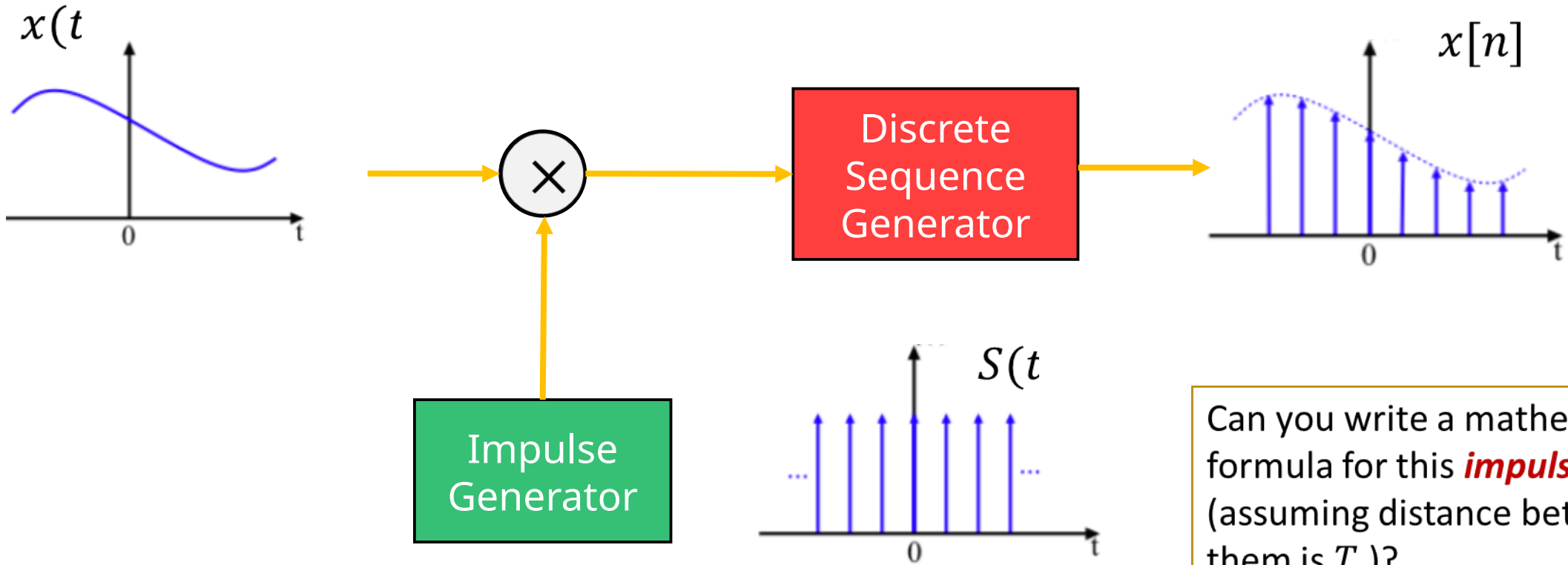
Impulse as an **IDEAL SAMPLER**

Multiplying a signal by a sequence of shifted impulses discretizes the signal in time.



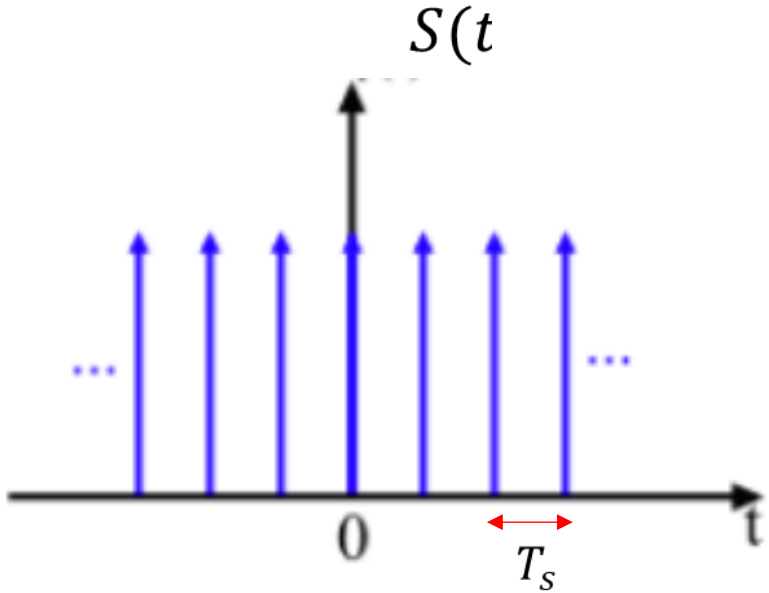
Impulse as an **IDEAL SAMPLER**

Multiplying a signal by a sequence of shifted impulses discretizes the signal in time.



Can you write a mathematical formula for this **impulse train** (assuming distance between them is T_s)?

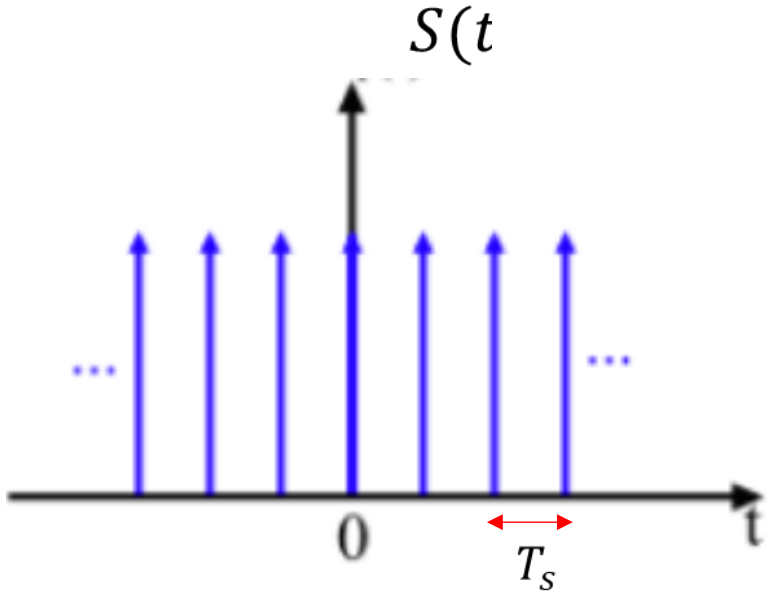
Here comes the **IMPULSE TRAIN**



Called "sampling time"

$$S(t) = ?$$

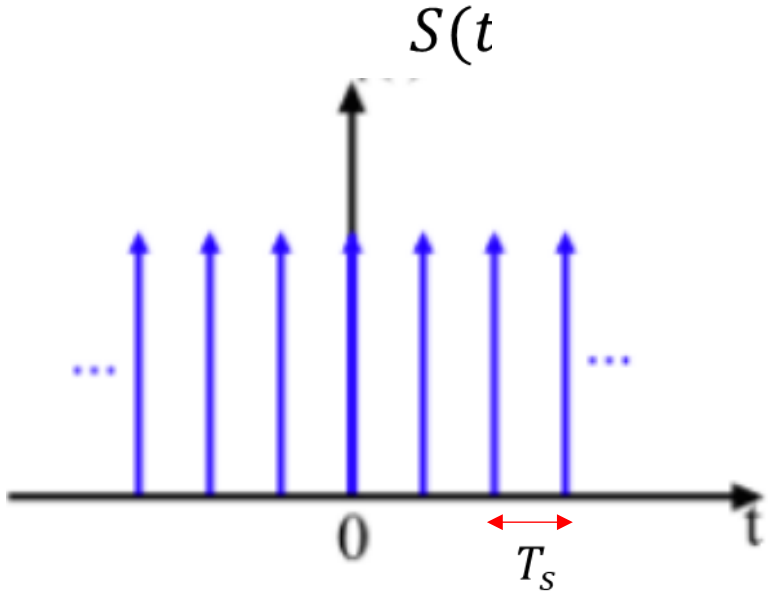
Here comes the **IMPULSE TRAIN**



Called "sampling time"

$$S(t) = \delta(t) + \delta(t - T_s) + \delta(t + T_s) \\ + \delta(t - 2T_s) + \delta(t + 2T_s) \dots$$

Here comes the **IMPULSE TRAIN**



Called "sampling time"

$$S(t) = \delta(t) + \delta(t - T_s) + \delta(t + T_s) \\ + \delta(t - 2T_s) + \delta(t + 2T_s) \dots$$



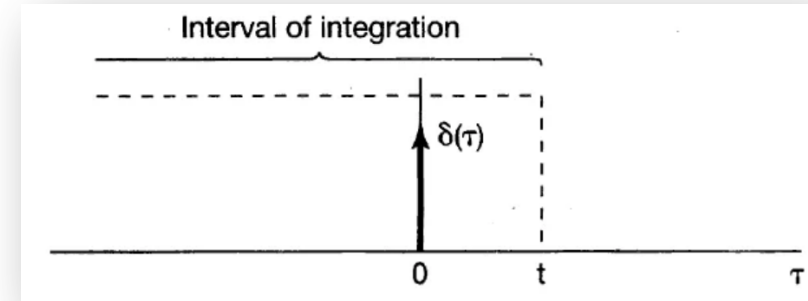
$$S(t) = \sum_{n \in \mathbb{Z}} \delta(t - nT_s)$$

Can you find the area under
the impulse from $-\infty$ to t ?

$$f(t) = \int_{-\infty}^t \delta(\tau) d\tau = ?$$

Can you find the area under the impulse from $-\infty$ to t ?

$$f(t) = \int_{-\infty}^t \delta(\tau) d\tau \quad \rightarrow \quad = 1 \text{ for } t \geq 0$$

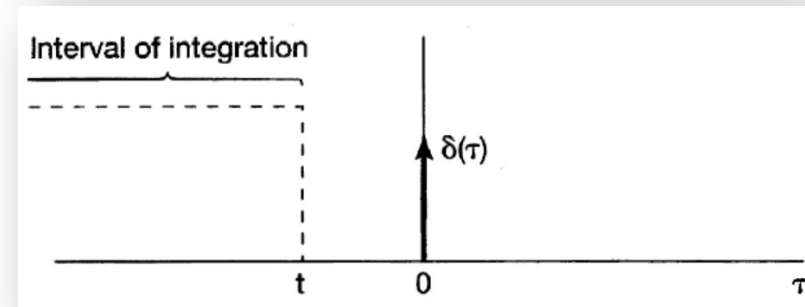
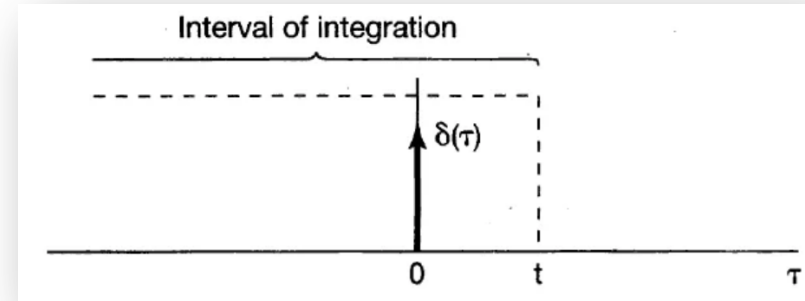


Can you find the area under the impulse from $-\infty$ to t ?

$$f(t) = \int_{-\infty}^t \delta(\tau) d\tau$$

$= 1 \text{ for } t \geq 0$

$= 0 \text{ for } t < 0$

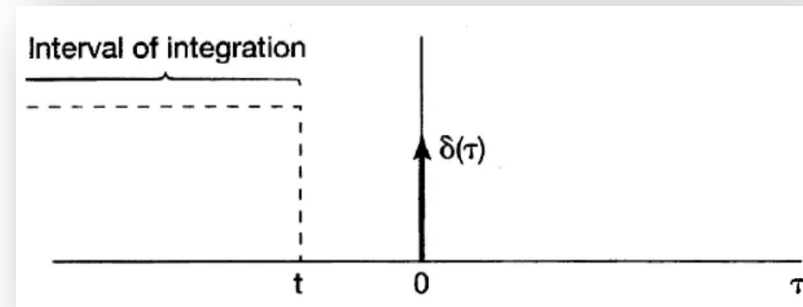
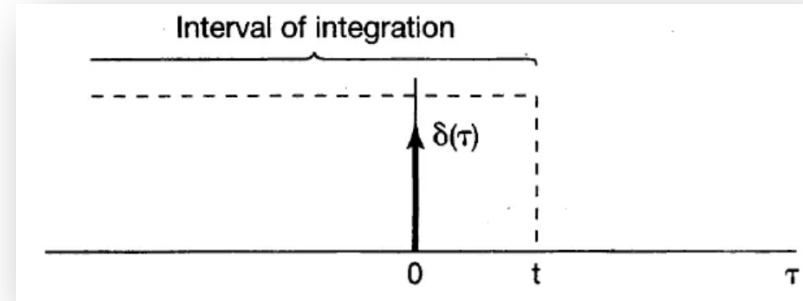


Can you find the area under the impulse from $-\infty$ to t ?

$$f(t) = \int_{-\infty}^t \delta(\tau) d\tau$$

$= 1 \text{ for } t \geq 0$

$= 0 \text{ for } t < 0$



$$f(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

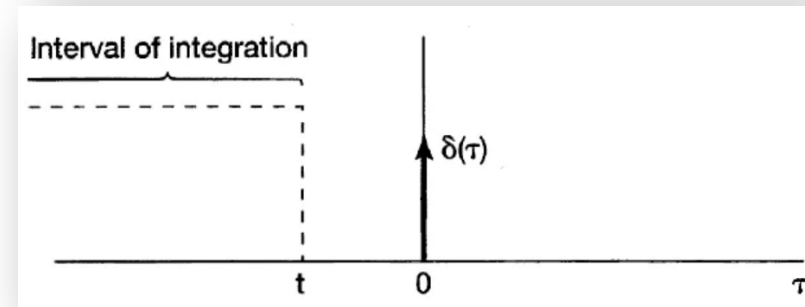
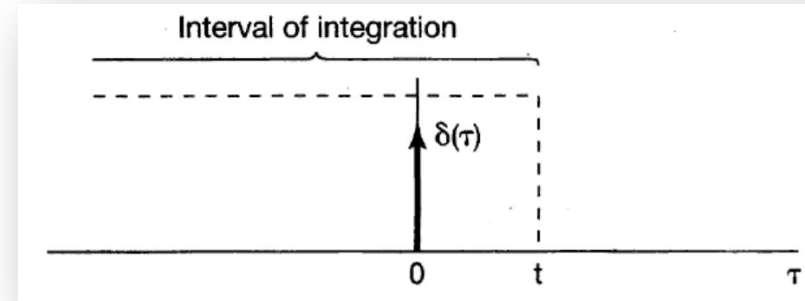
*Do we know
such a
function?*

Can you find the area under the impulse from $-\infty$ to t ?

$$f(t) = \int_{-\infty}^t \delta(\tau) d\tau$$

$= 1 \text{ for } t \geq 0$

$= 0 \text{ for } t < 0$



$$f(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

*Do we know
such a
function?*

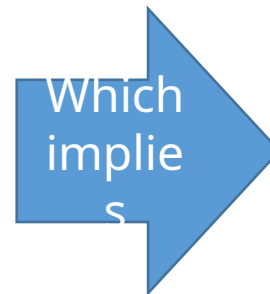
$$u(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

So unit step is
the integral of
impulse!

$$u(t) = \int_{-\infty}^t \delta(\tau) d\tau$$

So unit step is
the integral of
impulse!

$$u(t) = \int_{-\infty}^t \delta(\tau) d\tau$$



$$\frac{du(t)}{dt} =$$

So unit step is
the integral of
impulse!

$$u(t) = \int_{-\infty}^t \delta(\tau) d\tau$$

Which
implies

$$\frac{du(t)}{dt} = \delta(t)$$

*Next, we look
at*

**COMPLEX
SINUSOID**

$$e^{j\omega t}$$

*Next, we look
at*

**COMPLEX
SINUSOID**

$$e^{j\omega t}$$

And it's bigger
cousin

**COMPLEX
EXPONENTIAL**

$$e^{st} = e^{(\sigma + j\omega)t}$$

Next, we look
at

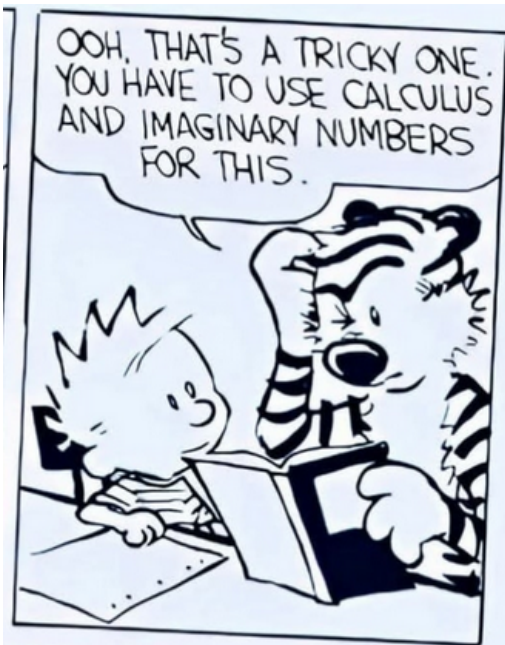
COMPLEX SINUSOID

$$e^{j\omega t}$$

And it's bigger
cousin

COMPLEX EXPONENTIAL

$$e^{st} = e^{(\sigma + j\omega)t}$$



1999: nineteen ninety nine
1888: eighteen eighty eight
1777: seventeen seventy seven
1111 :

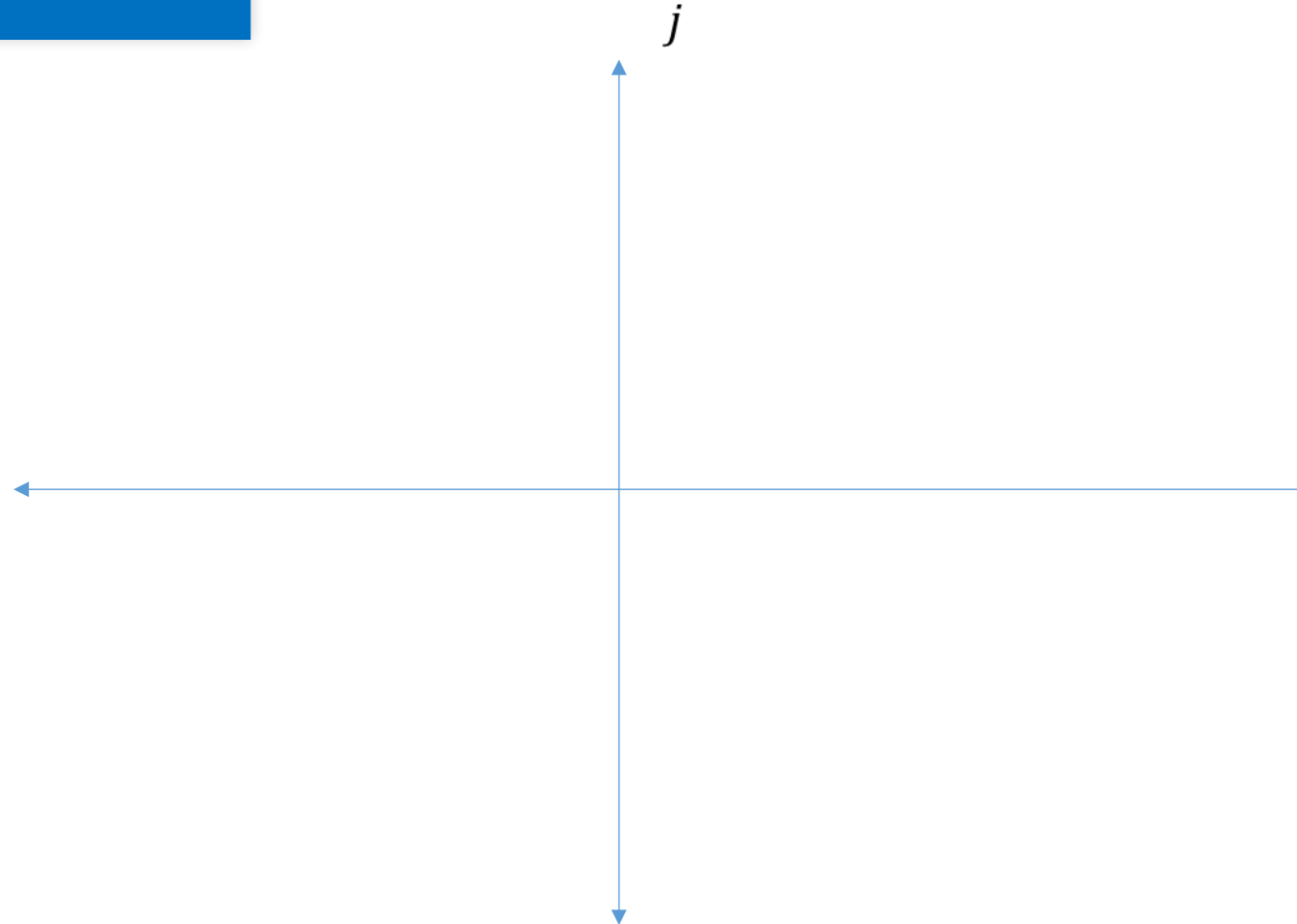


*An extremely short
review of complex
numbers...*

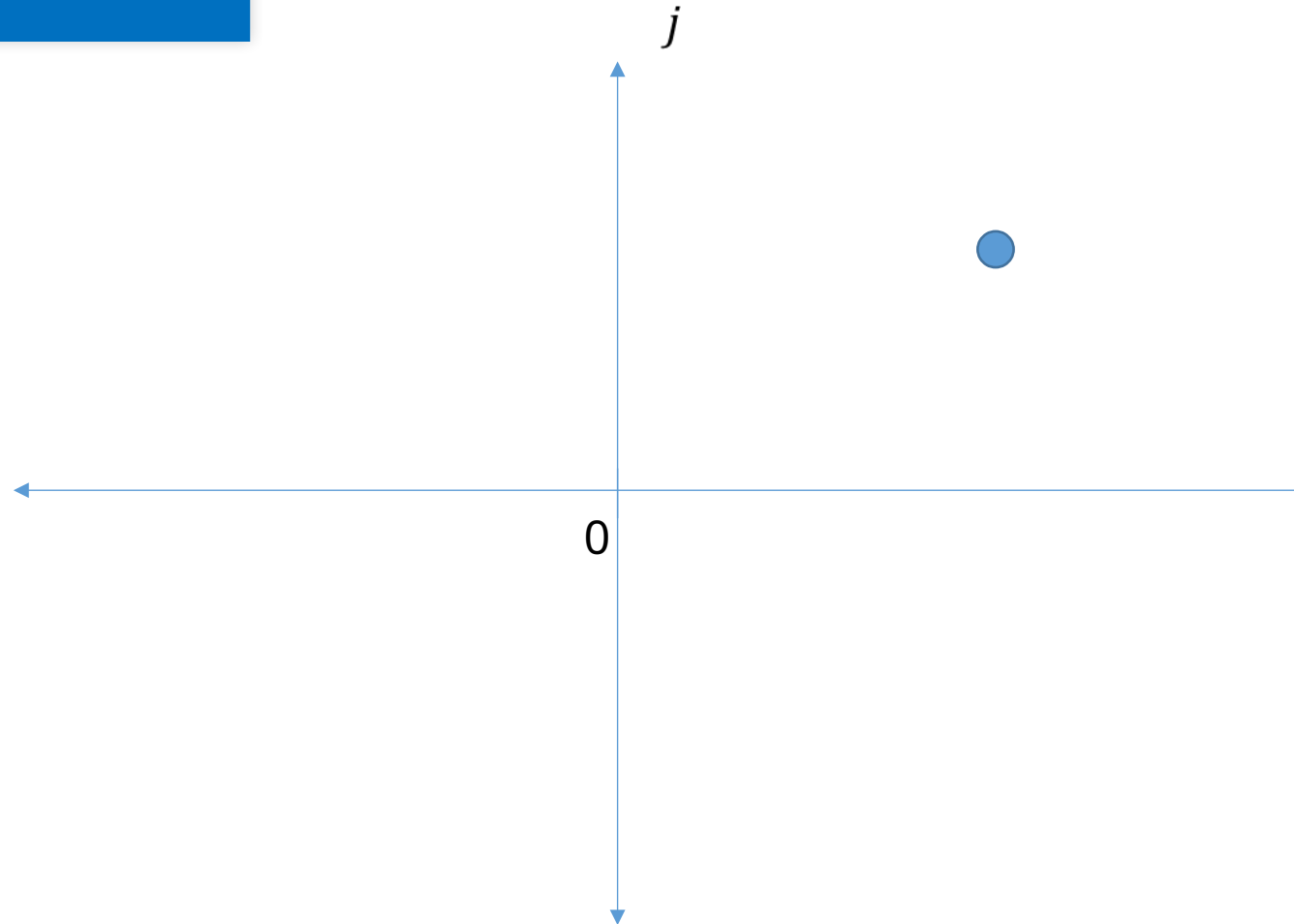
Plotting a Complex Number



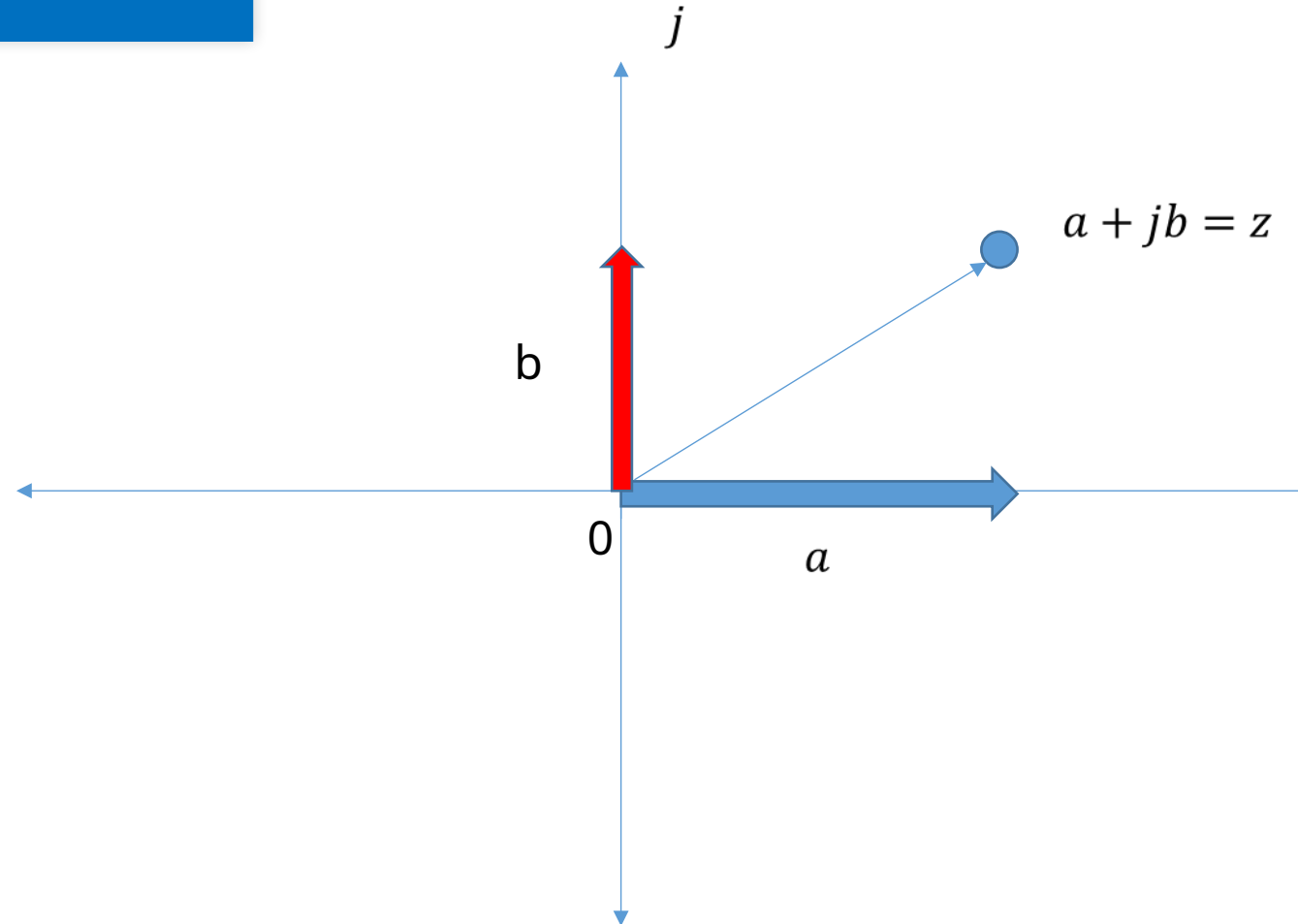
Plotting a Complex Number



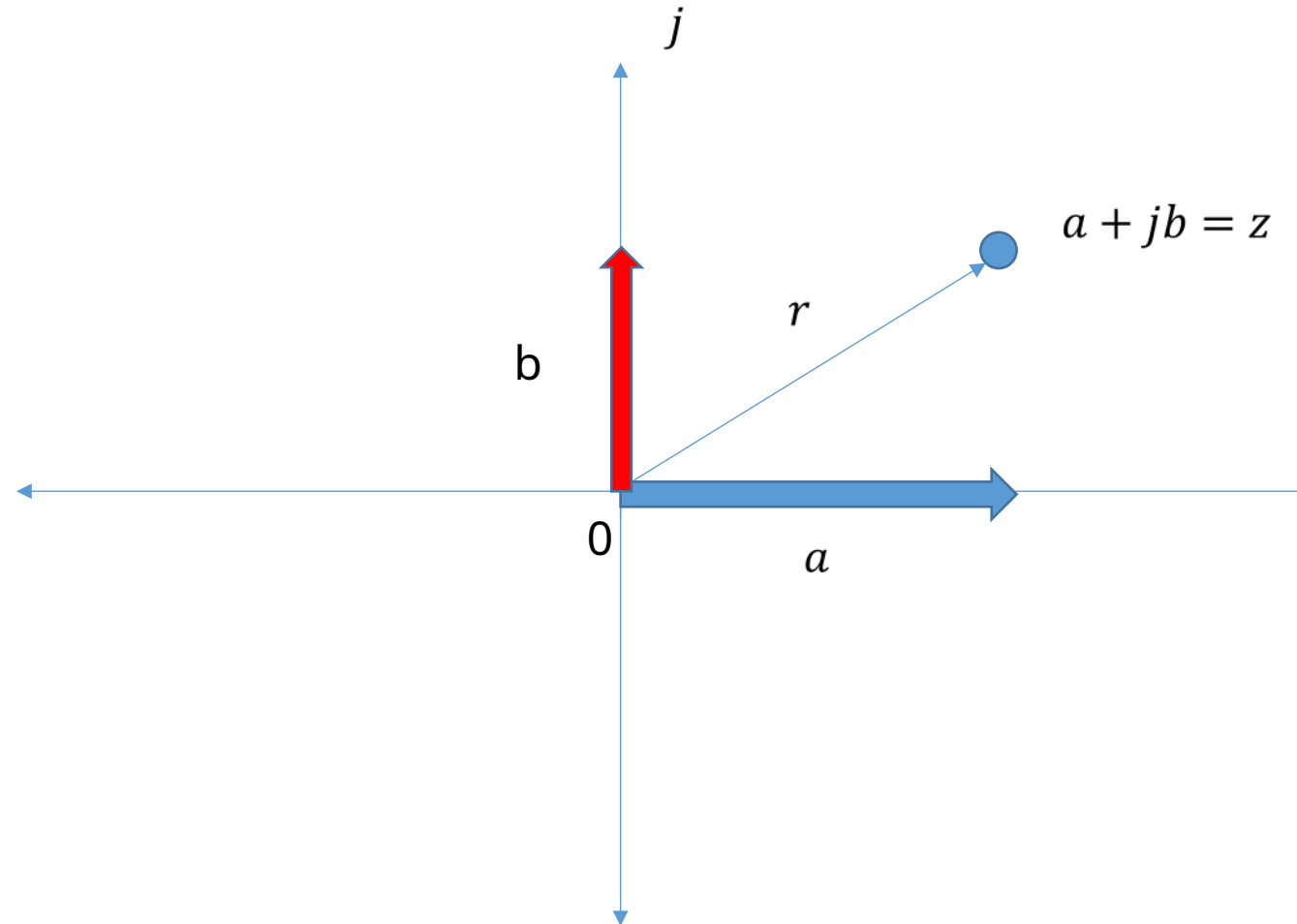
Plotting a Complex Number



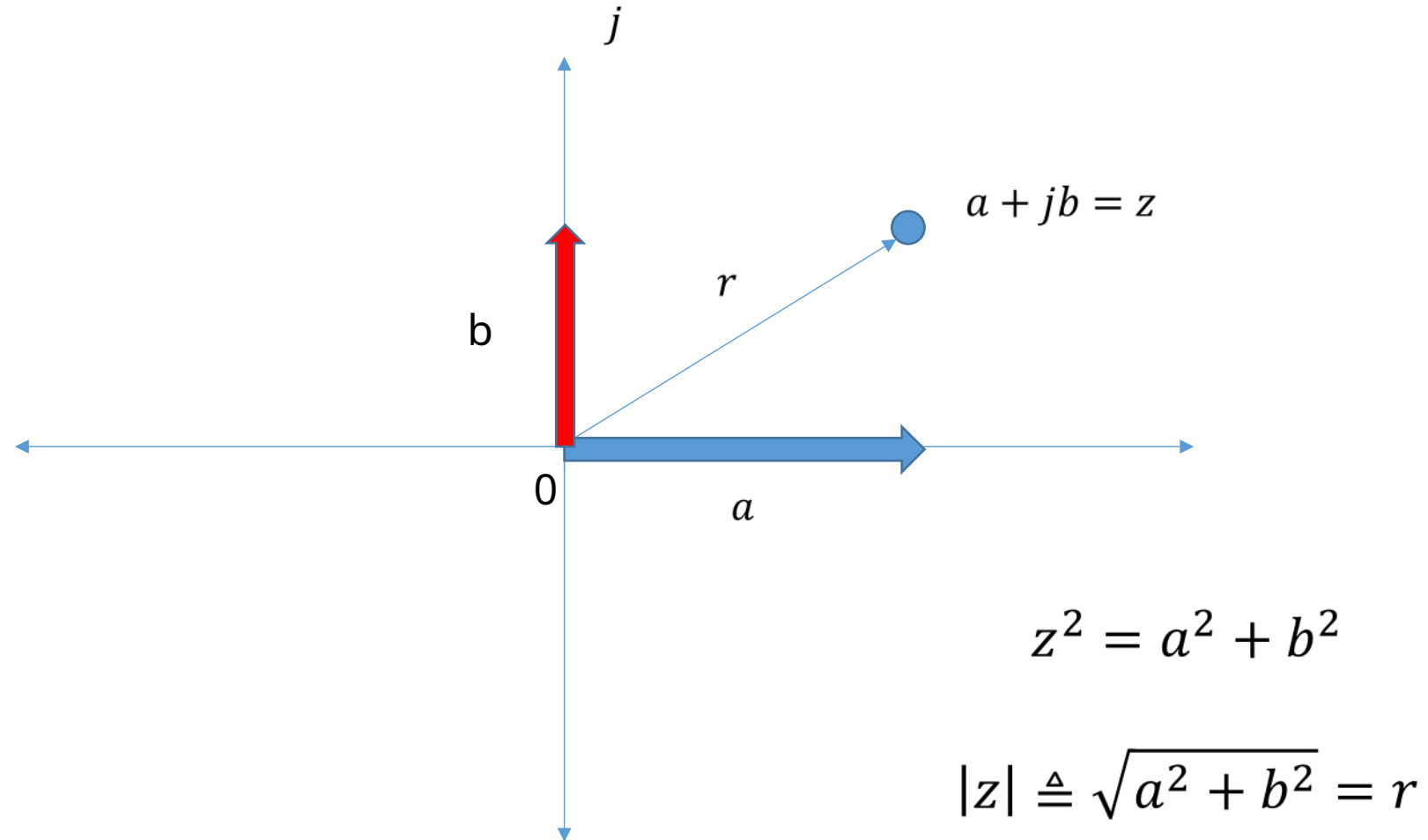
Plotting a Complex Number



What is the length of z ?

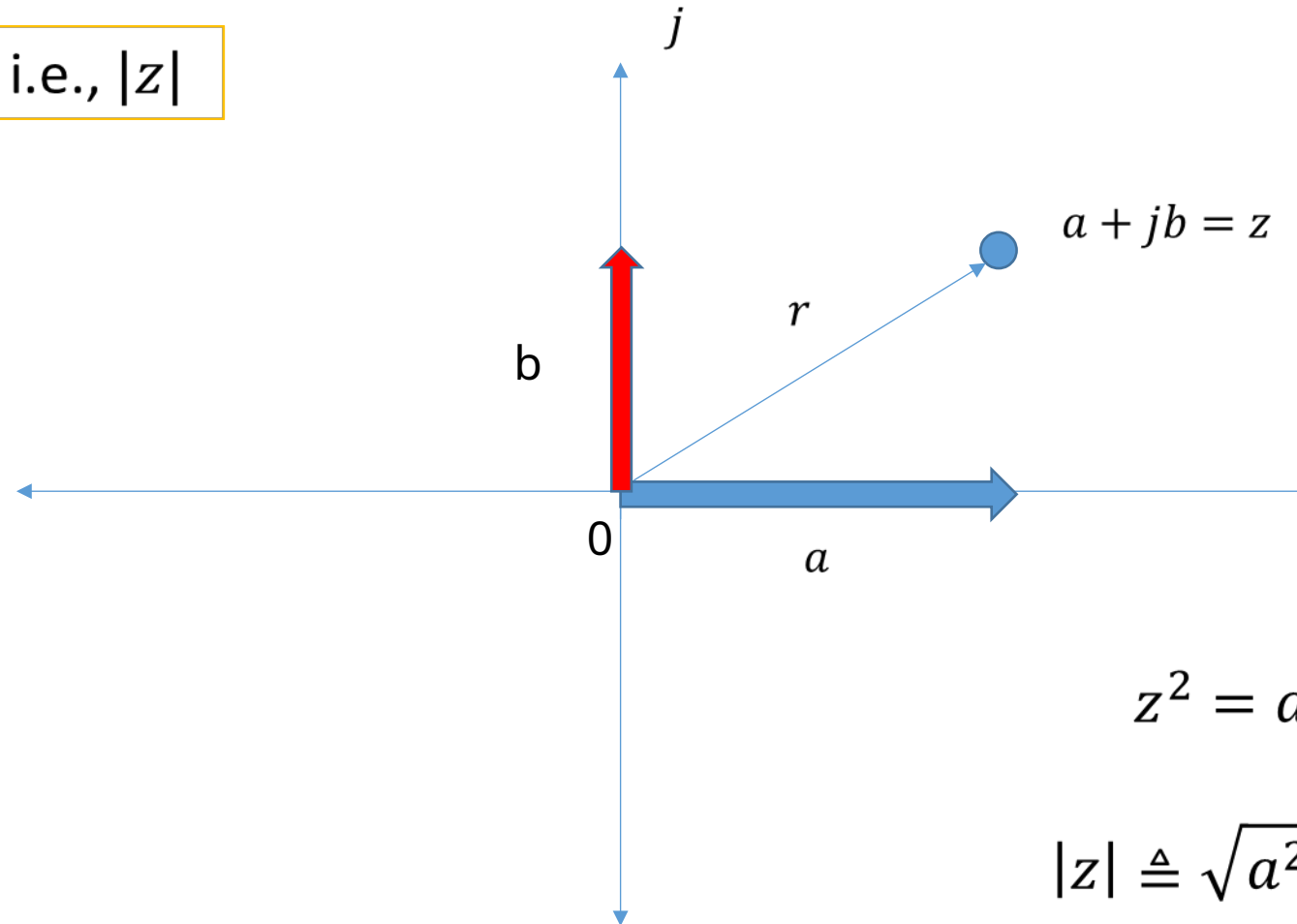


What is the length of z ?



What is the length of z ?

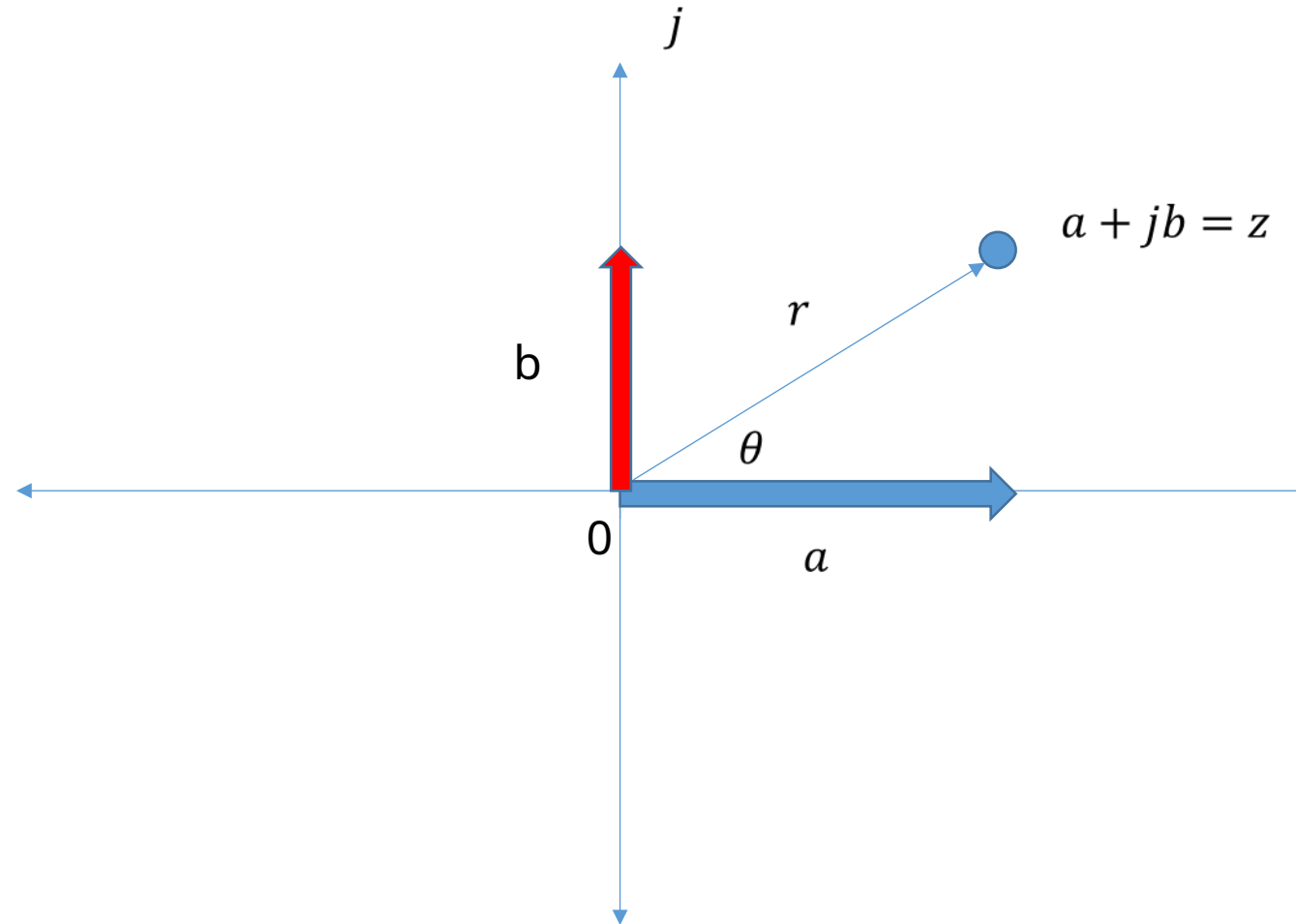
Modulus of z , i.e., $|z|$



$$z^2 = a^2 + b^2$$

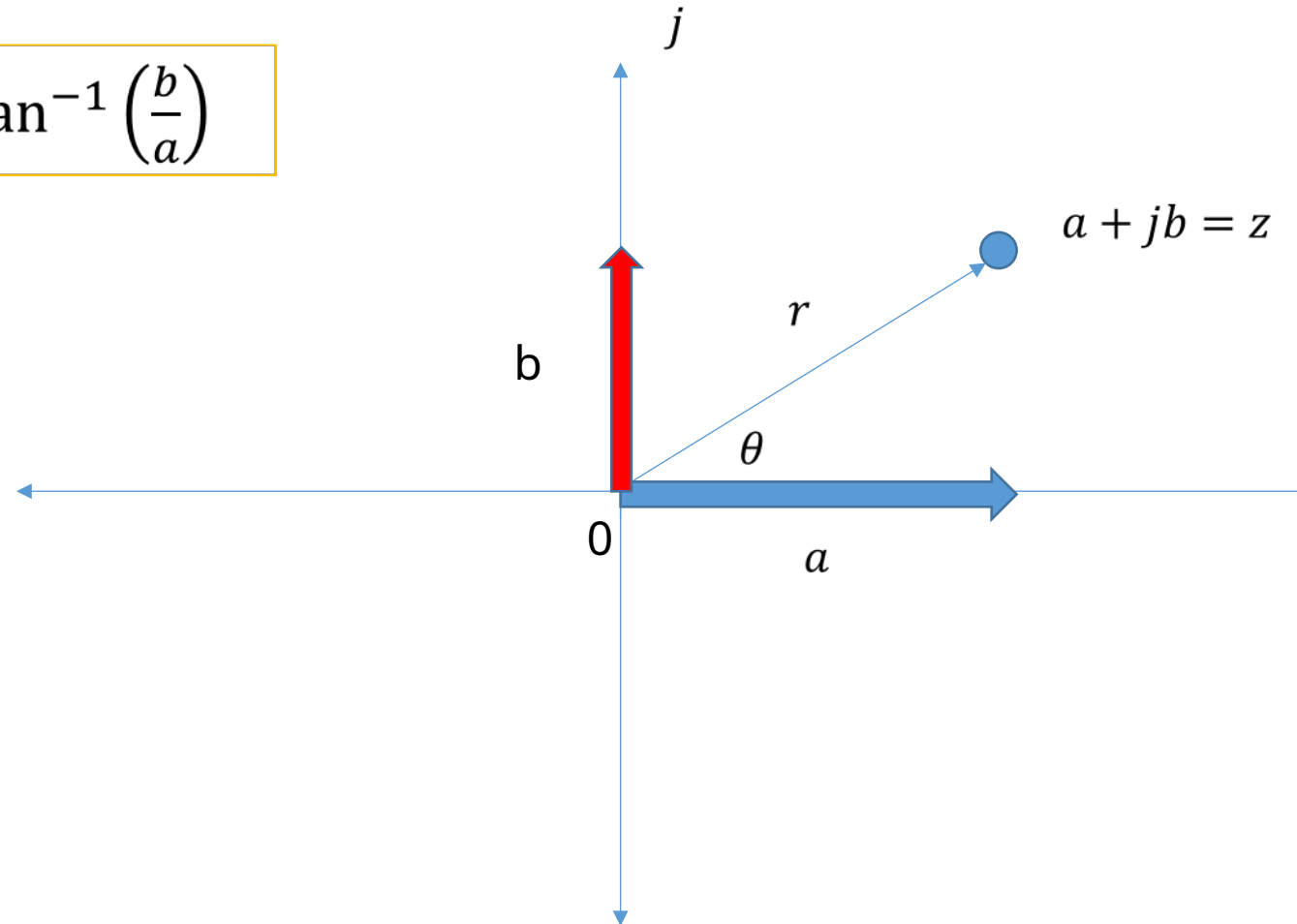
$$|z| \triangleq \sqrt{a^2 + b^2} = r$$

What is the angle of z ?

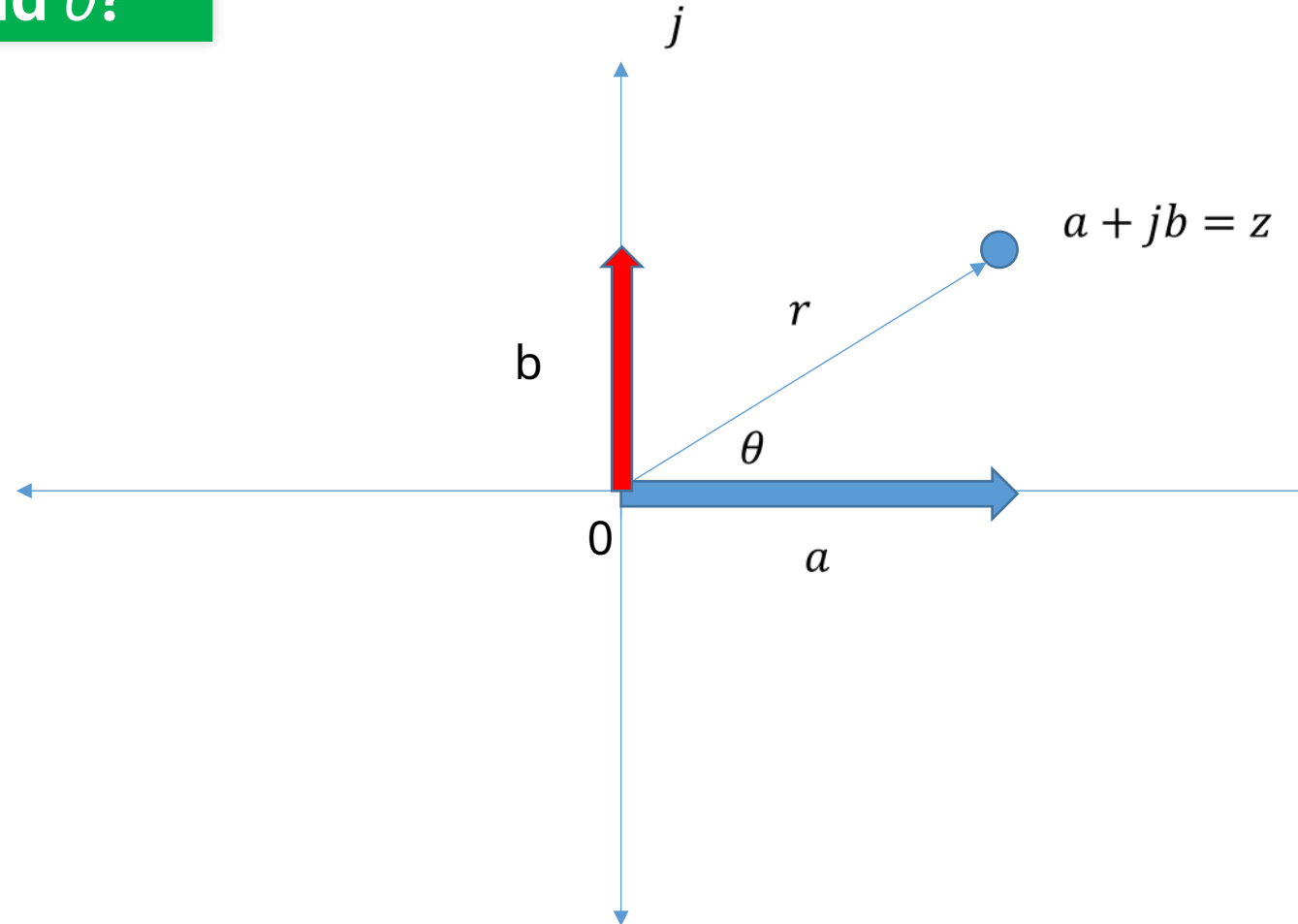


What is the angle of z ?

$$\theta = \tan^{-1} \left(\frac{b}{a} \right)$$

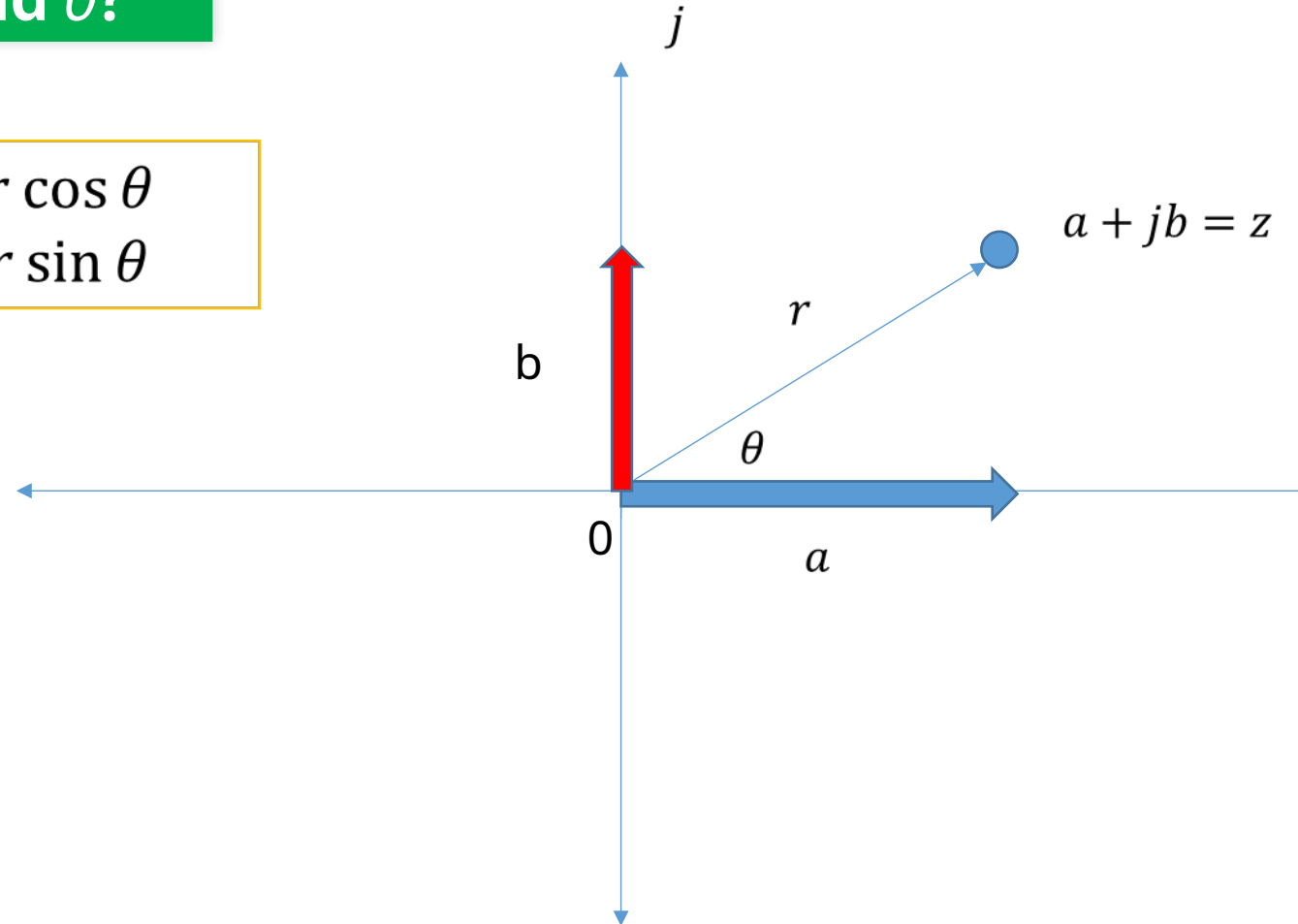


Can we write a and b
in terms of r and θ ?

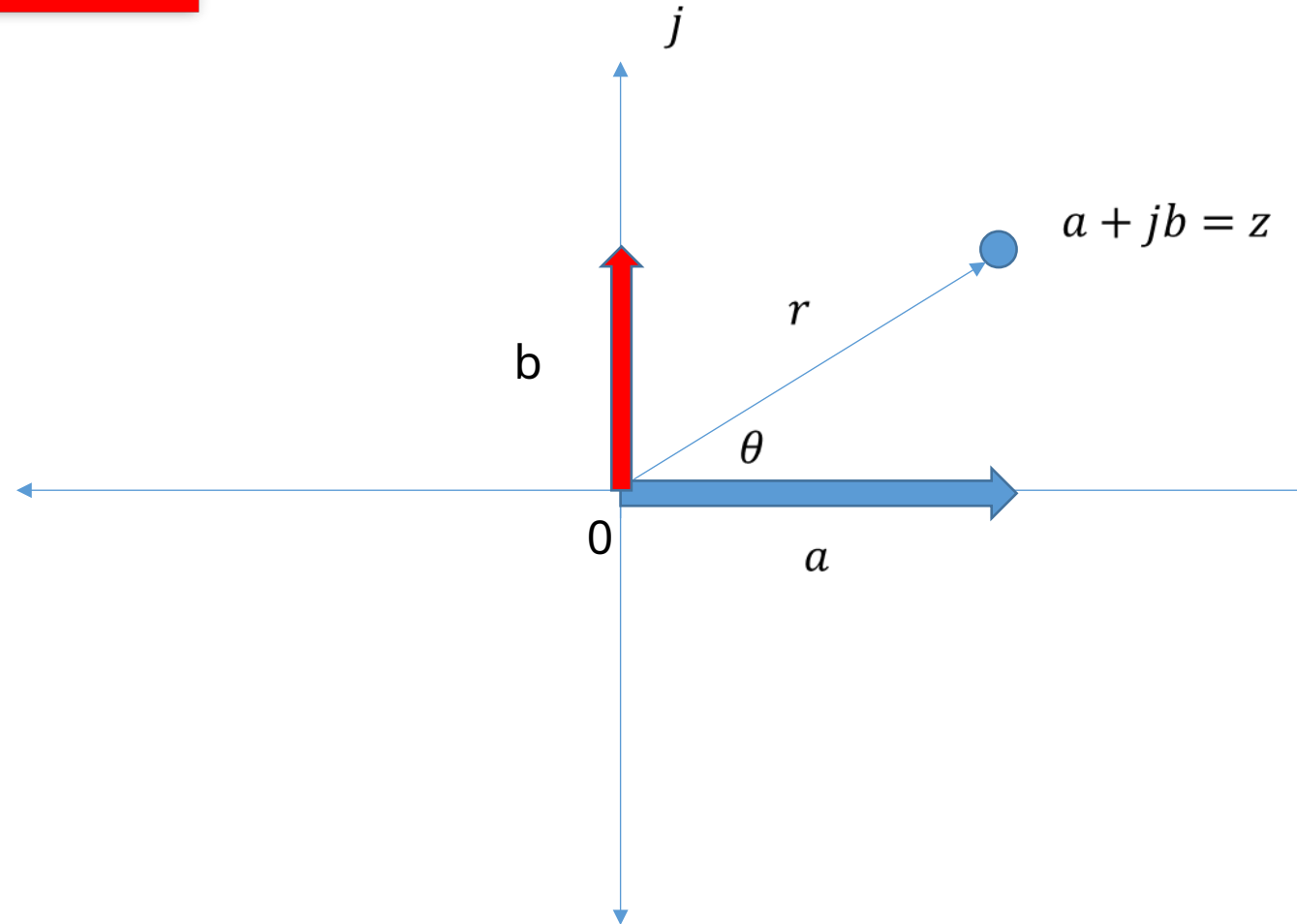


Can we write a and b
in terms of r and θ ?

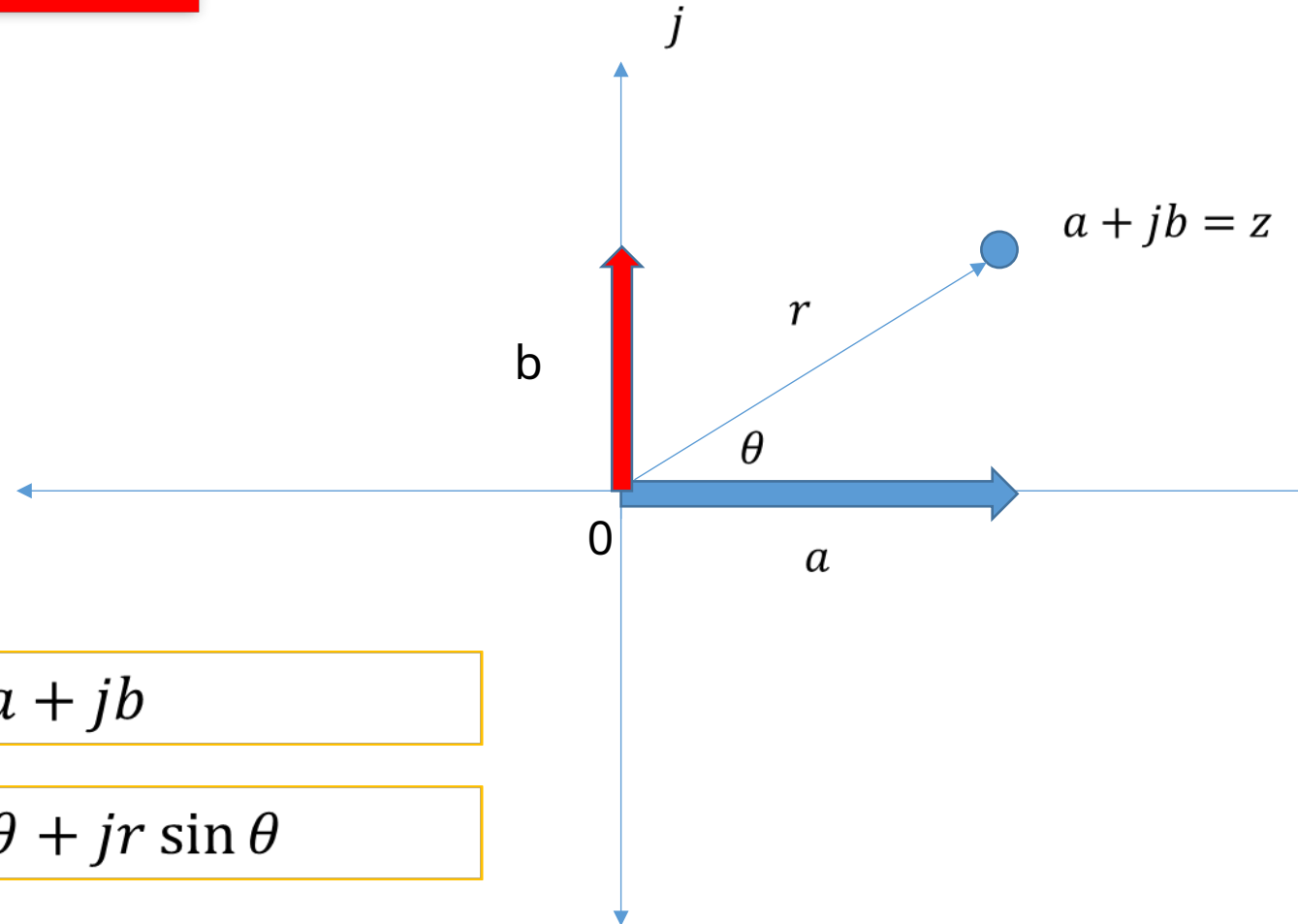
$$a = r \cos \theta$$
$$b = r \sin \theta$$



Can we write z in terms of r and θ ?



Can we write z in terms of r and θ ?

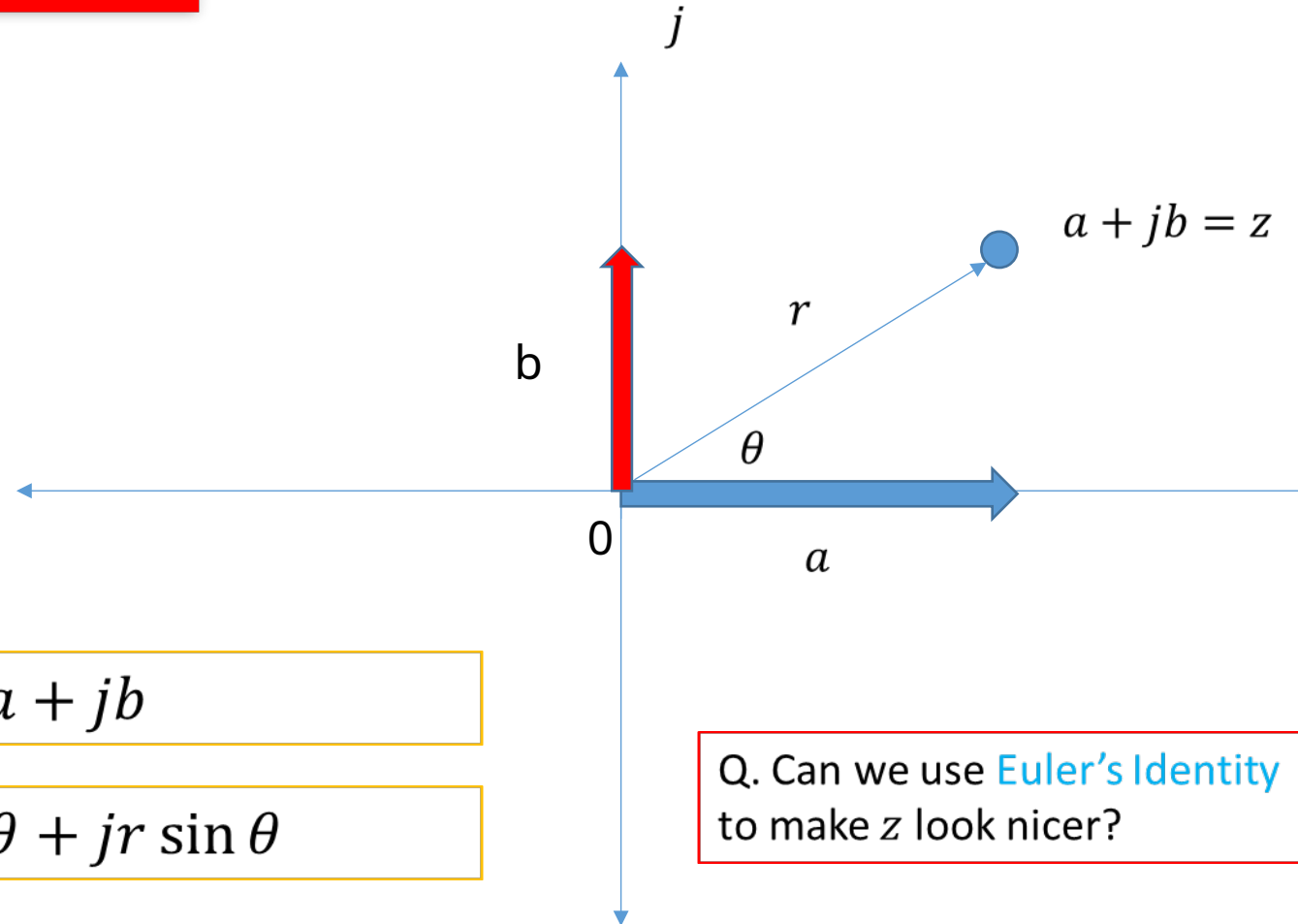


$$z = a + jb$$

$$z = r \cos \theta + jr \sin \theta$$

$$z = r(\cos \theta + j \sin \theta)$$

Can we write z in terms of r and θ ?



$$z = a + jb$$

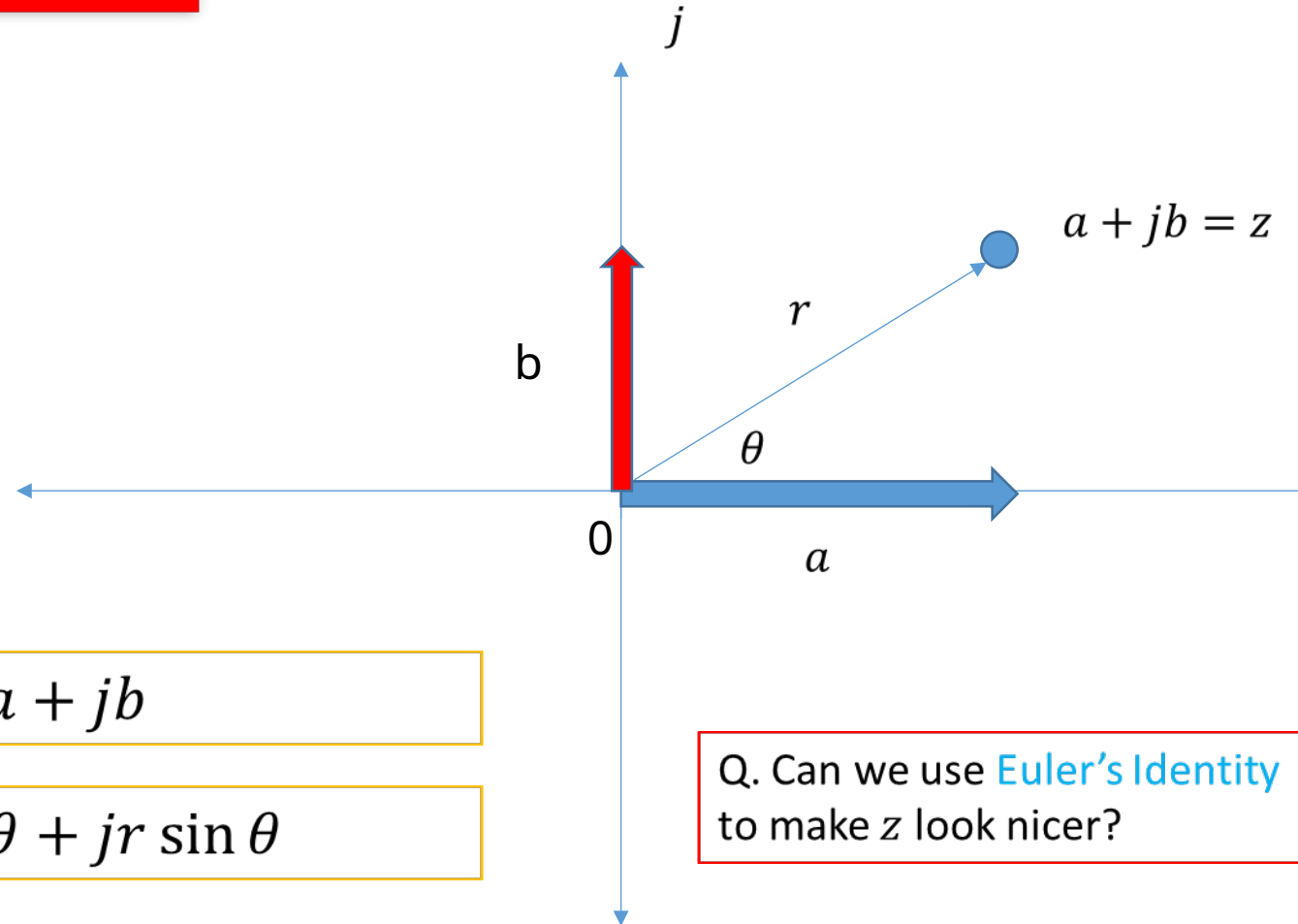
$$z = r \cos \theta + jr \sin \theta$$

$$z = r(\cos \theta + j \sin \theta)$$

Q. Can we use **Euler's Identity** to make z look nicer?

$$e^{j\theta} = \cos \theta + j \sin \theta$$

Can we write z in terms of r and θ ?



$$z = a + jb$$

$$z = r \cos \theta + jr \sin \theta$$

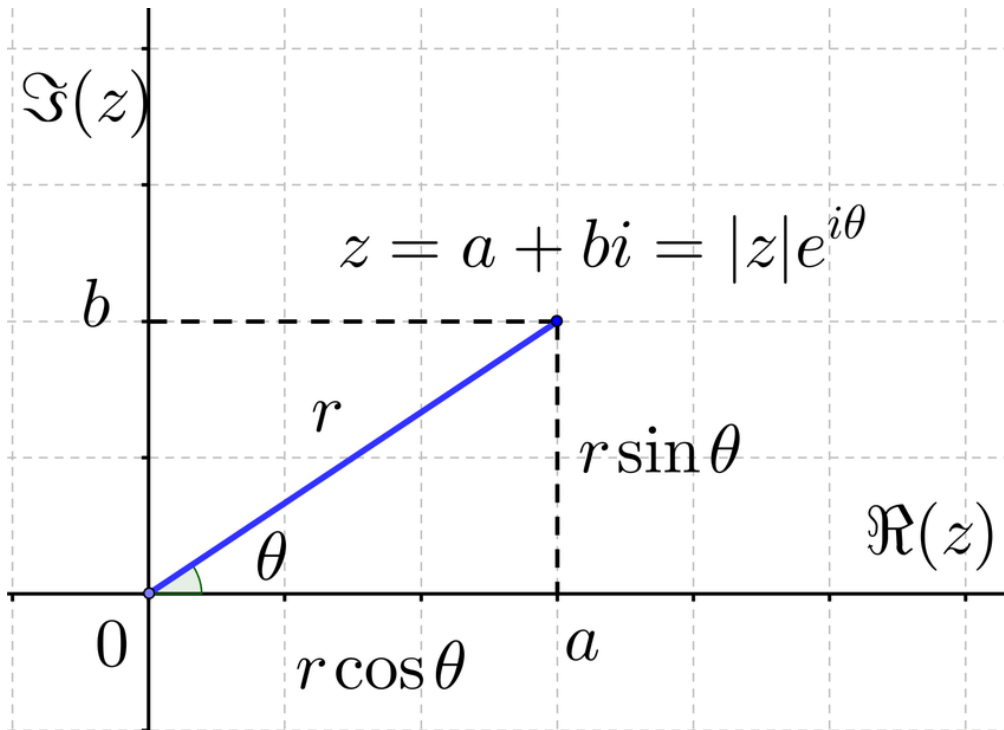
$$z = r(\cos \theta + j \sin \theta)$$

Q. Can we use **Euler's Identity** to make z look nicer?

$$e^{j\theta} = \cos \theta + j \sin \theta$$

$$z = re^{j\theta}$$

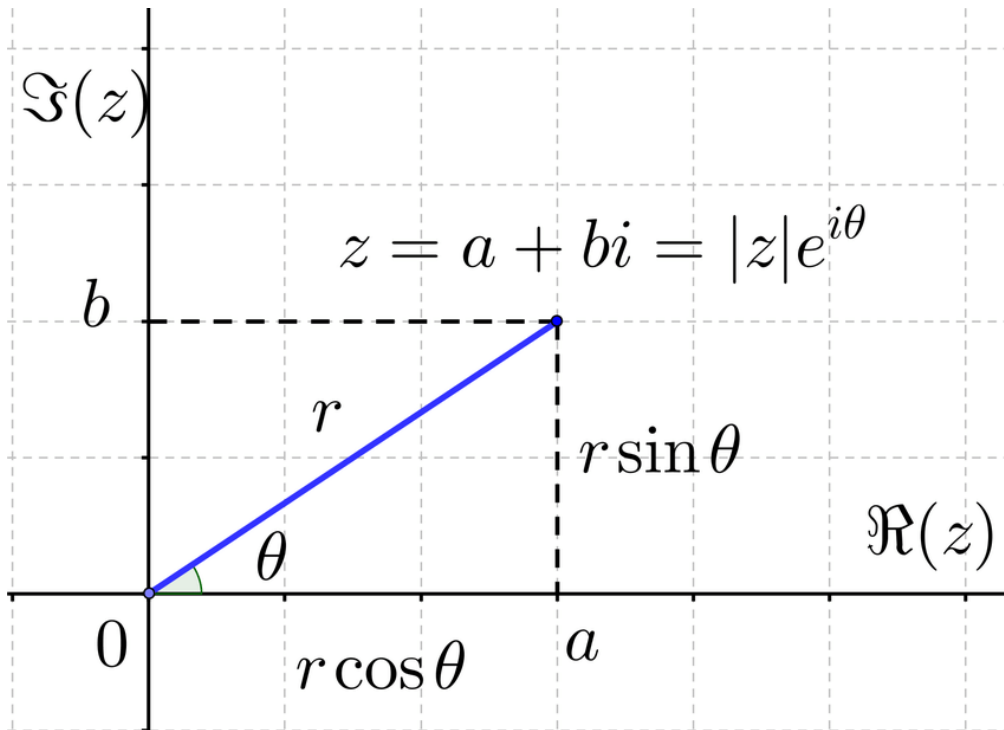
Let us
summarize...



Cartesian form $z = a + jb$

Polar form $z = |z|e^{j\theta} = re^{j\theta}$

Let us
summarize...



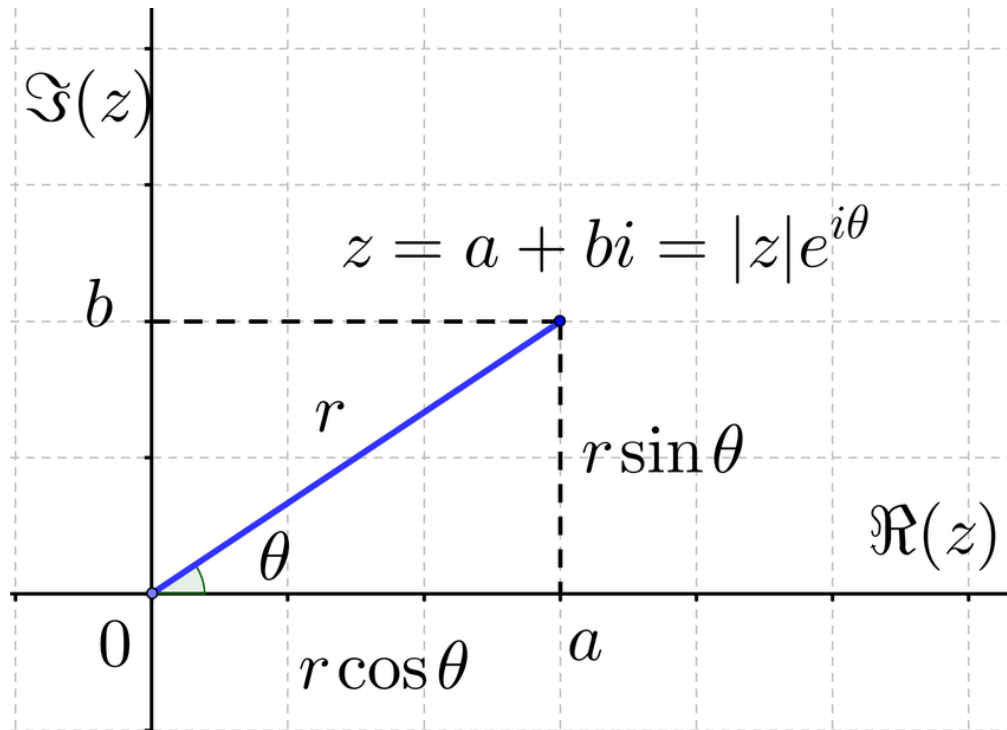
Cartesian form $z = a + jb$

Polar form $z = |z|e^{j\theta} = re^{j\theta}$

Cartesian to Polar

$$\left\{ \begin{array}{l} r = |z| = \sqrt{a^2 + b^2} \\ \theta = \tan^{-1} \frac{b}{a} \end{array} \right.$$

Let us
summarize...



Cartesian form $z = a + jb$

Polar form $z = |z|e^{j\theta} = re^{j\theta}$

Cartesian to
Polar

$$r = |z| = \sqrt{a^2 + b^2}$$

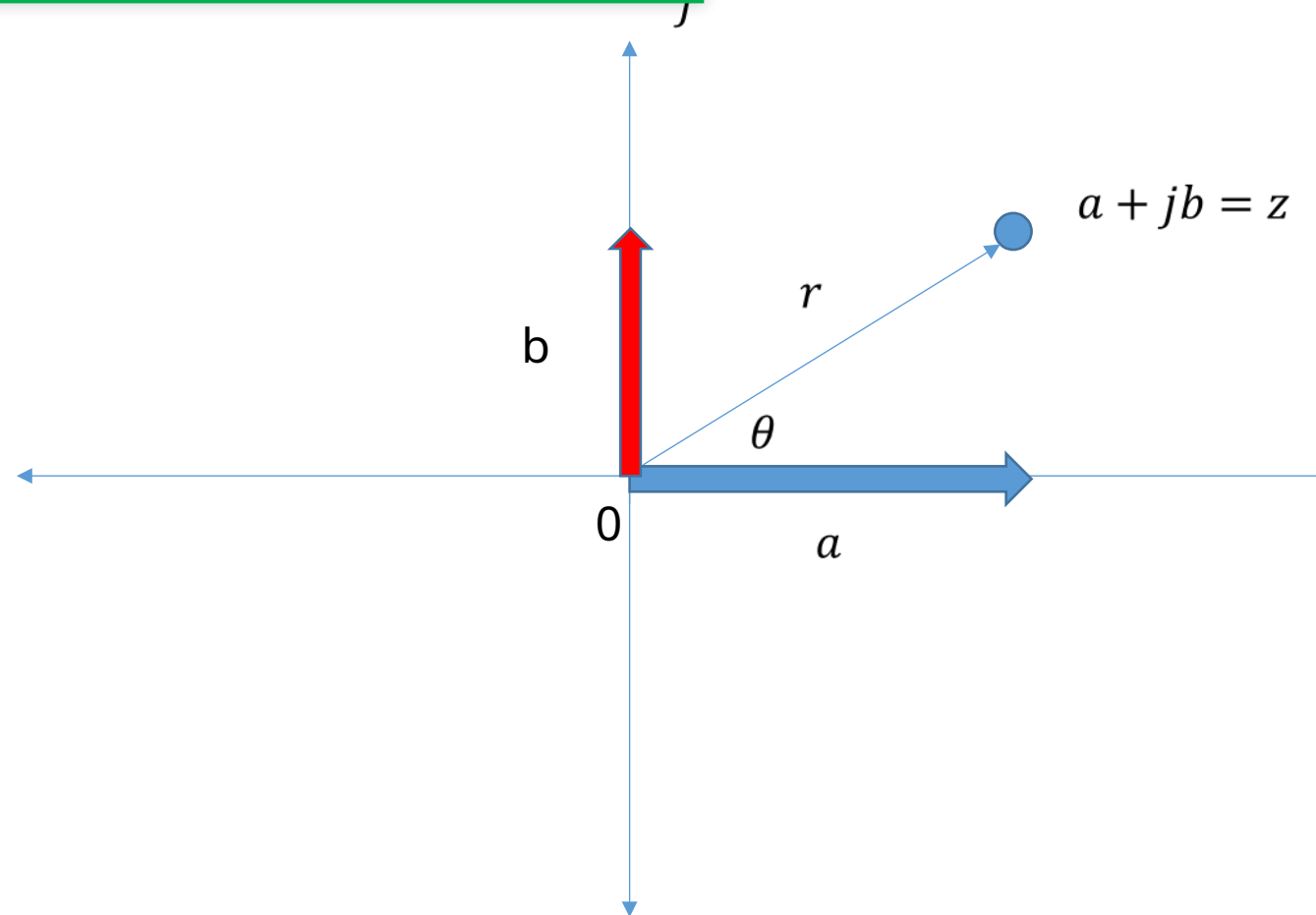
$$\theta = \tan^{-1} \frac{b}{a}$$

Polar to
Cartesian

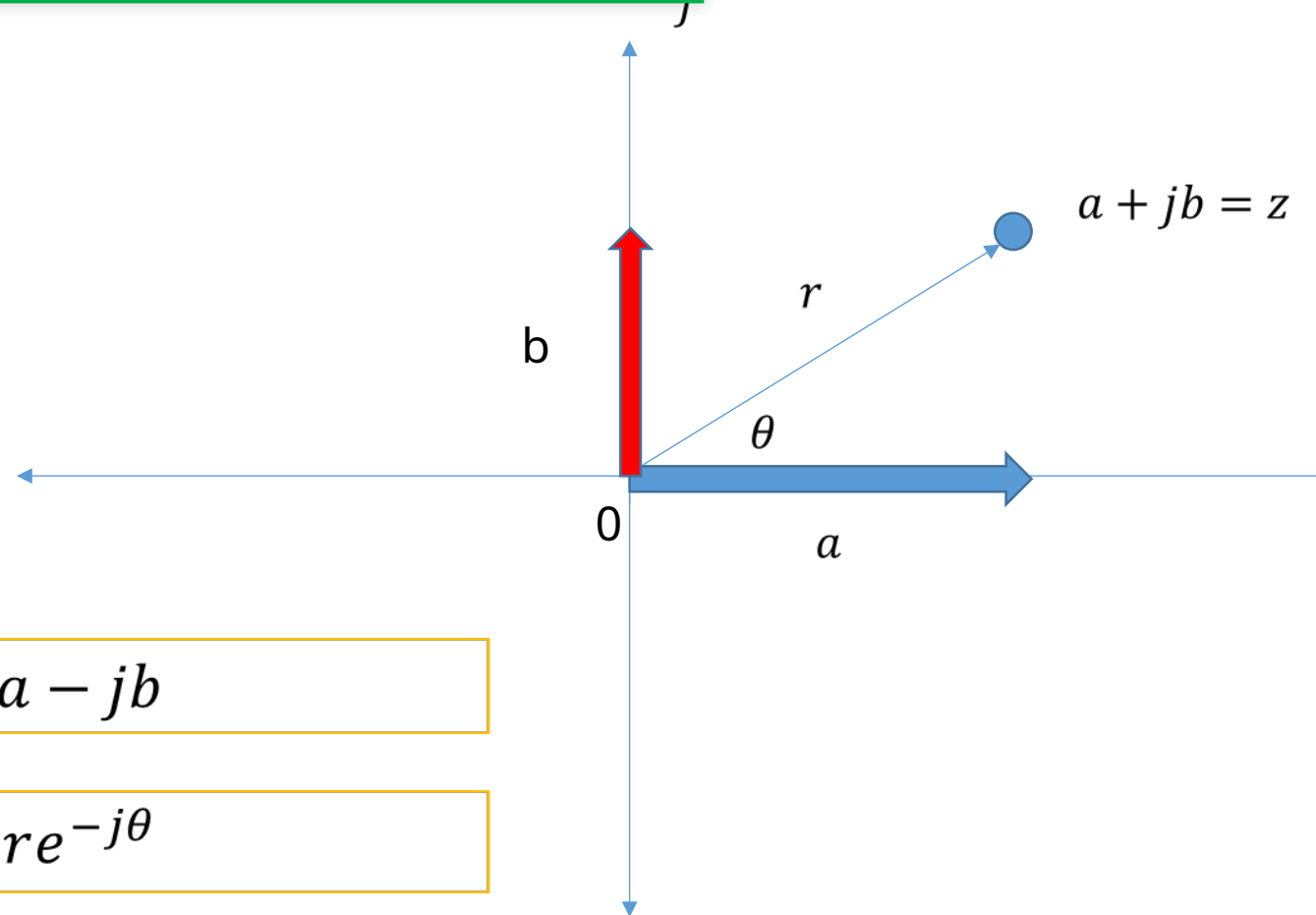
$$a = r \cos \theta$$

$$b = r \sin \theta$$

Q. What happens if we change the sign of j ?



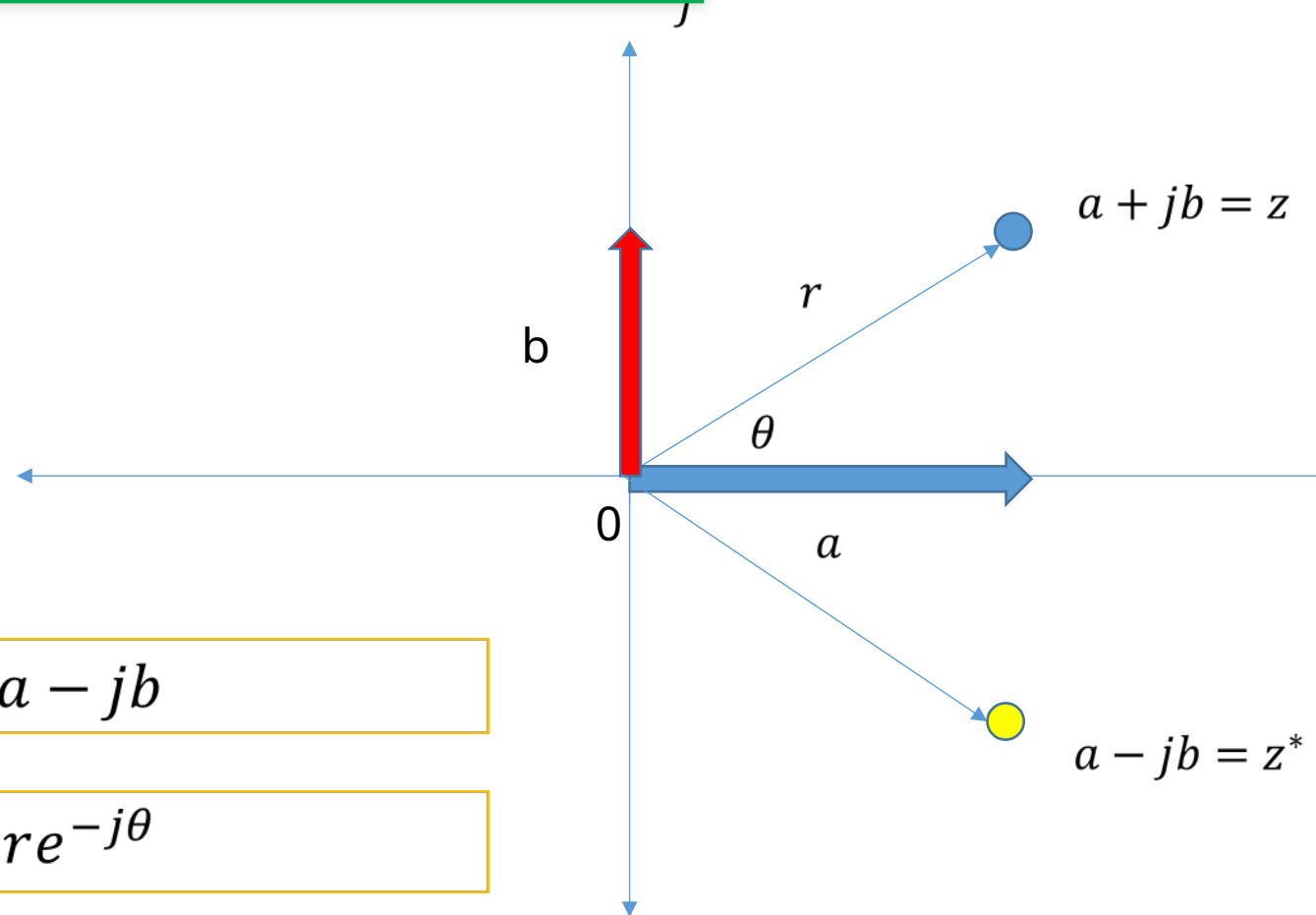
Q. What happens if we change the sign of j ?



$$z^* = a - jb$$

$$z^* = r e^{-j\theta}$$

Q. What happens if we change the sign of j ?

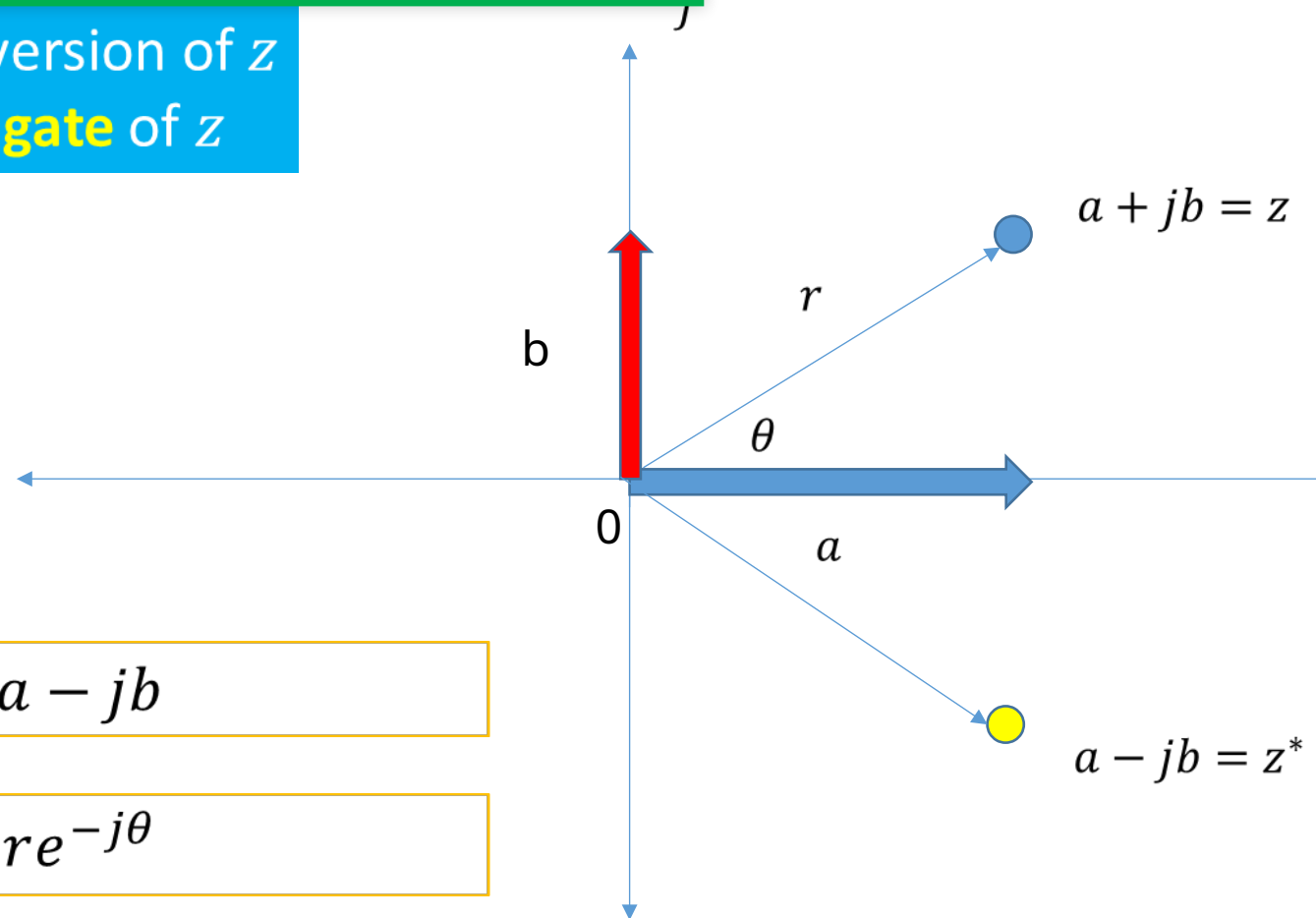


$$z^* = a - jb$$

$$z^* = re^{-j\theta}$$

Q. What happens if we change the sign of j ?

We get a flipped version of z that we call **conjugate** of z



$$z^* = a - jb$$

$$z^* = r e^{-j\theta}$$

Complex Number Arithmetic

Quick Review

Complex Numbers

Addition

$$\begin{aligned}(a + bi) + (c + di) \\&= a + c + bi + di \\&= (a + c) + (b + d)i\end{aligned}$$

Multiplication

$$\begin{aligned}(a + bi)(c + di) \\&= c(a + bi) + di(a + bi) \\&= ac + bci + adi + bdi^2 \\&= ac - bd + (bc + ad)i\end{aligned}$$

Complex Number Arithmetic

Quick Review

Using both polar and Cartesian forms, determine $z_1 z_2$ and z_1/z_2 for the numbers

$$z_1 = 3 + j4 = 5e^{j53.1^\circ} \quad \text{and} \quad z_2 = 2 + j3 = \sqrt{13}e^{j56.3^\circ}$$

Multiplication: Cartesian Form

$$z_1 z_2 = (3 + j4)(2 + j3) = (6 - 12) + j(8 + 9) = -6 + j17$$

Multiplication: Polar Form

$$z_1 z_2 = (5e^{j53.1^\circ})(\sqrt{13}e^{j56.3^\circ}) = 5\sqrt{13}e^{j109.4^\circ}$$

Complex Number Arithmetic

Quick Review

Division: Cartesian Form

$$\frac{z_1}{z_2} = \frac{3 + j4}{2 + j3}$$

$$\frac{z_1}{z_2} = \frac{(3 + j4)(2 - j3)}{(2 + j3)(2 - j3)} = \frac{18 - j1}{2^2 + 3^2} = \frac{18 - j1}{13} = \frac{18}{13} - j\frac{1}{13}$$

Division: Polar Form

$$\frac{z_1}{z_2} = \frac{5e^{j53.1^\circ}}{\sqrt{13}e^{j56.3^\circ}} = \frac{5}{\sqrt{13}}e^{j(53.1^\circ - 56.3^\circ)} = \frac{5}{\sqrt{13}}e^{-j3.2^\circ}$$

Some Interesting Results...

Can you evaluate the following?

$$(z^*)^* = ?$$

$$zz^* = ?$$

$$|e^{j\theta}| = ?$$

$$e^{j2\pi n} = ? \quad (n = \text{integer})$$

Some Interesting Results...

Can you evaluate the following?

$$(z^*)^* = z$$

$$zz^* = |z|^2$$

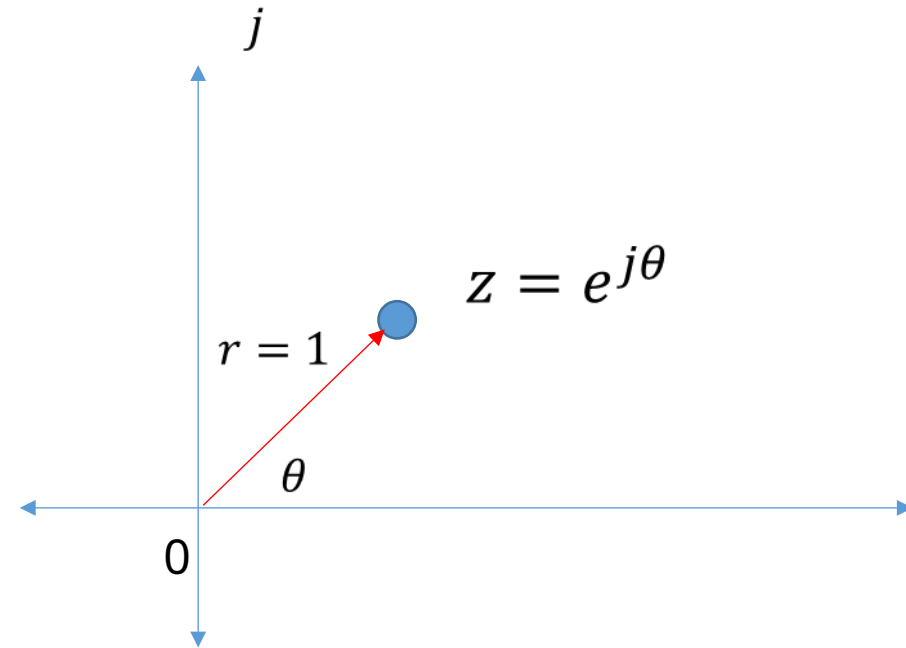
$$|e^{j\theta}| = \sqrt{\cos^2 \theta + \sin^2 \theta} = 1$$

$$e^{j2\pi n} = 1 \quad (n = \text{integer})$$

COMPLEX SINUSOID

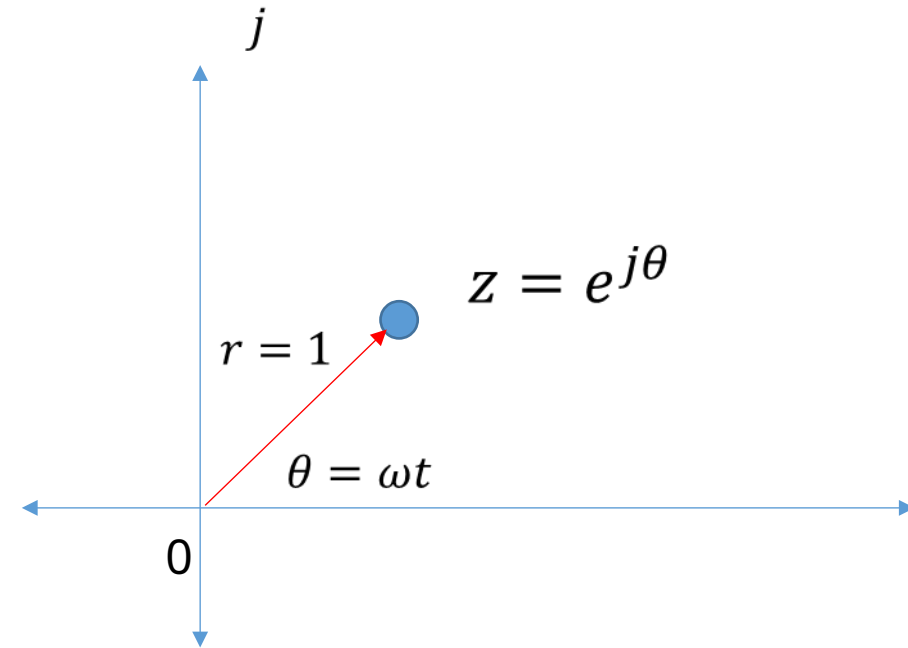
COMPLEX SINUSOID

- Consider a complex number with modulus 1 ($|z| = r = 1$)



COMPLEX SINUSOID

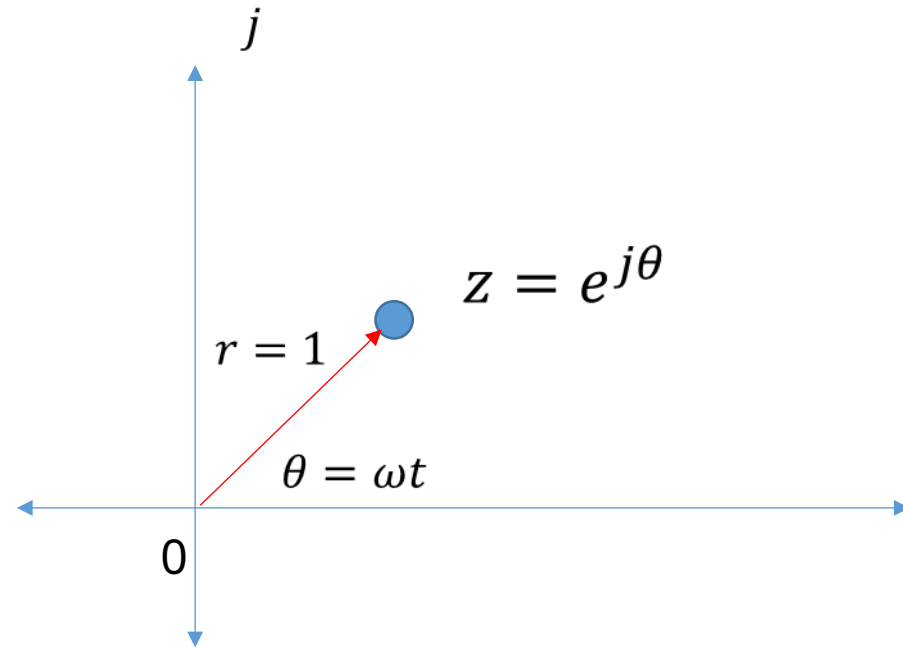
- Consider a complex number with modulus 1 ($|z| = r = 1$)
- What happens if we make its angle time-varying, $\theta = \omega t$?



COMPLEX SINUSOID

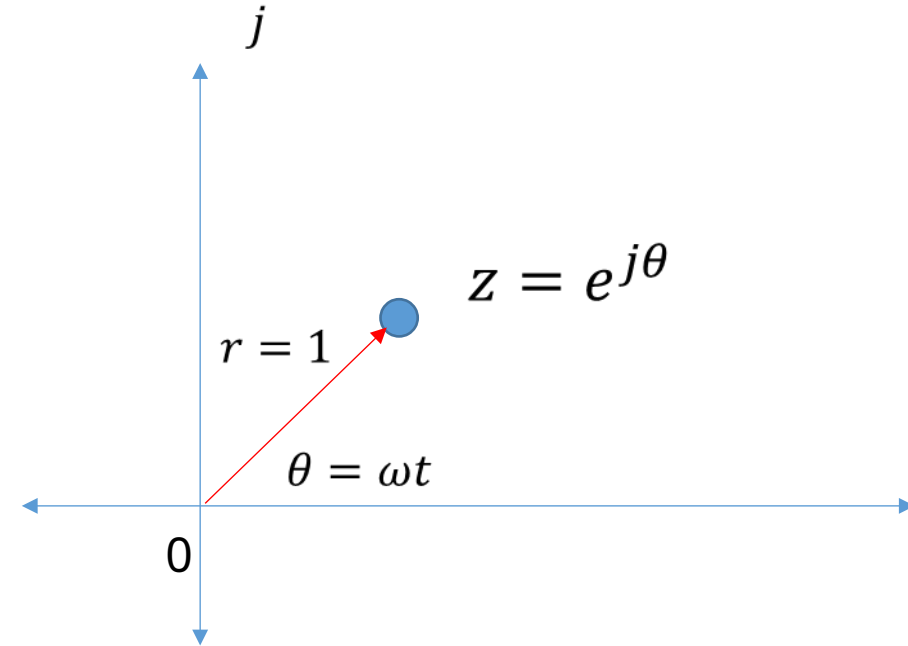
- Consider a complex number with modulus 1 ($|z| = r = 1$)
- What happens if we make its angle time-varying, $\theta = \omega t$?

$$e^{j6} \longrightarrow e^{j\omega t}$$



COMPLEX SINUSOID

- Consider a complex number with modulus 1 ($|z| = r = 1$)
- What happens if we make its angle time-varying, $\theta = \omega t$?

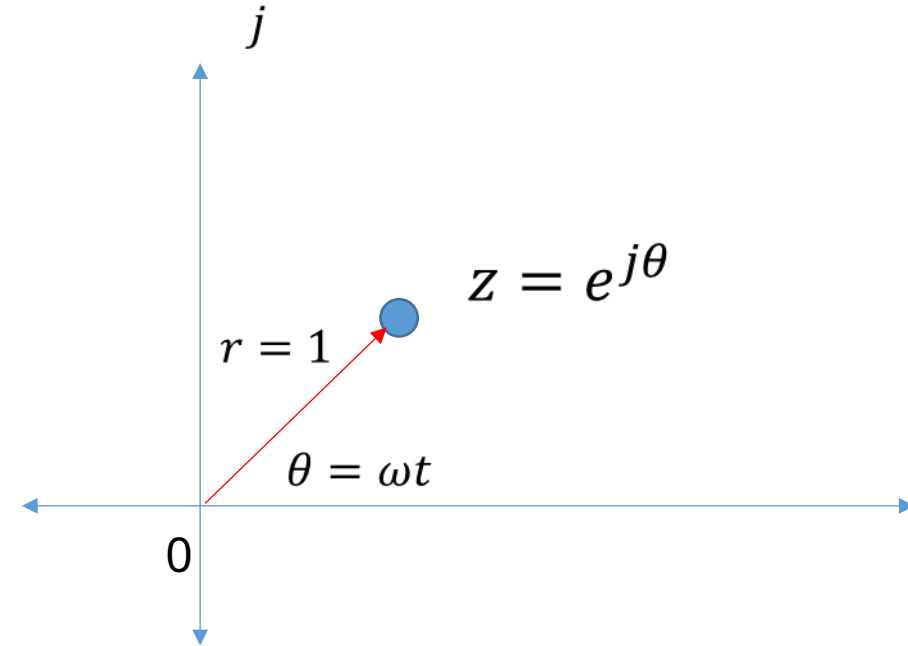


$$e^{j6} \rightarrow e^{j\omega t}$$

We get a complex-valued signal called the **COMPLEX SINUSOID**

COMPLEX SINUSOID

- Consider a complex number with modulus 1 ($|z| = r = 1$)
- What happens if we make its angle time-varying, $\theta = \omega t$?



$$e^{j\theta} \rightarrow e^{j\omega t}$$

We get a complex-valued signal called the **COMPLEX SINUSOID**

*Why
"Sinusoid"?*

A Sine and a Cosine Walk Into an Imaginary Bar...

A Sine and a Cosine Walk Into an Imaginary Bar...

... and Euler
says

$$e^{j\omega t} = \cos(\omega t) + j \sin(\omega t)$$

When $\theta = \omega t$ and t is continuous

When $\theta = \omega t$ and t is continuous

*Mathematically we
get*

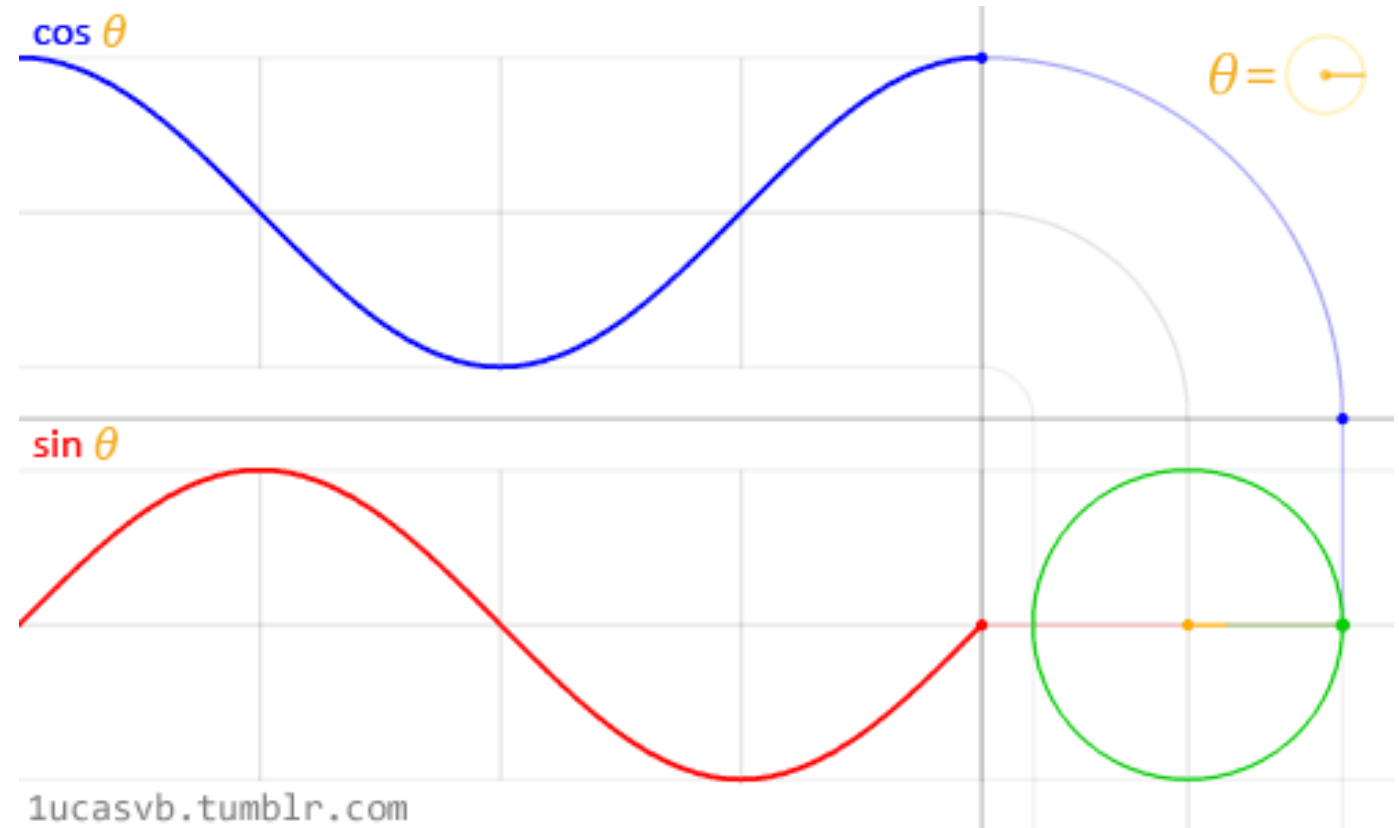
$$e^{j\theta} = e^{j\omega t} = \cos(\omega t) + j \sin(\omega t)$$

When $\theta = \omega t$ and t is continuous

Mathematically we get

$$e^{j\theta} = e^{j\omega t} = \cos(\omega t) + j \sin(\omega t)$$

And graphically we get



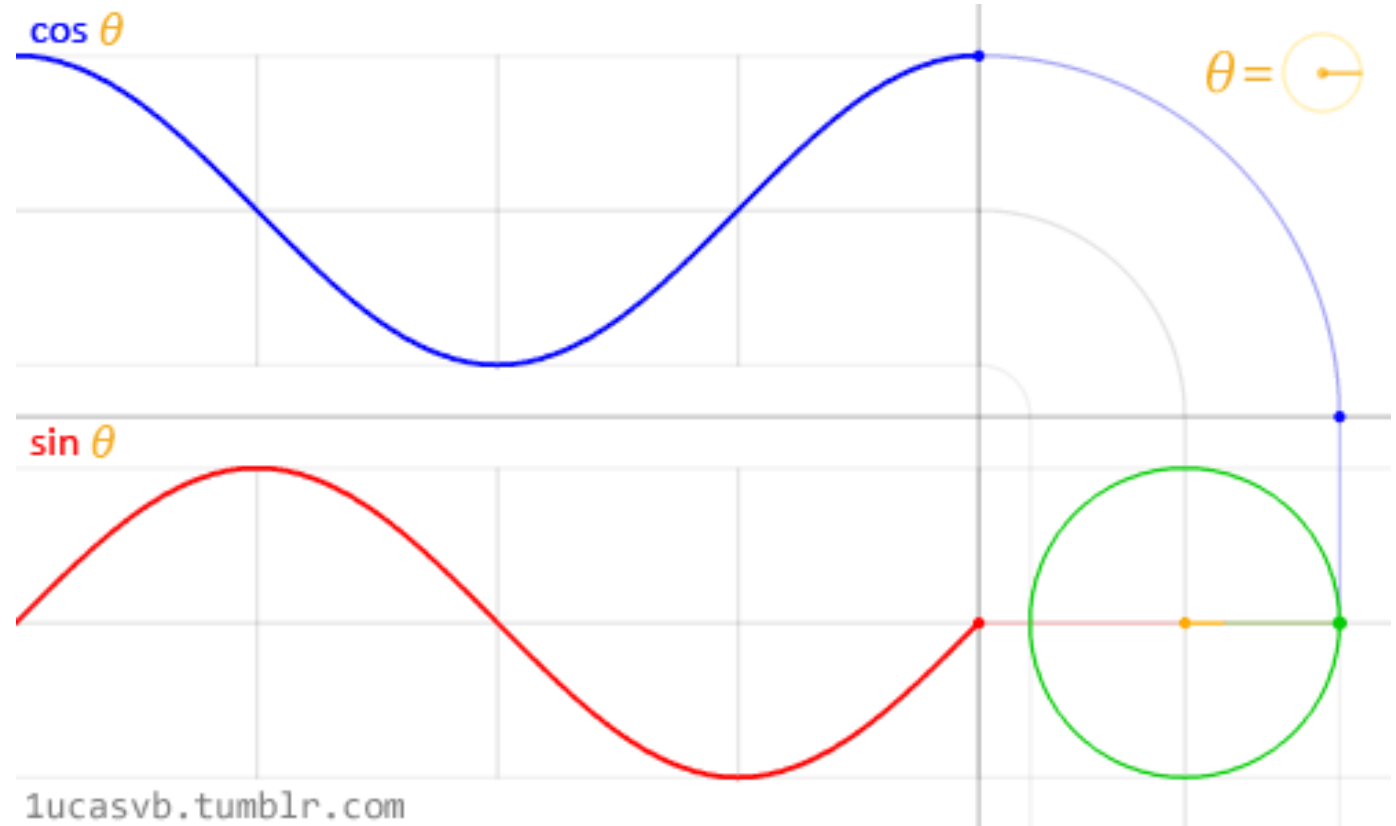
When $\theta = \omega t$ and t is continuous

Mathematically we get

$$e^{j\theta} = e^{j\omega t} = \cos(\omega t) + j \sin(\omega t)$$

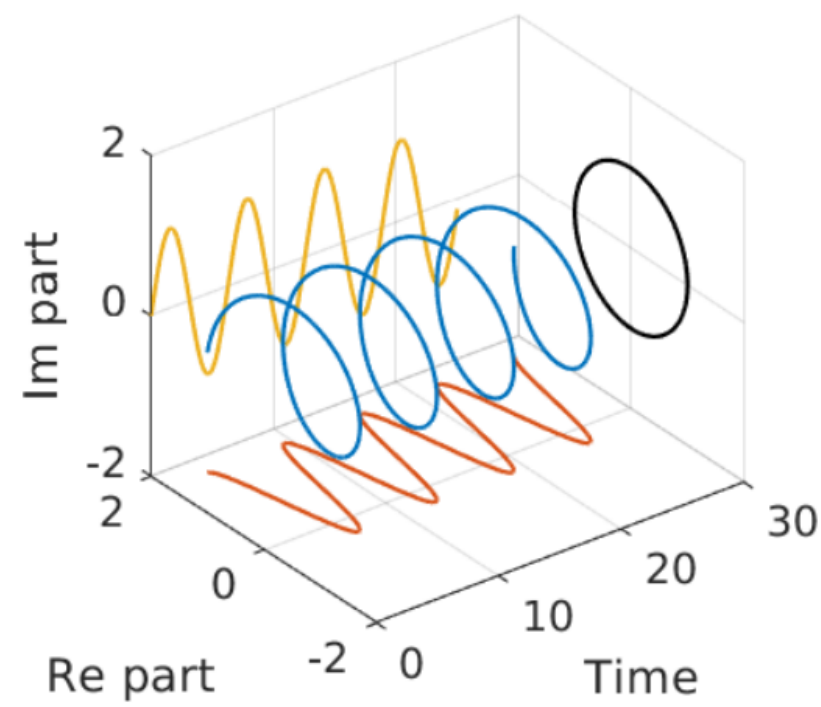
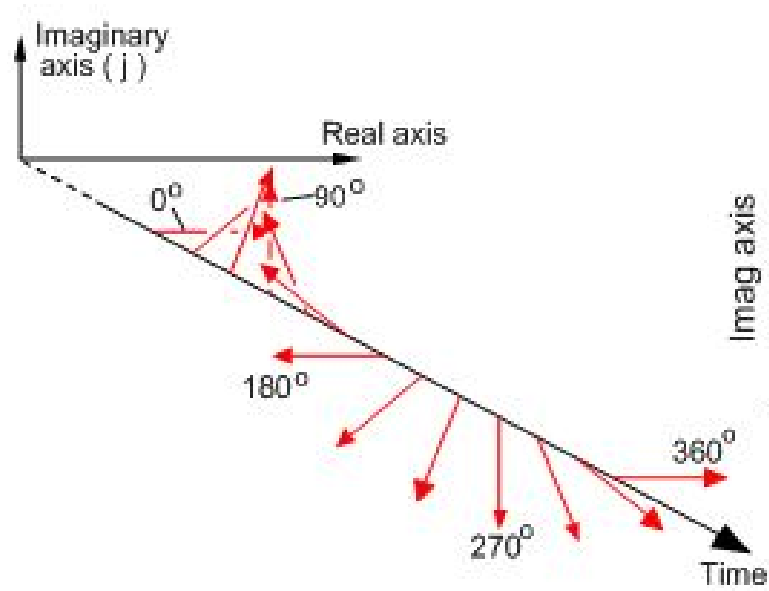
And graphically we get

- The complex number $z^t = e^{j\omega t}$ traces a unit circle in the complex plane.
- Its real part traces a cosine function.
- And its imaginary part traces a sine function.



$$e^{j\theta} = e^{j\omega t} = \cos(\omega t) + j \sin(\omega t)$$

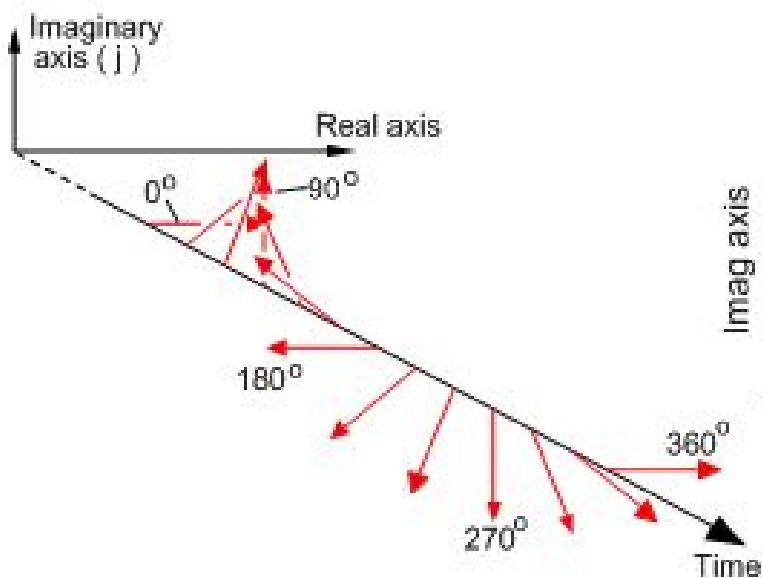
If we include the time as an axis...



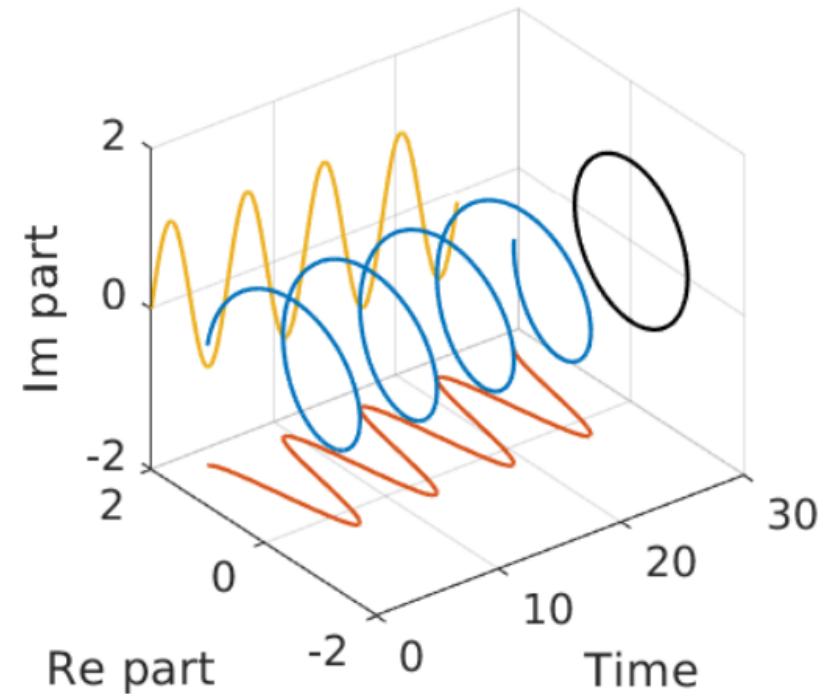
$$e^{j\theta} = e^{j\omega t} = \cos(\omega t) + j \sin(\omega t)$$

If we include the time as an axis...

- The complex sinusoid as a rotating vector.



- The complex sinusoid as a spiral, whose real and imaginary parts trace cosine and sine.



From

**COMPLEX
SINUSOID**

$$e^{j\omega t}$$

To

**COMPLEX
EXPONENTIAL**

$$e^{st} = e^{(\sigma + j\omega)t}$$

From

COMPLEX
SINUSOID

$$e^{j\omega t}$$

To

COMPLEX
EXPONENTIAL

$$e^{st} = e^{(\sigma + j\omega)t}$$

- The complex sinusoid traces a unit circle in the complex plane as its modulus is always 1.
- **What if we allow the modulus to also grow or decrease with time?**

From

COMPLEX
SINUSOID

$$e^{j\omega t}$$

To

COMPLEX
EXPONENTIAL

$$e^{st} = e^{(\sigma + j\omega)t}$$

- The complex sinusoid traces a unit circle in the complex plane as its modulus is always 1.
- **What if we allow the modulus to also grow or decrease with time?**
 - *We get the more general signal – Complex Exponential!*

$$e^{(\sigma+j\omega)t} = e^{\sigma t} e^{j\omega t}$$

Complex Exponential is basically a Complex Sinusoid with possibly decaying or growing amplitude.

$$e^{(\sigma+j\omega)t} = e^{\sigma t} e^{j\omega t}$$

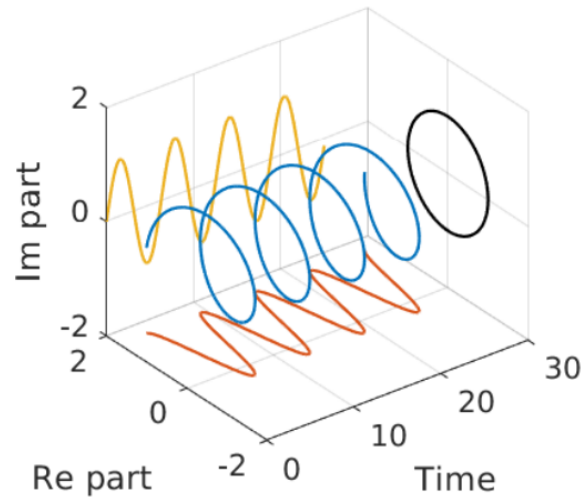
Complex Exponential is basically a Complex Sinusoid with possibly decaying or growing amplitude.

Graphically...

$$\sigma = 0$$

$$\sigma < 0$$

$$\sigma > 0$$

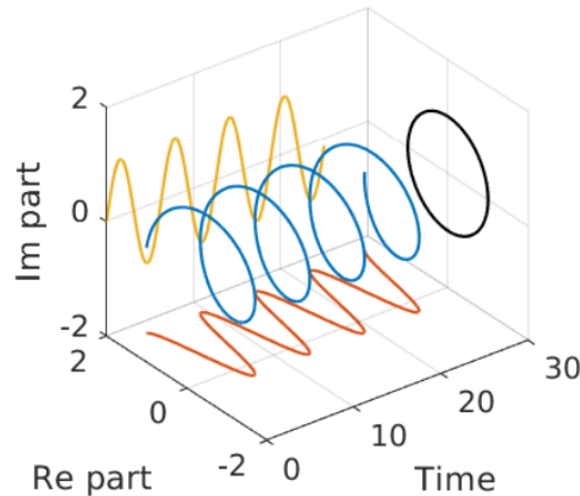


$$e^{(\sigma+j\omega)t} = e^{\sigma t} e^{j\omega t}$$

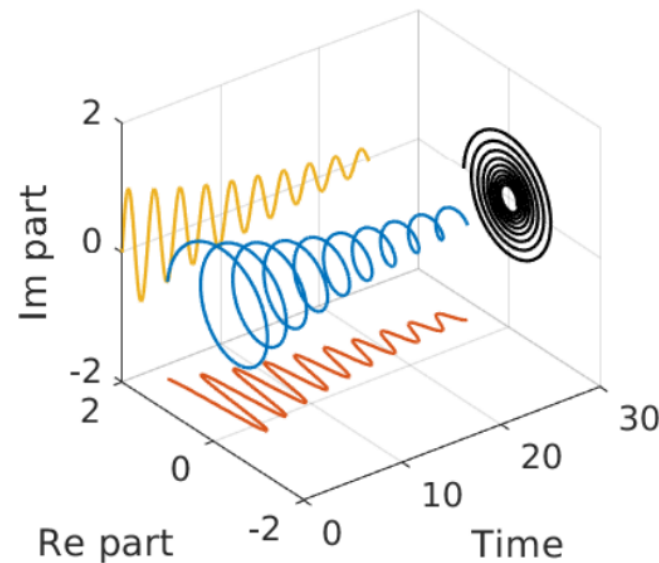
Complex Exponential is basically a Complex Sinusoid with possibly decaying or growing amplitude.

Graphically...

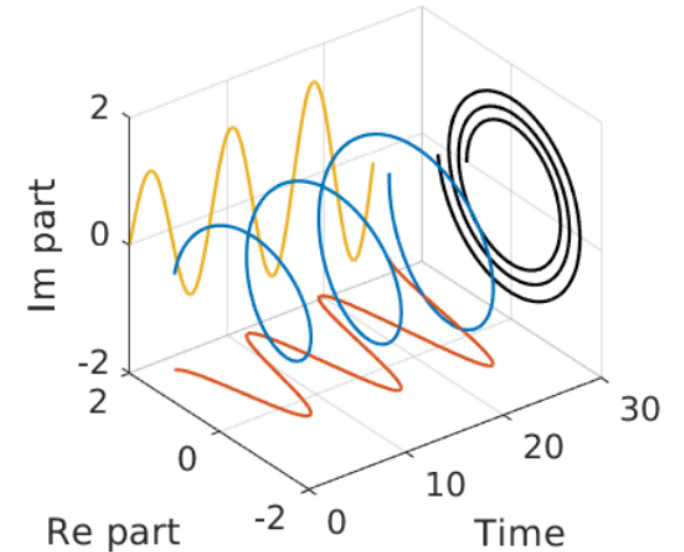
$$\sigma = 0$$



$$\sigma < 0$$



$$\sigma > 0$$

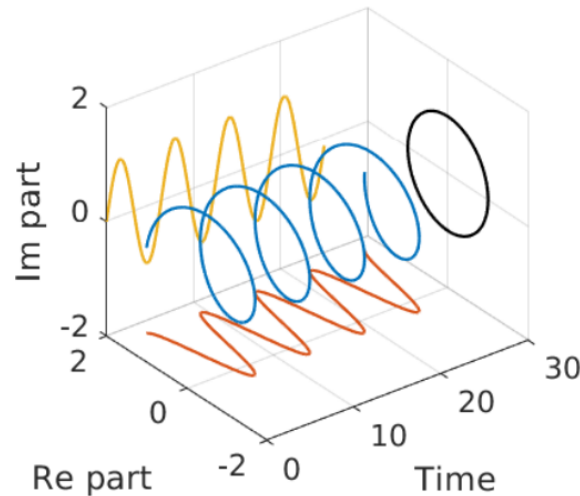


$$e^{(\sigma+j\omega)t} = e^{\sigma t} e^{j\omega t}$$

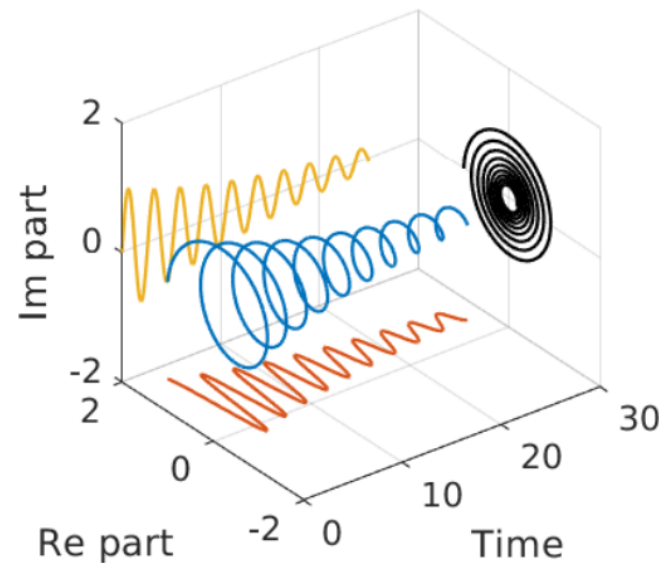
Complex Exponential is basically a Complex Sinusoid with possibly decaying or growing amplitude.

Graphically...

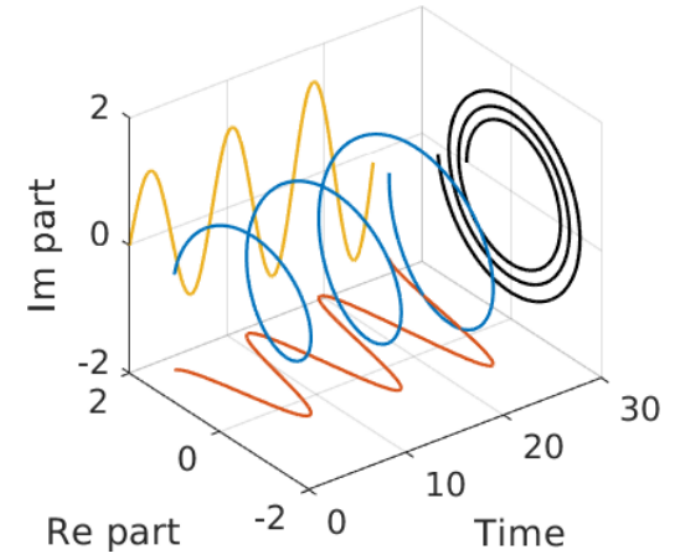
$$\sigma = 0$$



$$\sigma < 0$$



$$\sigma > 0$$



HW: how do these look for $\omega = 0$?

What if the time variable in the complex exponential is not continuous (t) but discrete (n) instead?

What if the time variable in the complex exponential is not continuous (t) but discrete (n) instead?

$$e^{\sigma n} e^{j\omega n} = |z|^n e^{j\omega n}$$

What if the time variable in the complex exponential is not continuous (t) but discrete (n) instead?

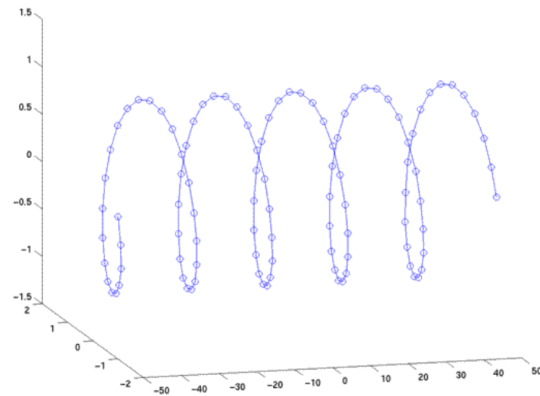
$$e^{\sigma n} e^{j\omega n} = |z|^n e^{j\omega n}$$

Graphically...

$$|z| = 1$$

$$|z| < 1$$

$$|z| > 1$$

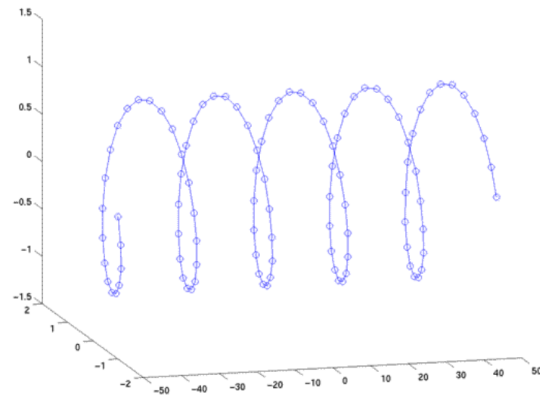


What if the time variable in the complex exponential is not continuous (t) but discrete (n) instead?

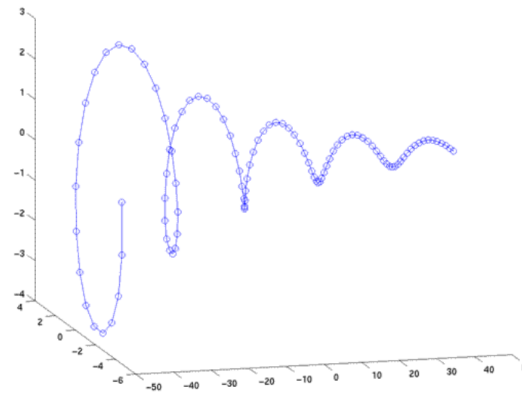
$$e^{\sigma n} e^{j\omega n} = |z|^n e^{j\omega n}$$

Graphically...

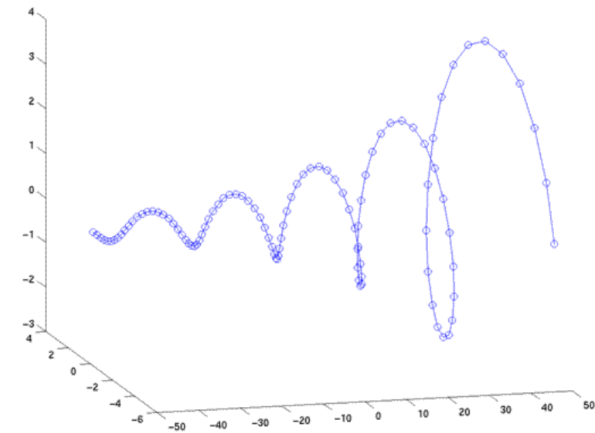
$$|z| = 1$$



$$|z| < 1$$



$$|z| > 1$$

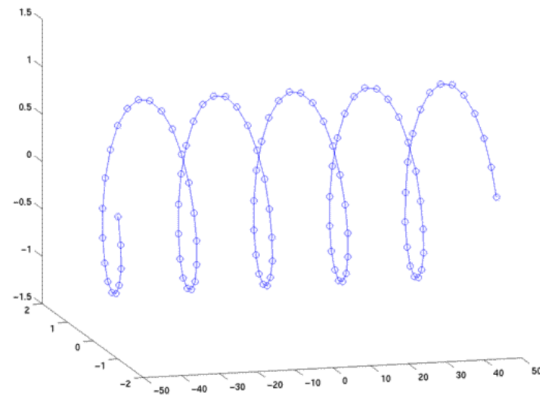


What if the time variable in the complex exponential is not continuous (t) but discrete (n) instead?

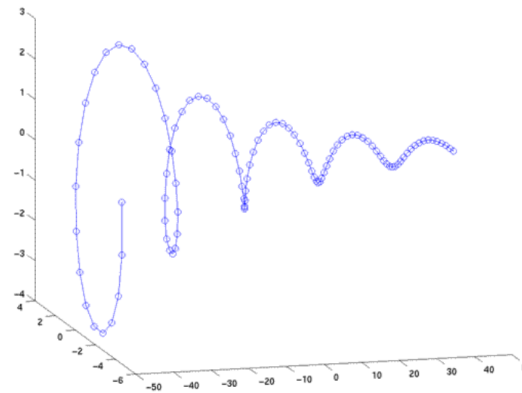
$$e^{\sigma n} e^{j\omega n} = |z|^n e^{j\omega n}$$

Graphically...

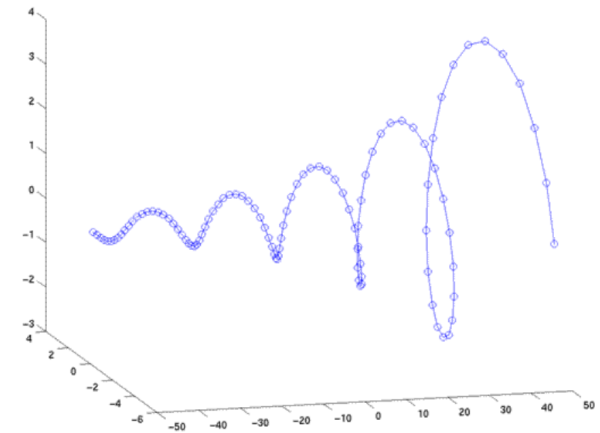
$$|z| = 1$$



$$|z| < 1$$



$$|z| > 1$$



We get discretized versions of the complex exponential!

- HW Activity
 - Prove the assertions below.

**COMPLEX
SINUSOID**

$$e^{j\omega t}$$



Periodic with period $T_0 = \frac{2\pi}{\omega}$

**COMPLEX
EXPONENTIAL**

$$e^{st} = e^{(\sigma + j\omega)t}$$



Not periodic, unless $\sigma = 0$

Questions?? Thoughts??

